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AIUB, Math 3 Note.

#Final Term

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Complex Integrations

Parametric Representation

Formula:— For Rectangular form:—

① if $z = x + iy$ the

Parametric representation of $z(t) = x(t) + iy(t)$

Example 1:—

⇒ Find and sketch the path whose orientation is given by

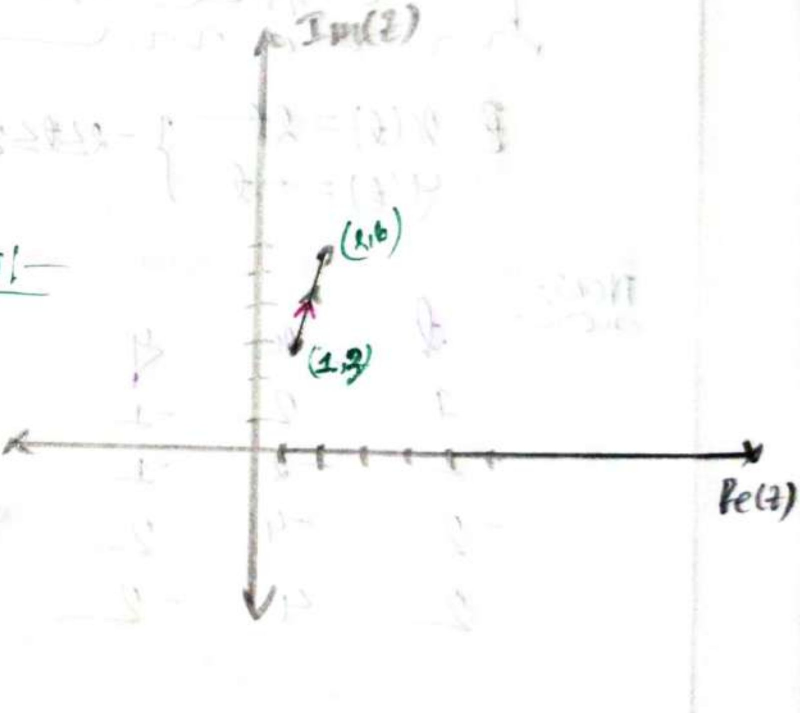
$z(t) = (1+3i)t \; ; \; 1 \le t \le 2$

⇒ In given:—

$$z(t) = (1+3i)t = t + i3t$$

So, The parametric equations:—

$$\left.\begin{aligned} x(t) &= t \\ y(t) &= 3t \end{aligned}\right\} 1 \le t \le 2$$



Now:—

t	x	y
1	1	3
2	2	6

So, $z(t) = (1+3i)t \; ; \; 1 \le t \le 2$ represents the line segment form $(1,3)$ to $(2,6)$ in complex plane.

Example: 1.2 :-

⇒ Find and Sketch the path whose orientation is given by $z(t) = (2-i)t$; $-2 \le t \le 2$

⇒ In given:-

$$z(t) = (2-i)t$$
$$= 2t - it \quad \ldots\ldots (1)$$

we know:-

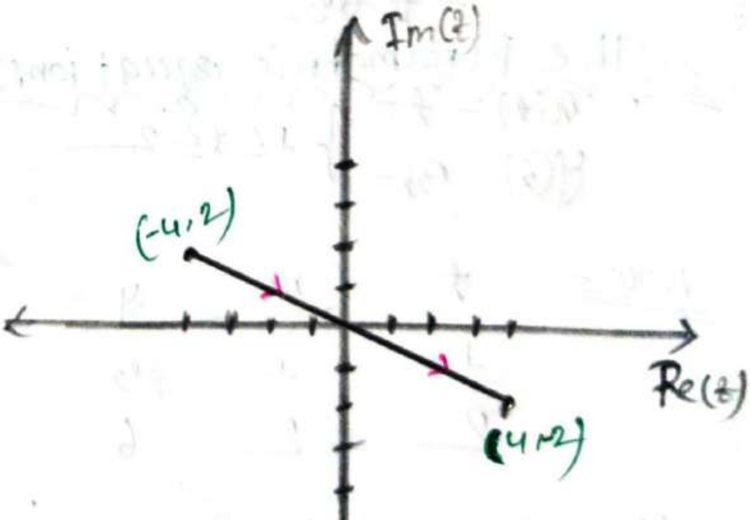
Paramatric equation of $z(t) = x(t) + iy(t) \quad \ldots (11)$

$\therefore$ from (1) And (11) eq :-

$$\left.\begin{aligned} x(t) &= 2t \\ y(t) &= -t \end{aligned}\right\} -2 \le t \le 2$$

Now:-

t	x	y
1	2	-1
-1	-2	1
-2	-4	2
2	4	-2



So, $z(t) = (2-i)t$; $-2 \le t \le 2$ represents the line segment from $(-4,2)$ to $(4,-2)$ in the complex plane.

Formula For polar form:-

(1) if $z(t) = \cos t + i\sin t$ then

Parametric representation of $z(t) = x(t) + iy(t)$

Unit Circle:-

⇒ For Unit circle Center (0,0) and radius = 1

And equation of Unit circle :- $x^2+y^2=1^2$

Now polar form:-

$$x = r\cos t$$
$$y = r\sin t \quad \text{and}$$
$$\cos^2 t + \sin^2 t = 1$$

Example : 4:-

⇒ Sketch and represent Unit circle parametrically. For $|z|=1$

⇒ In given:-

$$|z| = 1$$
$$\Rightarrow |x+iy| = 1$$
$$\Rightarrow \sqrt{x^2+y^2} = 1$$
$$\therefore x^2+y^2 = 1^2 \quad \ldots\ldots (1)$$

∴ Center (0,0)

∴ radius = 1

We know:-

$$x = r\cos t = \cos t$$
$$y = r\sin t = \sin t$$

put the value of x and y in eq (1) ⇒

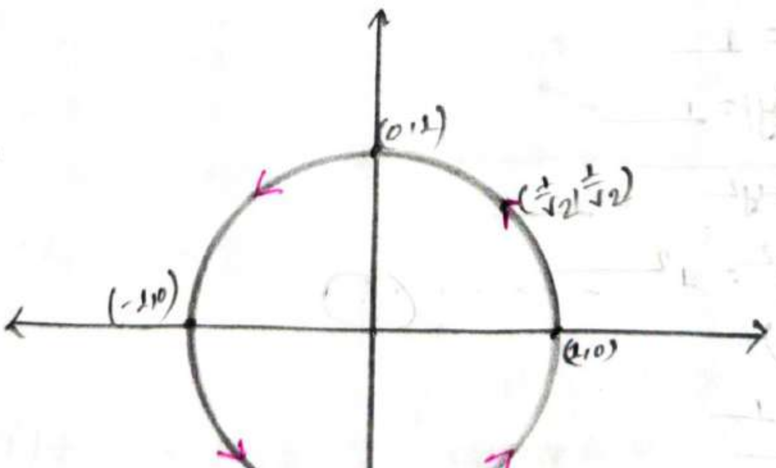
$$\therefore \cos^2 t + \sin^2 t = 1^2 \quad ; \quad 0 \le t \le 2\pi$$

Sol Parametric equations:-

$$x(t) = \cos t, \quad y(t) = \sin t \quad \Big\} \; 0 \le t \le 2\pi$$

Now:-

t	x	y
0	1	0
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$
$\frac{\pi}{2}$	0	1
π	-1	0
$\frac{3\pi}{2}$	0	-1
2π	1	0



Example:2:—

⇒ Find and sketch the path whose orientation is given by $z(t) = 2e^{it}$; $0 \le t \le \pi$

⇒ Is given:—

$$z(t) = 2e^{it}$$

$$z(t) = 2\cos t + i2\sin t$$

So, The parametric equations:—

$\therefore x(t) = 2\cos t$ ...... (i)

$y(t) = 2\sin t$ ...... (ii)

From (i) eq:—

$$\frac{x}{2} = \cos t$$

From (ii) eq:—

$$\frac{y}{2} = \sin t$$

We know:—

$$\cos^2 t + \sin^2 t = 1$$

$$\therefore \frac{x^2}{2^2} + \frac{y^2}{2^2} = 1$$

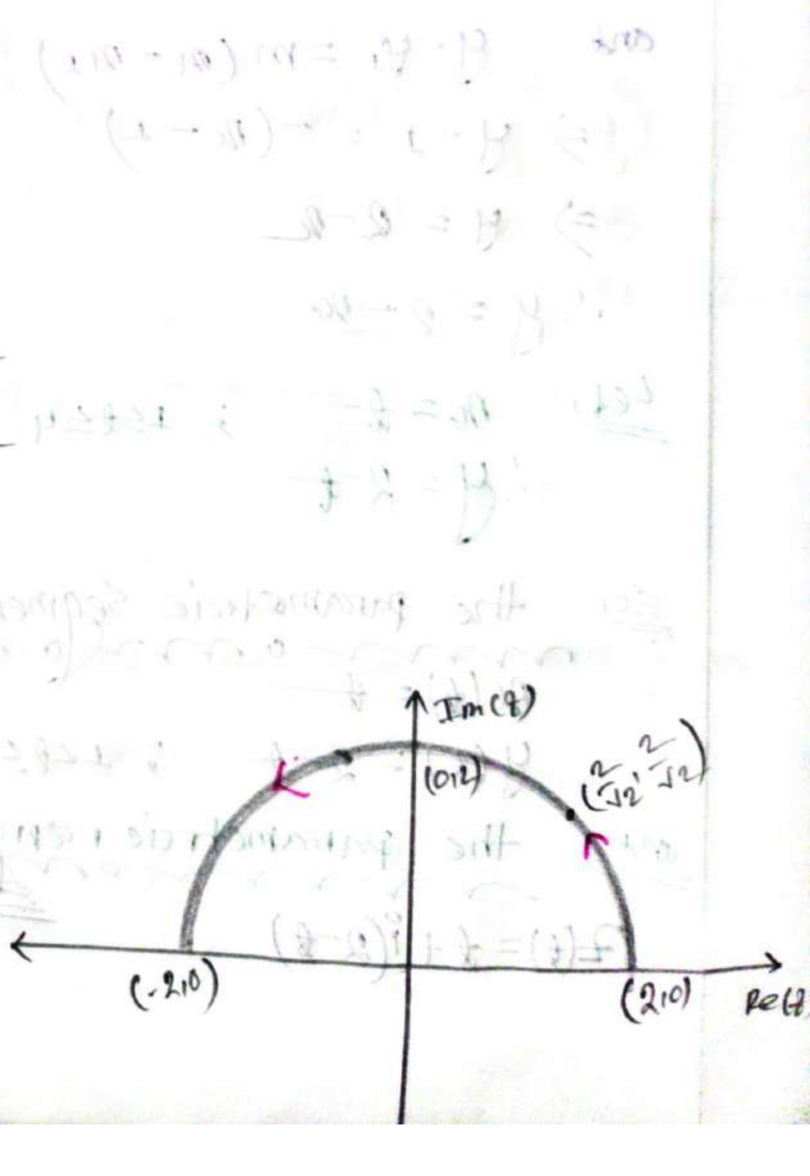
$$\Rightarrow x^2 + y^2 = 2^2$$

$\therefore$ Center $(x, y) = (0, 0)$

$r = 2$

Now:

t	x	y
0	2	0
$\frac{\pi}{4}$	$\frac{2}{\sqrt{2}}$	$\frac{2}{\sqrt{2}}$
$\frac{\pi}{2}$	0	2
π	-2	0



Example: 3:-

$\Rightarrow$ Sketch and represent the line segment from $1+i$ to $4-2i$ parametrically

$\Rightarrow$ In given:-

The straight line passing $(1,1)$ and $(4,-2)$ points

So we know:-

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-2-1}{4-1}$$

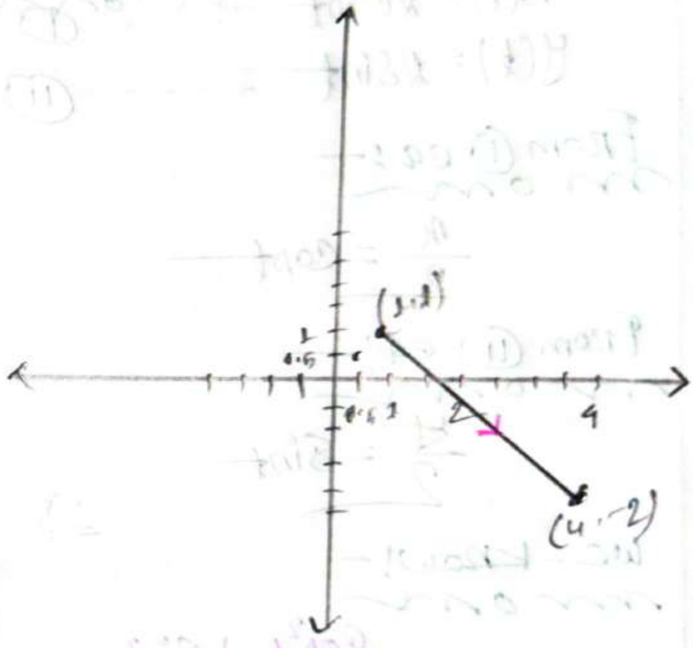
$$= -1$$

and $\quad y - y_1 = m(x - x_1)$

$\Rightarrow y - 1 = -(x-1)$

$\Rightarrow y = 2 - x$

$\therefore y = 2 - x$



Let, $\quad x = t \quad ; \; 1 \le t \le 4$

$\therefore y = 2 - t$

So the parametric segment of $1+i$ to $4-2i$ is

$x(t) = t$

$y(t) = 2 - t \quad ; \; 1 \le t \le 4$

and the parametric representation:-

$z(t) = t + i(2-t)$

Ans:-

Sample Exercises

Q:1. Find and sketch the path and its orientation. Also classify whether the indicated points are interior, exterior or boundary of the following curves.

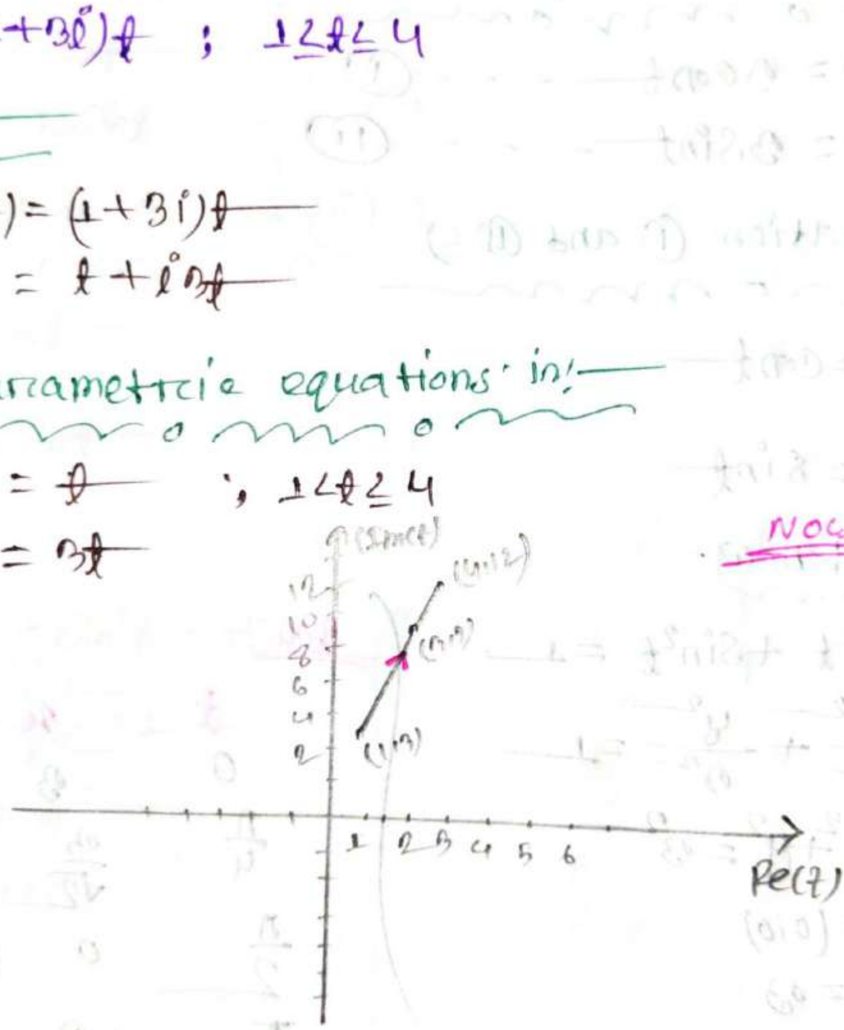
(i) $z(t) = (1+3i)t \;;\; 1 \le t \le 4$

$\Rightarrow$ In given:

$$z(t) = (1+3i)t$$
$$= t + i3t$$

So, the parametric equations is:

$$x(t) = t \quad ; \; 1 \le t \le 4$$
$$y(t) = 3t$$



Now,

t	x	y
1	1	3
3	3	9
4	4	12

So, $z = (1+3i)t \;;\; (1 \le t \le 4)$ represent the line segment from $(1,3)$ to $(4,12)$ in complex plane.

(iii) $z(t) = 3e^{it} \quad (0 \le t \le \pi)$

$\Rightarrow$ is given:-

$$z(t) = 3e^{it}$$

$$\therefore z(t) = 3\cos t + i\,3\sin t$$

So, the parrametric equation:-

$$x(t) = 3\cos t \quad \text{--- (i)}$$

$$y(t) = 3\sin t \quad \text{--- (ii)}$$

From equation (i) and (ii) $\Rightarrow$

$$\frac{x}{3} = \cos t$$

$$\frac{y}{3} = \sin t$$

Now, we know-

$$\cos^2 t + \sin^2 t = 1$$

$$\therefore \frac{x^2}{3^2} + \frac{y^2}{3^2} = 1$$

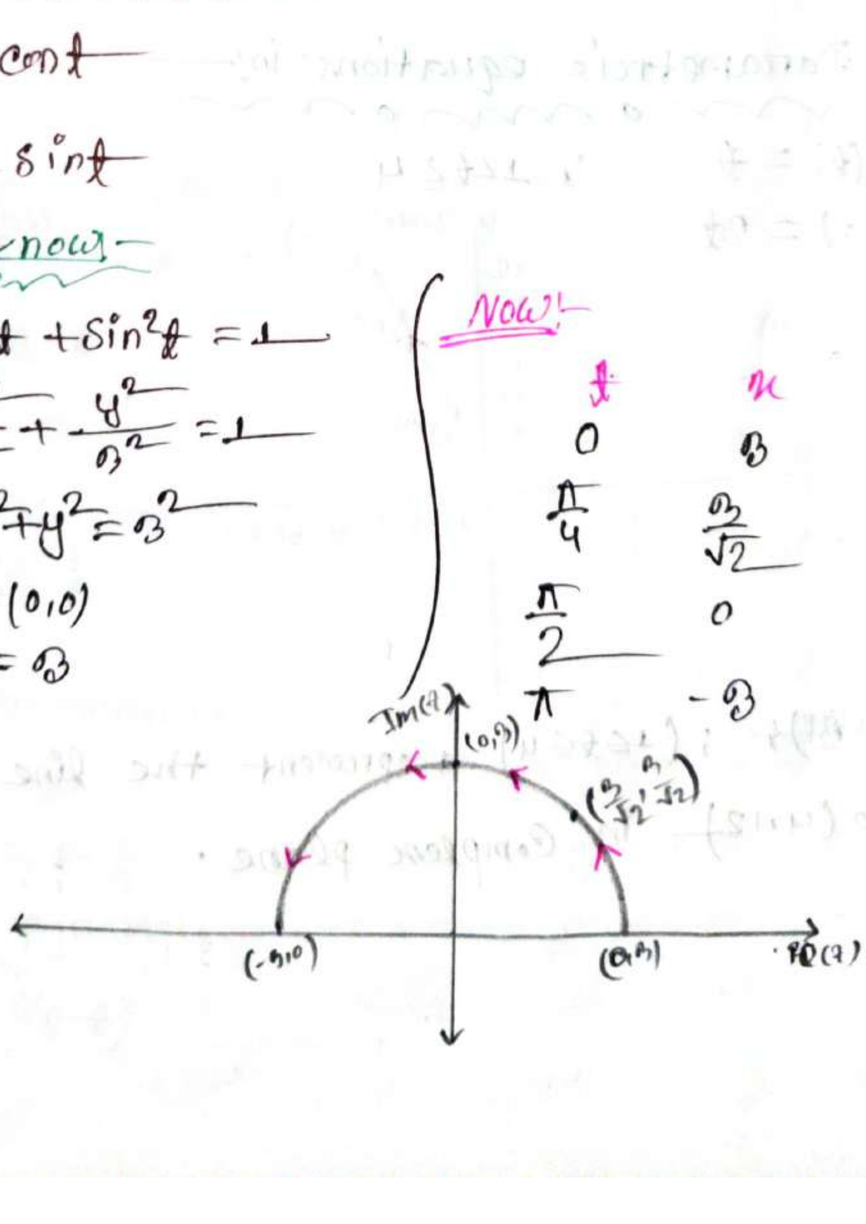
So, $x^2 + y^2 = 3^2$

$\therefore (x, y) = (0, 0)$

$r = 3$

Now:-

t	x	y
0	3	0
$\frac{\pi}{4}$	$\frac{3}{\sqrt{2}}$	$\frac{3}{\sqrt{2}}$
$\frac{\pi}{2}$	0	3
π	-3	0



(iv) $z(t) = 5e^{-it}$, $\left(0 \le t \le \frac{\pi}{2}\right)$

$\Rightarrow$ In given:—

$$z(t) = 5e^{-it}$$

$$\therefore z(t) = 5\cos t - i\,5\sin t$$

So, the parametric equation is:—

$$x(t) = 5\cos t \quad \text{---} \quad (i)$$

$$y(t) = -5\sin t \quad \text{---} \quad (ii)$$

From equation (i) and (ii) $\Rightarrow$

$$\frac{x}{5} = \cos t$$

$$\frac{y}{5} = -\sin t$$

We know:—

$$\cos^2 t + \sin^2 t = 1$$

$$\Rightarrow \frac{x^2}{5^2} + \frac{y^2}{5^2} = 1$$

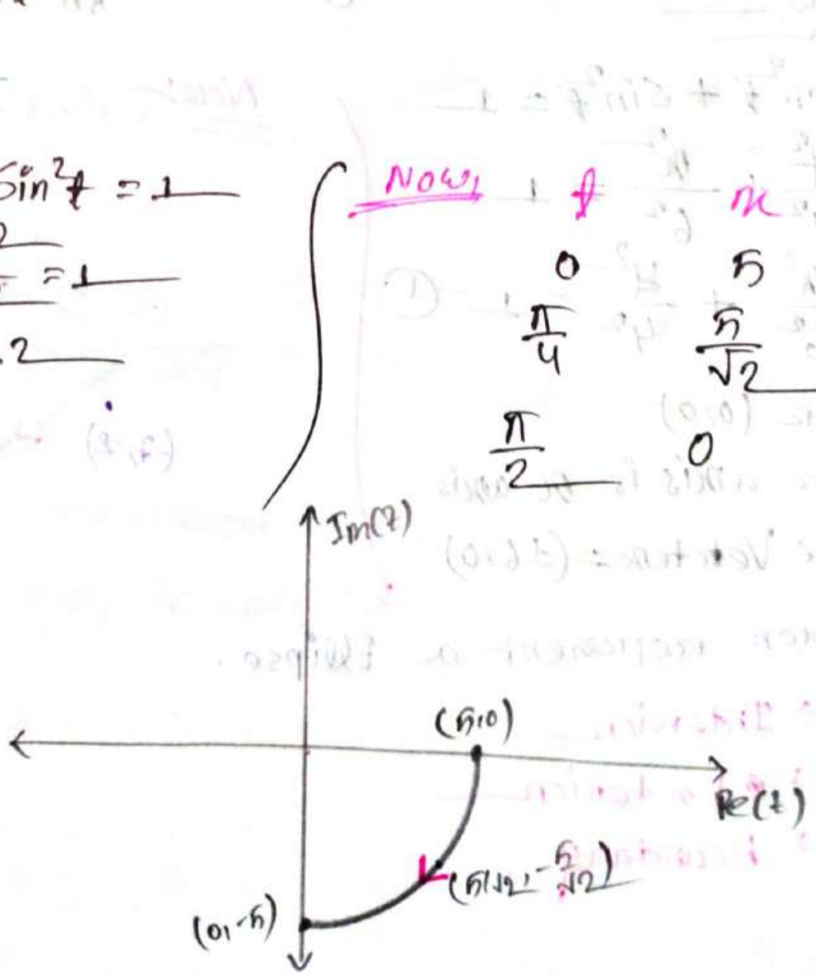
$$\therefore x^2 + y^2 = 5^2$$

$$\therefore (x, y) = (0, 0)$$

$$r = 5$$

Now,

t	x	y
0	5	0
$\frac{\pi}{4}$	$\frac{5}{\sqrt{2}}$	$-\frac{5}{\sqrt{2}}$
$\frac{\pi}{2}$	0	-5



(V) $z(t) = 6\sin(t) + i4\cos(t)$: $(0 \le t \le 2\pi)$; $(5,1), (-7,-2); (4,0)$

$\Rightarrow$ In given:—

$$z(t) = 6\sin(t) + i4\cos(t)$$

So, the parametric equation is:—

$$x(t) = 6\sin(t) \quad \text{------} \quad (i)$$

$$y(t) = 4\cos(t) \quad \text{------} \quad (ii)$$

from (i) and (ii) $\Rightarrow$

$$\frac{x}{6} = \sin t$$

$$\frac{y}{4} = \cos t$$

We know:—

$$\cos^2 t + \sin^2 t = 1$$

$$\frac{y^2}{4^2} + \frac{x^2}{6^2} = 1$$

$$\Rightarrow \frac{x^2}{6^2} + \frac{y^2}{4^2} = 1 \quad \text{—} (1)$$

$\therefore$ Center $(0,0)$

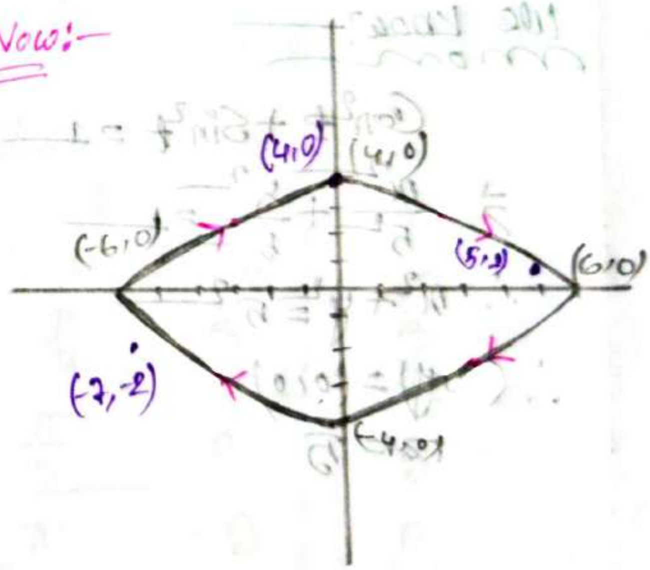
major axis is x axis

so, the Vertex $= (\pm 6, 0)$

Now:—

t	x	y
0	0	4
$\frac{\pi}{2}$	6	0
π	$+0$	-4
2π	-6	$+0$

Now:—



The equation represent a Ellipse.

$(5,1) \rightarrow$ Interior

$(-7,-2) \rightarrow$ Exterior

$(4,0) \rightarrow$ boundary

(vi) $z(t) = 2\cos(t) + i\sin(t)$; $(0 \le t \le 2\pi)$; $(6,5)$

=> In given:—

$$z(t) = 2\cos(t) + i\sin t$$

So, the parametric equation is:—

$$x(t) = 2\cos(t) \quad \dots\dots (i)$$
$$y(t) = \sin t \quad \dots\dots (ii)$$

From eq (i) and (ii) =>

$$\frac{x}{2} = \cos t$$

$$\frac{y}{1} = \sin t$$

We know:—

$$\cos^2 t + \sin^2 t = 1$$

$$\therefore \frac{x^2}{2^2} + \frac{y^2}{1^2} = 1$$

Center $(x,y) = (0,0)$

Major axis is x axis

& The Vertex $(\pm 2, 0)$

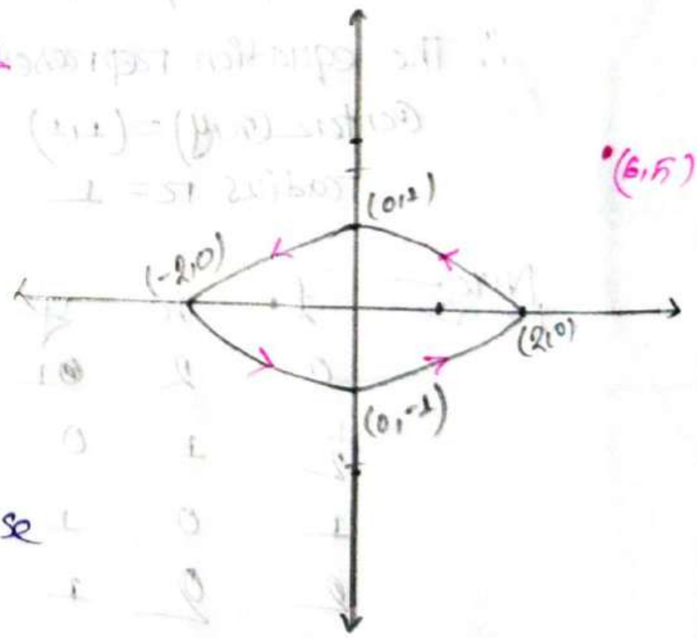
The equation represent a Ellipse

So, The point $(6,5)$ is exterior.

Now

t	x	y
0	2	0
$\frac{\pi}{2}$	0	1
π	-2	0
2π	0	-1

So,



(vii) $z(t) = 1 + i + e^{-\pi i t} \quad (0 \le t \le 2)$

⇒ In given:

$$z(t) = 1 + i + e^{-\pi i t}$$
$$= 1 + i + \cos\pi t - i\sin\pi t$$
$$= (1 + \cos\pi t) + i(1 - \sin\pi t)$$

So, the parametric eqs —

$$x(t) = 1 + \cos\pi t \quad \ldots\ldots (\mathrm{I})$$
$$y(t) = 1 - \sin\pi t \quad \ldots\ldots (\mathrm{II})$$

from eq (I) and (II) ⇒

$$\cos\pi t = x - 1$$
$$-\sin\pi t = y - 1$$

We know:

$$\cos^2\pi t + \sin^2\pi t = 1$$
$$\therefore (x-1)^2 + (y-1)^2 = 1$$

∴ The equation represent a circle

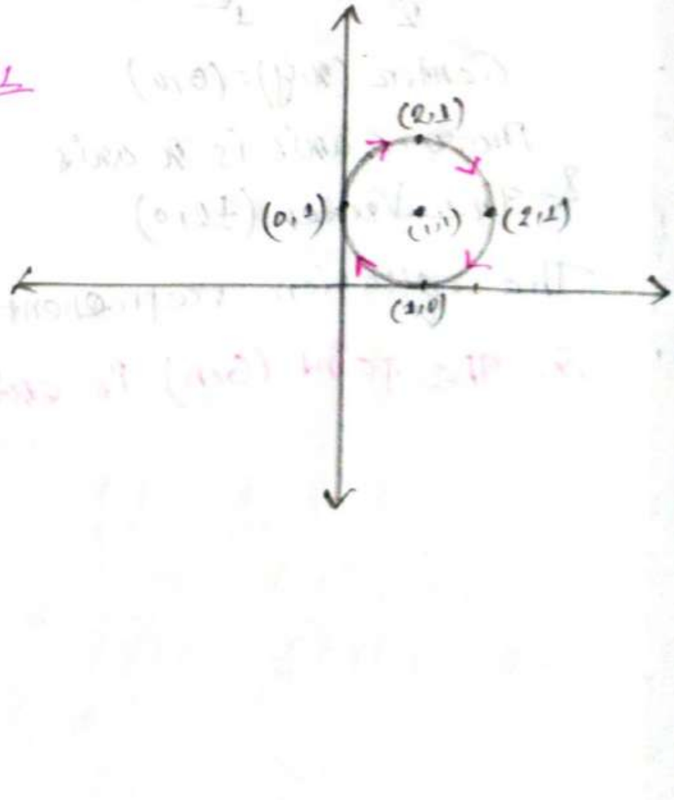
Center $(x, y) = (1, 1)$

radius $r = 1$

Now —

t	x	y
0	2	1
$\frac{1}{2}$	1	0
1	0	1
2	2	1

So,



(vii) $z(t) = 3+4i + 5\cosh t + i2\sinh t$

$\Rightarrow$ In given:-

$$z(t) = 3+4i+5\cosh t + i2\sinh t$$

$$= (3+5\cosh t) + i(4+2\sinh t)$$

So, The parametric equation is:-

$x(t) = 3+5\cosh t$ .... (i)

$y(t) = 4+2\sinh t$ .... (ii)

From the equation of (i), (ii) $\Rightarrow$

$$\cosh t = \frac{x-3}{5}$$

$$\sinh t = \frac{y-4}{2}$$

We know:-

$$\cosh^2 t - \sinh^2 t = 1$$

$$\therefore \frac{(x-3)^2}{5^2} - \frac{(y-4)^2}{2^2} = 1$$

The equation represent a Hyperbola

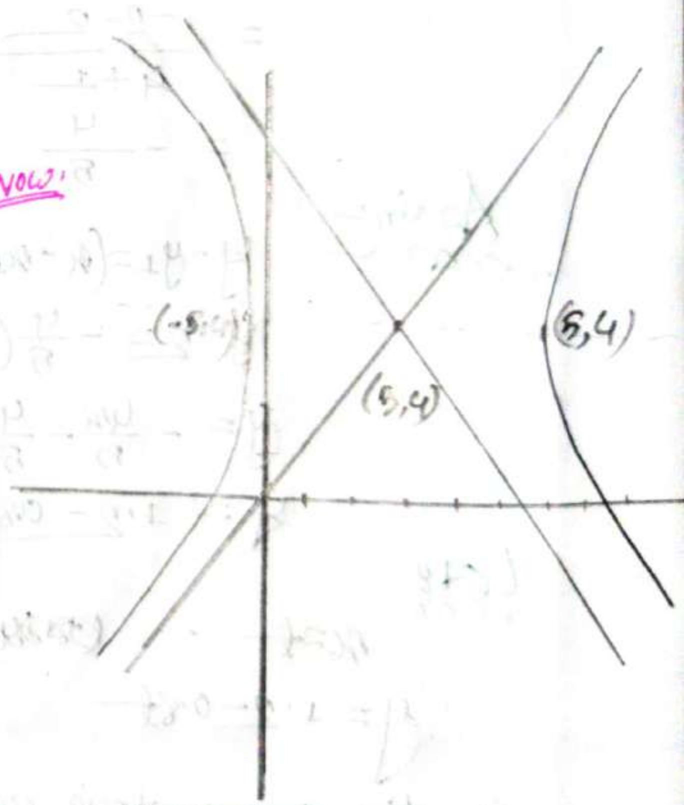
Center = (3, 4)

major axis is x axis

Vertices = (± 5, 4)

and Asymptes, $y-4 = \pm \frac{2(x-3)}{5}$

Now:



Q:2: Sketch and represent them parametrically; Also classify whether the indicate points are interior, exterior or boundary of the following curves

(i) Line Segment from $-1+2i$ to $4-2i$

$\Rightarrow$ Is given:—

The Straight line passing through the point $(-1,2)$ to $(4,-2)$

we know—

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-2-2}{4+1}$$

$$= -\frac{4}{5}$$

Again:—

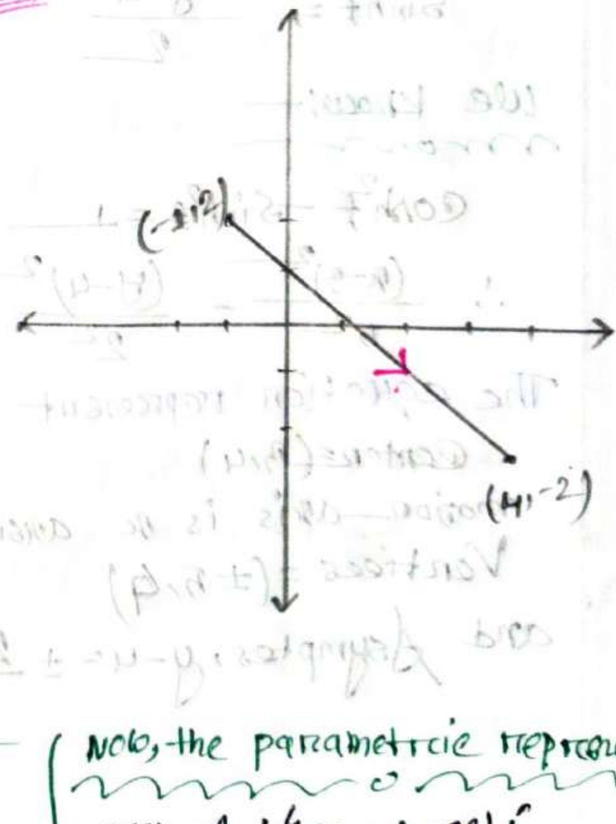
$$y - y_1 = (x - x_1) \times m$$

$$y - 2 = -\frac{4}{5}(x+1)$$

$$y = -\frac{4x}{5} - \frac{4}{5} + 2$$

$$y = 1.2 - 0.8x$$

Sol



Let:—

$x = t \quad \dots \qquad (-1 \le t \le 4)$

$\therefore y = 1.2 - 0.8t$

So the parametric equation is:—

$$x(t) = t$$

$$y(t) = 1.2 - 0.8t \qquad (1 \le t \le 4)$$

Now, the parametric representation

$$z(t) = t + (1.2 - 0.8t)\,i$$

$$\Rightarrow z(t) = t + i(1.2 - 0.8t)$$

Ans:—

(ii) Unit circle $|z|=1$ (clockwise)

$\Rightarrow$ In given:—

$$|z|=1$$
$$\Rightarrow |x+iy|=1$$
$$\Rightarrow \sqrt{x^2+y^2}=1$$
$$\Rightarrow x^2+y^2=1 \quad \cdots\cdots (1)$$
$$\therefore (x,y)=(0,0)$$
$$r=1$$

We know:—

$$x=r\cos t=\cos t$$
$$y=r\sin t=\sin t$$

put it in eq (1) $\Rightarrow$

$$x^2+y^2=1$$
$$\therefore \cos^2 t+\sin^2 t=1 \quad ; (0\le t\le 2\pi)$$

So, the parametric equations:—

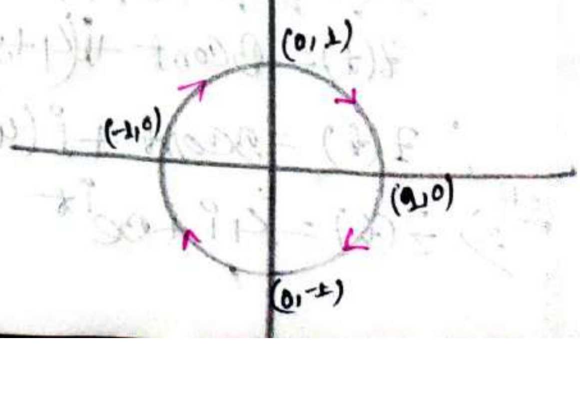
$$x(t)=\cos t \qquad 0\le t\le 2\pi$$
$$y(t)=\sin t$$

and the parametric representation:—

$$z(t)=\cos(t)+i\sin(t)$$

Now: In the case, the circle is clockwise, so we can use $-t$ instead of t in the above representation.

$$\therefore z(t)=\cos(-t)+i\sin(-t)$$
$$\Rightarrow z(t)=\cos t-i\sin t$$



Then the parametric equation:—

$$x(t)=\cos t$$
$$y(t)=-\sin t$$

(iii) $|z-4i|=3$ (counter clockwise) ; $(1,6)$

$\Rightarrow$ In given:—

$$|z-4i|=3$$
$$\Rightarrow |x+iy-4i|=3$$
$$\Rightarrow |x+i(y-4)|=3$$
$$\Rightarrow \sqrt{x^2+(y-4)^2}=3$$
$$\Rightarrow x^2+(y-4)^2=3^2 \quad (0\le t\le 2\pi)$$

$\therefore$ Center $(x,y)=(0,4)$
$r=3$

We know:—

$x=r\cos t$
$\therefore x=3\cos t$
$\Rightarrow \cos t=\frac{x}{3}$

$y-4=r\sin t$
$\therefore y-4=3\sin t$
$\Rightarrow \frac{y-4}{3}=\sin t$

Again:—

$$\cos^2 t+\sin^2 t=1$$
$$\frac{x^2}{3^2}+\frac{(y-4)^2}{3^2}=1^2$$
$$x^2+(y-4)^2=3^2$$

$\therefore$ parametric equations:—

$x(t)=3\cos t \quad (0\le t\le 2\pi)$
$y(t)=4+3\sin t$

Now, Parametric representation:—

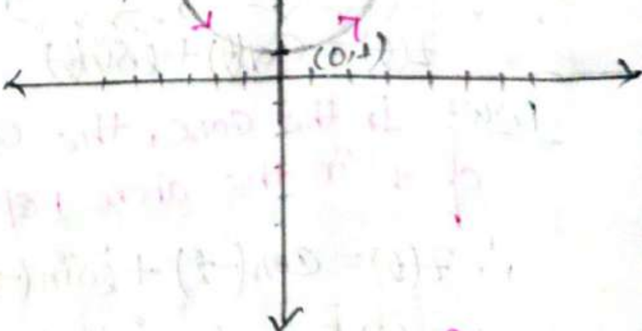
$z(t)=3\cos t+i(4+3\sin t)$

$\therefore z(t)=3\cos t+i(4+3\sin t)$

OR $\Rightarrow z(t)=4i+3e^{it}$

Now:—

t	x	y
0	3	4
$\frac{\pi}{2}$	0	7
π	-3	4
$3\pi/2$	0	+1
2π	3	4



So, The point $(1,6)$ is interior

(iv) $|z-5+i|=4$ (Counter clock wise); $(1,2)$

$\Rightarrow$ In given:—

$$|z-5+i|=4$$

$$\Rightarrow |x+iy-5+i|=4$$

$$\Rightarrow |(x-5)+i(y+1)|=4$$

$$\Rightarrow \sqrt{(x-5)^2+(y+1)^2}=4$$

$$\Rightarrow (x-5)^2+(y+1)^2=4^2$$

So, Center $(x,y)=(5,-1)$

$$r=4$$

We know:—

$x = r\cos t + 5$ | $y = r\sin t - 1$

$\therefore x = 4\cos t + 5$ | $\therefore y = 4\sin t - 1$

$\therefore \cos t = \dfrac{x-5}{4}$ | $\therefore \sin t = \dfrac{y+1}{4}$

Again we know:—

$$\cos^2 t + \sin^2 t = 1 \quad (0 \le t \le 2\pi)$$

$$\Rightarrow \frac{(x-5)^2}{4^2} + \frac{(y+1)^2}{4^2} = 1$$

$$\therefore (x-5)^2 + (y+1)^2 = 4^2$$

So, the parametric equation:—

$$x(t) = 4\cos t + 5$$

$$y(t) = 4\sin t - 1$$

Now, The parametric representation:—

$$z(t) = 5 + 4\cos t + i(4\sin t - 1)$$

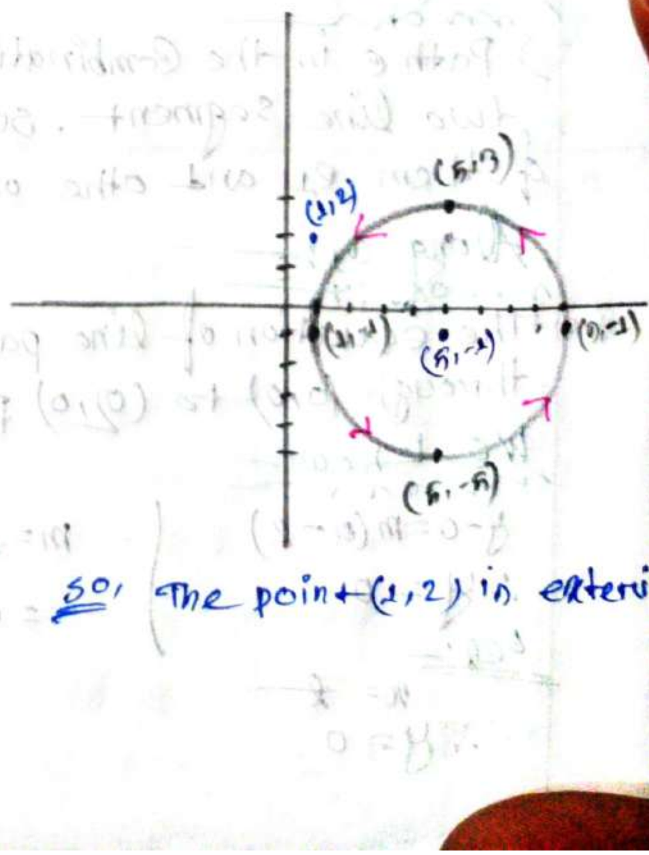
OR, $z(t) = 5 + (-i) + 4e^{it}$

$$z(t) = 5 - i + 4e^{it}$$

Now:—

t	x	y
0	9	-1
$\frac{\pi}{2}$	5	3
π	1	-1
$\frac{3\pi}{2}$	5	-5
2π	9	-1

So,



So, The point $(1,2)$ is exterior

Line Integral in the Complex plane

Formula:

① $\int_c f(z)\,dz = \int_c (x+iy)(dx+idy)$ ...→ if $f(z)=z$

② $\int_c f(z)\,dz = \int_{c_1} f(z)\,dz + \int_{c_2} f(z)\cdot dz$ → if c is the combination of c_1 and c_2

③ $\int_c f(z)\,dz = \int_c f(z(t)\,z'(t))\,dt$ ......→ if c represent by parametric representation $z(t) = x(t)+iy(t)$ $a \le t \le b$.

④ $\int \ln x = x\ln x - x + c$

Example: 5:

⇒ Sketch the path c consisting of two line segments, one from $z=0$ to $z=2$ and other from $z=2$ to $z=3+i$ hence evaluate $\int_c f(z)\,dz$, if $f(z)=z^2$.

⇒ In given:

⇒ Path c is the combination of two line segment. So let one of them c_1 and othe one c_2

Along c_1:

The equation of line passing through $(0,0)$ to $(2,0)$ points.

We know,

$y-0 = m(x-2)$ $\qquad m = \frac{0-0}{0-2} = 0$

$\therefore y = 0$

Let:

$x = t$

$\therefore y = 0$

So, The parametric equations:

$x(t) = t \quad 0 \le t \le 2$

$y(t) = 0$

Now, The parametric representation

$$\boxed{z(t) = t} \quad 0 \le t \le 2$$

We know,

$$\int_c f(z)\cdot dz = \int_{c_1} z^2\cdot dz$$

$$= \int_{c_1} (x+iy)^2 (dx+idy)$$

$$= \int_{c_1} (t+0)^2 \cdot dt \qquad \frac{dx}{dt} = 1,\ \therefore dx = dt$$

$$= \int_0^2 t^2\,dt$$

$$= \frac{1}{3}[t^3]_0^2 = \frac{8}{3}$$

Along c_2:—

⇒ The equation of line passing through (2,0) to (3,1) points.

We know:—

$$y - 0 = m(x-2) \qquad m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1-0}{3-2} = 1$$

$$\therefore y = x - 2$$

Let consider:—

$x = t \; ; \; 2 \le t \le 3$

$y = t - 2$

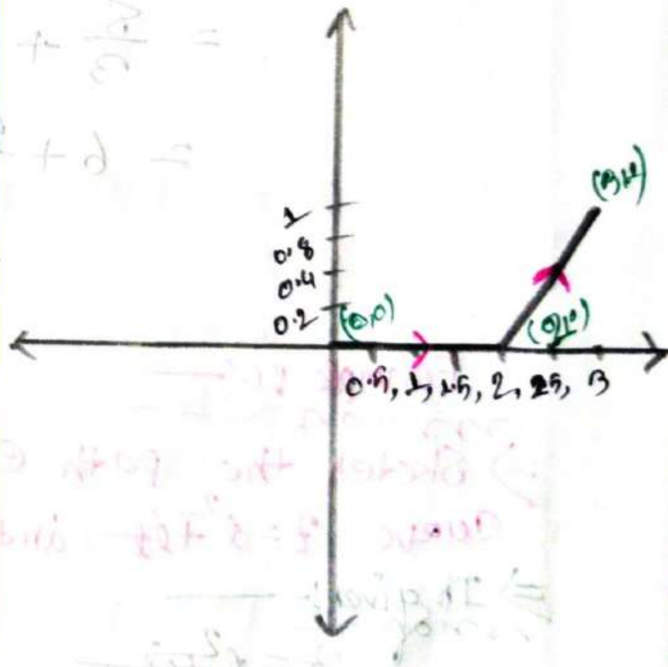
So, the parametric eq.:—

$x(t) = t$

$y(t) = t - 2$

Now, The parametric representation:—

$$\boxed{z(t) = t + i(t-2)}$$



So we know:—

$$\Rightarrow \int_c f(z)\cdot dz = \int_{c_2} z^2 \cdot dz$$

$$= \int_{c_2} (x+iy)^2 \cdot (dx + i\,dy)$$

$$= \int_{c_2} (t + i(t-2))^2 (dt + i\,dt)$$

$$= \int_2^3 (t + it - i2)^2 \cdot (1+i)\,dt$$

$$= (1+i)\int_2^3 (t^2 + i2t^2 - t^2) - 4it + 4t - 4)\,dt$$

$$= (1+i)\int_2^3 (i2t^2 - 4it + 4t - 4)\,dt$$

$\frac{dx}{dt} = t$

$\therefore dx = dt$

$\frac{dy}{dt} = t - 2$

$\therefore dy = dt$

P.T.O

$$= (1+i)\int_2^3 (i2t^2 - i4t + 4t - 4)\cdot dt$$

$$= (1+i)\left[i2\cdot\frac{1}{3}t^3 - i4\frac{1}{2}t^2 + 4\frac{1}{2}t^2 - 4t\right]_2^3$$

$$= (1+i)\left[\frac{2i}{3}(3^3-2^3) - \frac{4i}{2}(3^2-2^2) + \frac{4}{2}(3^2-2^2) - 4(3-2)\right]$$

$$= (1+i)\left[\frac{38i}{3} - 10i + 10 - 4\right]$$

$$= \frac{8i}{3} + 6 + \frac{8i^2}{3} + 6i$$

$$= \frac{10}{3} + \frac{26i}{3}$$

Now:— we know:—

$$\int_C f(z)\cdot dz = \int_{C_1} f(z)\cdot dz + \int_{C_2} f(z)\cdot dz$$

$$= \frac{8}{3} + \frac{10}{3} + \frac{26i}{3}$$

$$= 6 + \frac{26}{3}i$$

Ans:—

Example: 6:—

⇒ Sketch the path C from $z=0$ to $z=4+2i$ along the curve $z=t^2+it$ and hence evaulate $\int_C f(z)dz$ where $f(z)=\bar{z}$

⇒ In given:—

$$z = t^2+it$$

$$\therefore x+iy = t^2+it$$

∴ Parametric equation:—

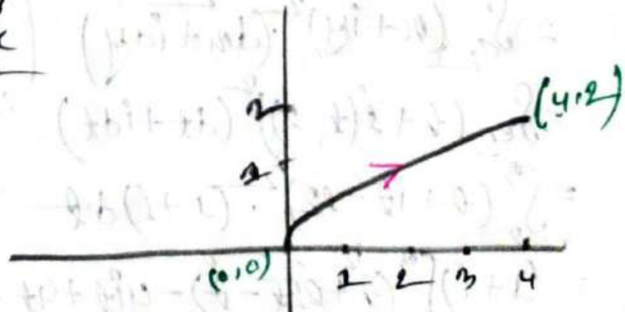
$$x(t) = t^2$$

$$y(t) = t \quad (0 \le t \le 2)$$

∴

Now, $y^2 = t^2 = x$

$\therefore x = y^2$



So, $f(z) = \bar{z} = t^2 - it$

and $z = t^2 + it$

$\therefore dz = 2t\,dt + i\,dt$

We know—

$$= \int_c f(z) \cdot dz$$

$$= \int_c \bar{z} \cdot dz$$

$$= \int_0^2 (t^2 - it)\,(2t \cdot dt + i\,dt)$$

$$= \int_0^2 (t^2 - it) \cdot (2t + i)\,dt$$

$$= \int_0^2 (2t^3 + it^2 - 2t^2 i - i^2 t)\,dt$$

$$= \int_0^2 (2t^3 - it^2 + t)\,dt$$

$$= \left[\frac{2}{4}t^4 - \frac{i}{3}t^3 + \frac{1}{2}t^2\right]_0^2$$

$$= \frac{2}{4}2^4 - \frac{i}{3}2^3 + \frac{1}{2}2^2$$

$$= 8 - \frac{8i}{3} + 2$$

$$= 10 - i\frac{8}{3}$$

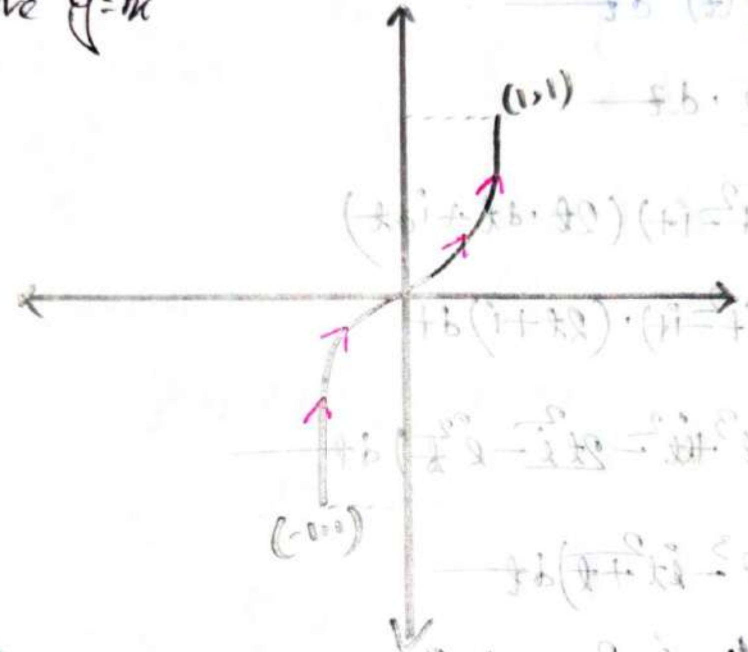
Ans:-

Ex: 7:-

∂ Sketch the path c from $z = -1 - i$ to $z = 1 + i$ along the curve $y = x^3$ and hence evaluate $\int_c f(z)\cdot dz$ where $f(z) = \begin{cases} y ; & y>0 \\ 2 ; & y<0 \end{cases}$

=> In given:-

the curve $y = x^3$



Let:- $x = t \quad (-1 \le t \le 1)$

$\therefore y = t^3$

$\therefore z = x + iy$
$= t + it^3$

$\therefore dz = dt + i3t^2 dt$
$= (1 + i3t^2) dt$

$\therefore f(z) = \begin{cases} t^3 ; & t>0 \\ 2 ; & t<0 \end{cases}$

We know:-

$$\int_c f(z)dz = \int_{c_1} f(z)\cdot dz + \int_{c_2} f(z)\cdot dz$$

$$= \int_{-1}^{0} 2(1+i3t^2)dt + \int_0^1 t^3(1+i3t^2)dt$$

$$= \int_{-1}^{0} (2+i6t^2)dt + \int_0^1 (t^3+i3t^5)dt$$

$$= \left[2t + \frac{i6}{3}t^3\right]_{-1}^{0} + \left[\frac{1}{4}t^4 + \frac{i3}{6}t^6\right]_0^1$$

$$= 2 + 2i + \frac{1}{4} + \frac{1}{2}i$$

$$= \frac{9}{4} + \frac{5i}{2}$$

Ans:-

Sample Exercise:

(8) Sketch the path c from $z=0$ to $z=3i$ and hence evaluate $\int_c z^2 dz$

⇒ In given:—

$z=0$
$z=3i$

That's means the equation of line passing through the points (0,0) to (0,3)

We know:—

$y-0=m(x-0)$ ∴ $m=\frac{y_2-y_1}{x_2-x_1}$

∴ $y-0=\frac{3}{0}(x-0)$ $m=\frac{3-0}{0-0}$

$3x=0$

∴ $x=0$

∴ $dx=0$

Now:— $\int_c z^2 \cdot dz=\int_c (x+iy)^2 \cdot (dx+idy)$

$=\int_c (0+iy)^2 \cdot (dx+idy)$

$=\int_0^3 i^2y^2 \times idy$

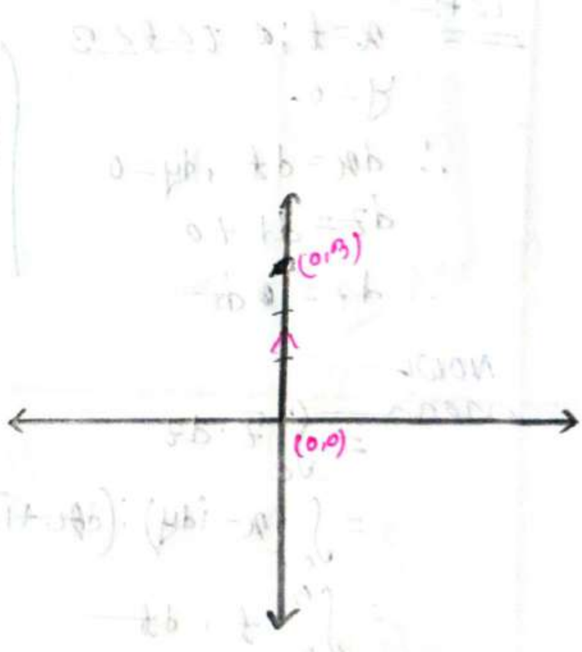
$=-i\int_0^3 y^2 \cdot dy$

$=-i\left[\frac{1}{3}y^3\right]_0^3$

$=-i\left[\frac{1}{3}\times 3^3\right]$

$=-9i$

Ans:—



④ Sketch the path C from $z=0$ to $z=3$ and hence evaluate $\int_C \bar{z} \cdot dz$.

⇒ In given:—

$z=0$; $z=3$

that's means the equation of line passing through the point $(0,0)$ to $(3,0)$

we know:—

$y-0=m(x-0)$

$\therefore y=0$

$\therefore m = \frac{y_2-y_1}{x_2-x_1}$

$= \frac{0-0}{3-0}$

$= 0$

let:—

$x=t$; $0 \le t \le 3$

$y=0$

$\therefore dx = dt$, $dy = 0$

$dz = dt + 0$

$\therefore dz = dt$

The parametric rep:—

$z = x(t) + y(t)$

$\therefore z = t + 0$

NOW—

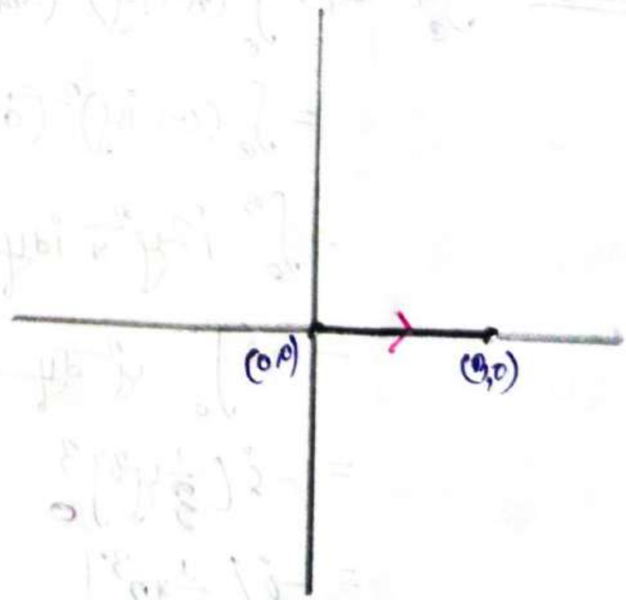
$= \int_C \bar{z} \cdot dz$

$= \int_C (x - iy) \cdot (dx + idy)$

$= \int_0^3 t \cdot dt$

$= \left[\frac{1}{2} t^2\right]_0^3$

$= \frac{9}{2}$ Ans:—



(b) Sketch the path C from $z = 1+i$ to $z = 3+3i$ and hence evaluate $\int_C Re(z) \cdot dz$

$\Rightarrow$ Is given:-

$z = 1+i$

$z = 3+3i$

that mean's the path passing through the points $(1,1)$ to $(3,3)$.

We know:-

$$\frac{y_2 - y_1}{x_2 - x_1} = m$$

$$\therefore m = \frac{3-1}{3-1}$$

$$\therefore m = 2/2 = 1$$

$$y - 1 = m(x-1)$$

$$y - 1 = x - 1$$

$$\therefore y = x$$

Let:- $x = t$; $1 \le t \le 3$

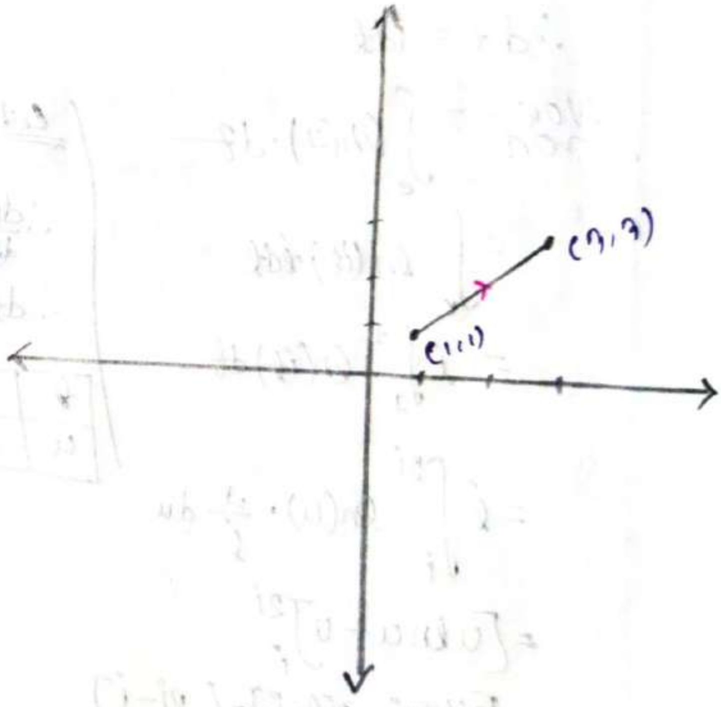
$\therefore y = t$

So, the parametric representation:-

$z = t + it$

$\therefore Re(z) = t$

$\therefore dz = dt + idt$



Now:- $= \int_C Re(z)\, dz$

$$= \int_1^3 t \cdot (dt + idt)$$

$$= \left[\frac{1}{2}t^2\right]_1^3 (1+i)$$

$$= 4+4i$$

Ans:-

⑥. $\int_C \ln(z)\,dz$ C is the shortest path of from i to $2i$

$\Rightarrow$ In given:—

$$z = i$$
$$z = 2i$$

that's means the path through the points $(0, 1)$ to $(0, 2)$

For the path:—

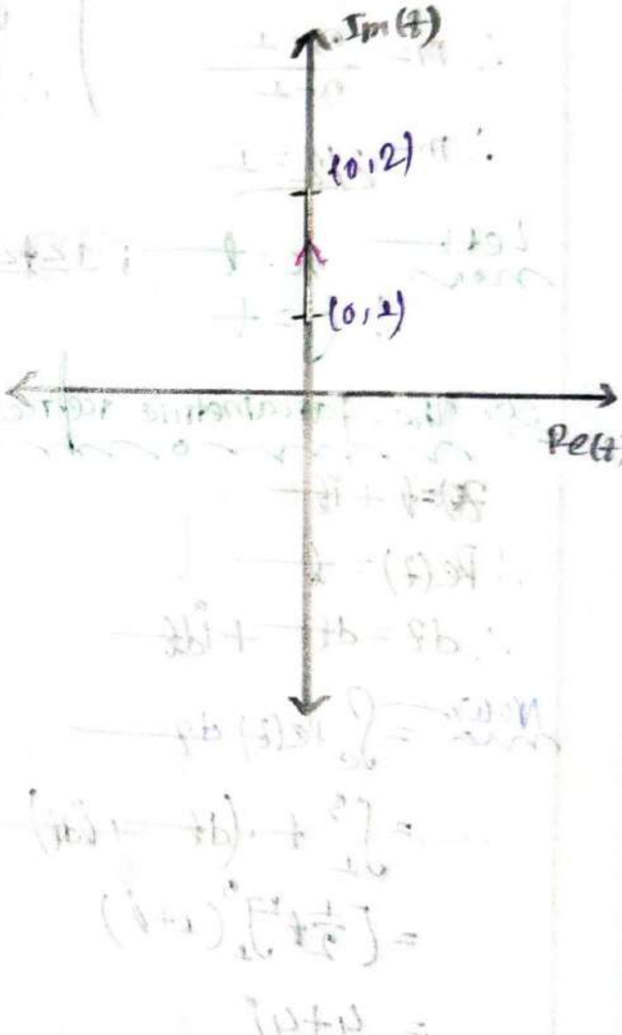
$$x = 0$$

Let:—

$$y = t \;;\; 1 \le t \le 2$$

so, the parametric repr:—

$$z(t) = 0 + it$$
$$\therefore dz = i\,dt$$



Now:— $\int_C \ln(z)\cdot dz$

$$= \int_C \ln(it)\cdot i\,dt$$

$$= i\int_1^2 \ln(it)\,dt$$

Let, $u = it$

$\therefore \frac{du}{dt} = i$

$\therefore dt = \frac{1}{i}du$

t	1	2
u	i	$2i$

$$= i\int_i^{2i} \ln(u)\cdot\frac{1}{i}\,du$$

$$= [u\ln u - u]_i^{2i}$$

$$= [2i\ln 2i - i\ln i] - [2i - i]$$

$$= i(2\ln 2i - \ln i - 1)$$

Ans:—

(7) Sketch the path C which is the circle $|z|=2$ and hence evalute $\int_C (z+z^{-1})dz$

$\Rightarrow$ In given:—

$|z|=2$

$\therefore |x+iy| = 2$

$\Rightarrow \sqrt{x^2+y^2} = 2$

$\Rightarrow x^2+y^2=2^2$

$(x,y) = (0,0)$

$r = 2$

We know:—

$x = r\cos t = 2\cos t$

$y = r\sin t = 2\sin t$

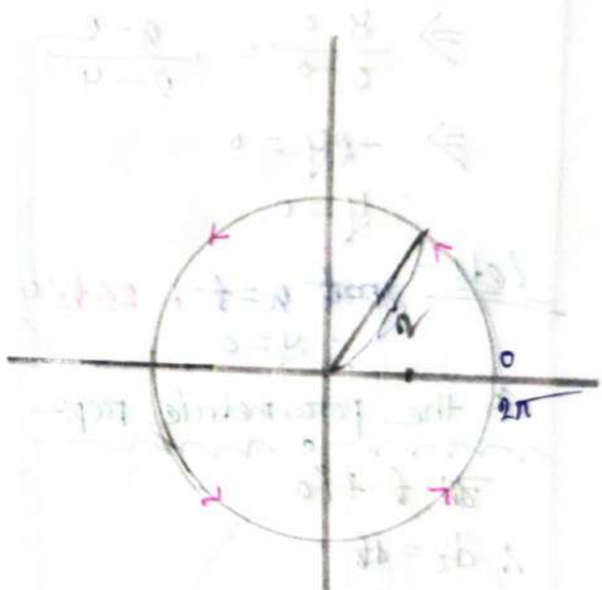
So, Parametric rep:—

$z = 2\cos t + i2\sin t \quad 0 \le t \le 2\pi$

$\therefore z = 2e^{it}$

$z^{-1} = \frac{1}{z} = \frac{1}{2}e^{-it}$

$\therefore dz = 2ie^{it}.dt$



Now:— $\int_C (z+z^{-1})\cdot dz$

$= \int_0^{2\pi} (2e^{it}+\frac{1}{2}e^{-it})\cdot 2ie^{it}\,dt$

$= \int_0^{2\pi} (2e^{it}\cdot 2ie^{it} + \frac{1}{2}\times 2e^{-it+it})\cdot dt$

$= \int_0^{2\pi} (4ie^{2it}+e^{0}i)\,dt$

$= \left[4i\times\frac{1}{2i}e^{2it} + it\right]_0^{2\pi}$

$= 2e^{4\pi i} - 2\pi i$ Ans:—

(8) $\int_c (e^{2z} + \cos z)\,dz$, c is the shortest path of $z=2$ to $z=4$

$\Rightarrow$ In given:-

$z = 2$

$z = 4$

So, The path in through the points $(2,0)$ to $(4,0)$

We know:-

$$\frac{y-y_1}{y_1-y_2} = \frac{x-x_1}{x_1-x_2}$$

$$\Rightarrow \frac{y-0}{0-0} = \frac{x-2}{2-4}$$

$$\Rightarrow -2y = 0$$

$$\therefore y = 0$$

Let:-

$x = t \;;\; 2 \le t \le 4$

$\therefore y = 0$

So, the parametric rep:-

$\cancel{z=}\ t + i0$

$\therefore dz = dt$

Now:- $\int_c (e^{2z} + \cos z)\cdot dz$

$$\Rightarrow \int_2^4 (e^{2t} + \cos t)\cdot dt$$

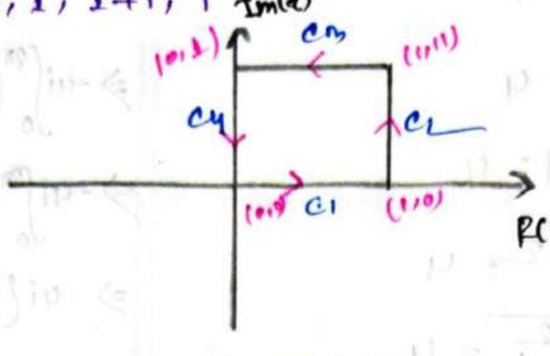
$$\Rightarrow \left[\frac{1}{2}e^{2t} + \sin t\right]_2^4$$

$$\Rightarrow \frac{1}{2}(e^8 - e^4) + \sin 4 - \sin 2$$

Ans:-

① $\int_c (z\cdot\bar{z})\,dz$ c in the path arround the square with Vertices $0, 1, 1+i, i$

$\Rightarrow$



Along C1:–

The line segment passing through the points $(0,0)$ to $(1,0)$.

We know:–

$$\frac{y-y_1}{y_1-y_2} = \frac{x-x_1}{x_1-x_2}$$

$$\Rightarrow \frac{y-0}{0-0} = \frac{x-0}{0-1}$$

$$\Rightarrow y=0,\ dy=0 \quad 0\le x\le 1$$

$$\therefore \int_{c_1} (x+iy)(x-iy)\cdot(dx+idy)$$

$$\Rightarrow \int_{c_1} (x+0)(x-0)(dx+i0)$$

$$\Rightarrow \int_0^1 x^2\,dx$$

$$\Rightarrow \frac{1}{3}\left[x^3\right]_0^1$$

$$\Rightarrow \frac{1}{3}$$

Along C2:–

the line segment passing through the points $(1,0)$ to $(1,1)$

we know,–

$$x=1 \quad 0\le y\le 1 \ /\ \therefore dx=0$$

$$\therefore \int_{c_2} (1+iy)(1-iy)(dx+idy)$$

$$= \int_0^1 (1+y^2)\,i\,dy$$

$$= i\left[y+\frac{1}{3}y^3\right]_0^1 = i+\frac{i}{3}$$

$$= \frac{4i}{3}$$

Along C3

The line segment passing through the points $(1,1)$ to $(0,1)$

We know:–

$$\therefore y=1,\ 1<x<0$$

$$dy=0$$

$$\therefore \int_c (x+i)(x-i)(dx+idy)$$

$$\Rightarrow \int_c (x^2+1)\,dx$$

$$\Rightarrow \int_1^0 (x^2+1)\,dx$$

$$\Rightarrow \left[\frac{1}{3}x^3+x\right]_1^0 = -\frac{1}{3}-1 = -\frac{4}{3}$$

Along C4:–

$(0,1)$ to $(0,0)$

We know:–

$$x=0 \quad 1<y\le 0$$

$$\therefore dx=0$$

$$= \int_c (x+iy)(x-iy)(dx+idy)$$

$$= \int_c y^2\,i\,dy = \int_1^0 y^2\,i\,dy = i\left[\frac{1}{3}y^3\right]_1^0 = -\frac{i}{3}$$

Sol $\int_c (z\cdot\bar{z})\,dz = \int_{c_1}(z\cdot\bar{z})\,dz + \int_{c_2}(z\cdot\bar{z})\,dz + \int_{c_3}(z\cdot\bar{z})\,dz + \int_{c_4}(z\cdot\bar{z})\,dz$

$$= \frac{1}{3}+\frac{4i}{3}-\frac{4}{3}-\frac{i}{3}$$

$$= -1+i$$

Ans.

10. $\int_c \left(\frac{5}{z-2i} - \frac{6}{(z-2i)^2}\right)dz$. c is the circle $|z-2i| = 4$ clockwise

⇒ In given:—

$$|z-2i| = 4$$

$$\Rightarrow |x+iy-2i| = 4$$

$$\Rightarrow \sqrt{x^2+(y-2)^2} = 4$$

$$\Rightarrow \sqrt{x^2+(y-2)^2} = 4$$

$$\therefore x^2+(y-2)^2 = 4^2$$

$\because$ Center $(x,y) = (0,2)$

radius $r = 4$

Parametric equation:—

$$x(t) = 4\cos t$$

$$y(t) = 2+4\sin t$$

So, Parametric rep:—

$$z(t) = 4\cos t + i(2+4\sin t)$$

$$z(t) = 2i + 4\cos t + i4\sin t$$

In the case the circle in clockwise so, we can use $-t$ instead of t in the above rep:—

$$\therefore z(t) = 2i + 4\cos(-t) + i4\sin(-t)\ ;\ 0 \le t \le 2\pi$$

$$= 2i + 4\cos t - i4\sin t$$

$$z(t) = 2i + 4e^{-it}$$

$$\therefore dz = -4ie^{-it}\,dt$$

Now:— $\int_c \left[\frac{5}{z-2i} - \frac{6}{(z-2i)^2}\right]dz$

$$\Rightarrow \int_0^{2\pi}\left[\frac{5}{2i+4e^{-it}-2i} - \frac{6}{(2i+4e^{-it}-2i)^2}\right]\times -4ie^{-it}dt$$

$$\Rightarrow -4i\int_0^{2\pi}\left[\frac{5}{4e^{-it}} - \frac{6}{(4e^{-it})^2}\right]\cdot e^{-it}\,dt$$

$$\Rightarrow -4i\int_0^{2\pi}\left(\frac{5}{4}e^{it-it} - \frac{3}{8}e^{2it-it}\right)dt$$

$$\Rightarrow -4i\left[\frac{5}{4}t - \frac{3}{8}\times\frac{1}{i}e^{it}\right]_0^{2\pi}$$

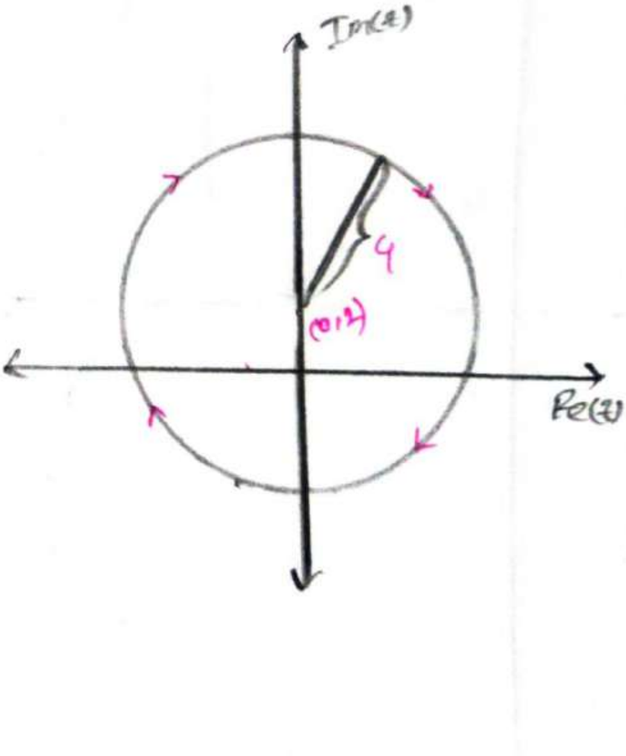
$$\Rightarrow -4i\left[\frac{5}{4}\times 2\pi - \frac{8}{3}\times\frac{1}{i}e^{2\pi i}\right]$$

Just put the Value 3/8 here
Then do the calculation
And also do the calculation of e^0

$$\Rightarrow -5i\times 2\pi + \frac{32}{3}e^{2\pi i}$$

$$\Rightarrow \frac{32}{3}e^{2\pi i} - 10\pi i$$

Ans:—



Sketch the path C from $z=-1$ to $z=1$ along the upper circle of $|z|=1$ and hence evaluate $\int_C f(z)\cdot dz$; $f(z)=\bar{z}$

$\Rightarrow$ In given:

$$|z|=1$$

$$\Rightarrow |x+iy|=1$$

$$\Rightarrow \sqrt{x^2+y^2}=1$$

$$\Rightarrow x^2+y^2=1^2$$

We know:

$$x = r\cos t = \cos t$$

$$y = r\sin t = \sin t$$

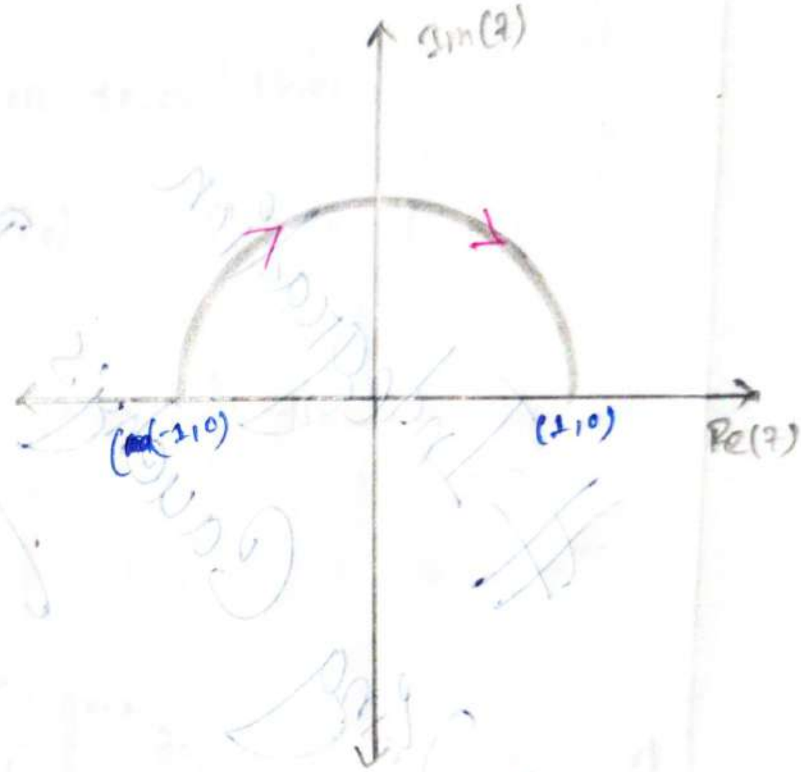
So Parametric rep.:

$$z(t) = \cos t + i\sin t$$

$$\Rightarrow z(t) = e^{it}$$

$$\therefore dz = ie^{it}\cdot dt$$

and $\bar{z} = e^{-it}$ $\quad \pi \le t \le 0$



Now $\int_C f(z)\cdot dz$

$$= \int_C \bar{z}\cdot dz$$

$$= \int_{\pi}^{0} e^{-it}\cdot ie^{it}\, dt$$

$$= i\,[t]_{\pi}^{0}$$

$$= -\pi i$$

Ans.

Integration Using Cauchy's Residue Theorem

Integration using Cauchy's Residue Theorem (CRT)

Formula:—

⇒ If a function $f(z)$ in analytic within and on a simple closed Contour C and

① if z_0 is any point interior to C then,

$$\oint_C \frac{f(z)}{z-z_0} \cdot dz = 2\pi i\, f(z_0)$$

② if z_0 is not an interior point of the Contour C then,

$$\oint_C \frac{f(z) \cdot dz}{z-z_0} = 0$$

③ $$\oint_C \frac{f(z)}{(z-z_0)^n}\, dz = \frac{2\pi i}{(n-1)!} f^{n-1}(z_0)$$

4. Residul Finding method:—

⇒ If $f(z)$ analytic inside and on a simple closed curve C except at a finite number or pole or has a singularity at $z=z_0$ of order 1 then

$$Res(z_0) = \lim_{z \to z_0} (z-z_0) f(z)$$

5. For m order:—

$$Re(z_0) = \lim_{z \to z_0} \frac{1}{(m-1)!} \times \frac{d^{m-1}}{dz^{m-1}} \left\{ (z-z_0)^m f(z) \right\}$$

6. $$\oint_C f(z) \cdot dz = 2\pi i \left[Res(z_1) + Res(z_2) + Res(z_3) + \cdots + Res(z_n) \right]$$

Example: 78:—

⇒ Evaluate the contour integral $\oint_C \frac{dz}{z^3}$ by CRT, where C is the circle $|z+1| = 3$

⇒ Is given:—

$$|z+1| = 3$$

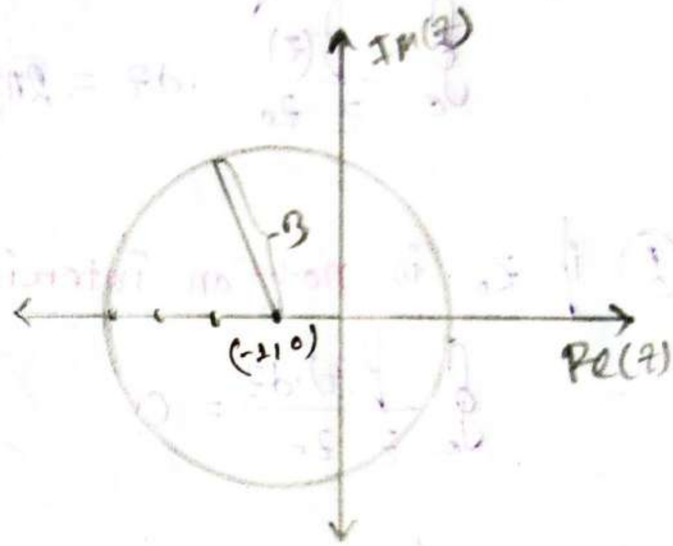
$$\Rightarrow |x+iy+1| = 3$$

$$\Rightarrow \sqrt{(x+1)^2+y^2} = 3$$

$$\Rightarrow (x+1)^2+y^2 = 3^2$$

$$\therefore \text{Center } (x,y) = (-1,0)$$

$$r = 3$$



Now:—

$$\oint_C \frac{dz}{z^3} = \frac{2\pi i}{(3-1)!} f^{3-1}(z_0)$$

$$= \frac{2\pi i}{2} \times f''(0)$$

$$= \frac{2\pi i}{2} \times 0$$

$$= 0$$

Sample Exercise set on Cauchy residue theorem:—

(2) State Cauchy's integral formula and CRT.

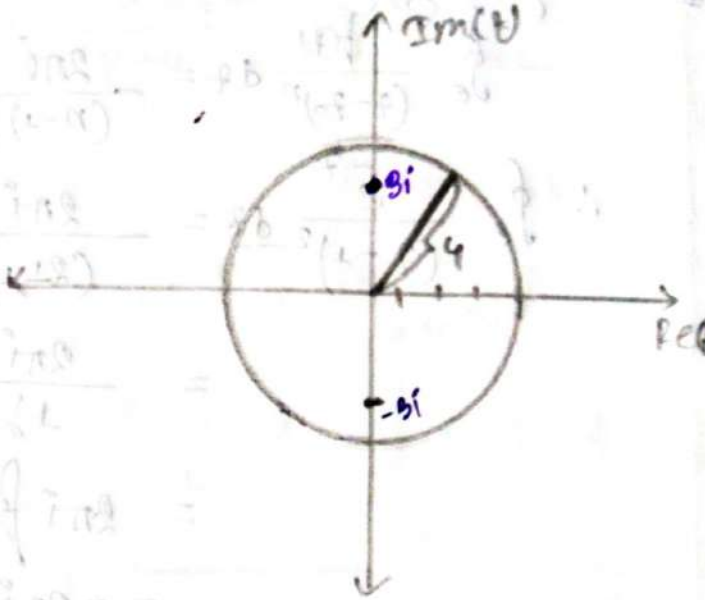
(a) $\oint_C \frac{dz}{z-3i}$, C is the circle $|z|=4$

⇒ In given:—

$$|z|=4$$
$$\Rightarrow \sqrt{x^2+y^2}=4$$
$$\therefore x^2+y^2=4^2$$

So, Center $(x,y)=(0,0)$

$r=4$



⇒ The z_0 is interior:—

we know:—

$$\oint_C \frac{f(z)}{z-z_0}\,dz = 2\pi i\, f(z_0)$$

here, $f(z)=1$

$z_0=3i$

$$\therefore \oint_C \frac{1}{z-3i}\,dz = 2\pi i\, f(3i)$$
$$= 2\pi i \times 1$$
$$= 2\pi i$$

Ans:—

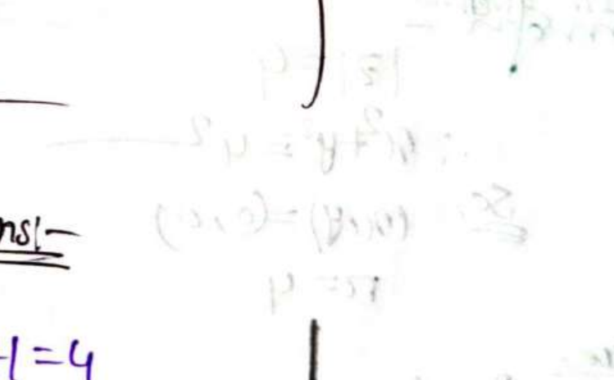
(b) $\oint \frac{e^{-z}\,dz}{(z-1)^2}$, C consists of $|z|=4$

⇒ In given:—

$$|z|=4$$
$$\therefore x^2+y^2=4^2$$

Center $(x,y)=(0,0)$

$r=4$



here :—

$z_0 = 1$

$f(z) = e^{-z}$

So, The z_0 is interrior:—

Now, we know:—

$$\oint_c \frac{f(z)}{(z-z_0)^n}\,dz = \frac{2\pi i}{(n-1)!} f^{n-1}(z_0)$$

$$\therefore \oint_c \frac{e^{-z}}{(z-1)^2}\,dz = \frac{2\pi i}{(2-1)!} f^{2-1}(z_0)$$

$$= \frac{2\pi i}{1!} f'(e^{-1})$$

$$= 2\pi i\, f(e^{-1})$$

$$= -2\pi i e^{-1}$$

Ans:—

(a) $\oint_c \frac{dz}{(z-6)^{10}}$; where c in the circle $|z|=4$

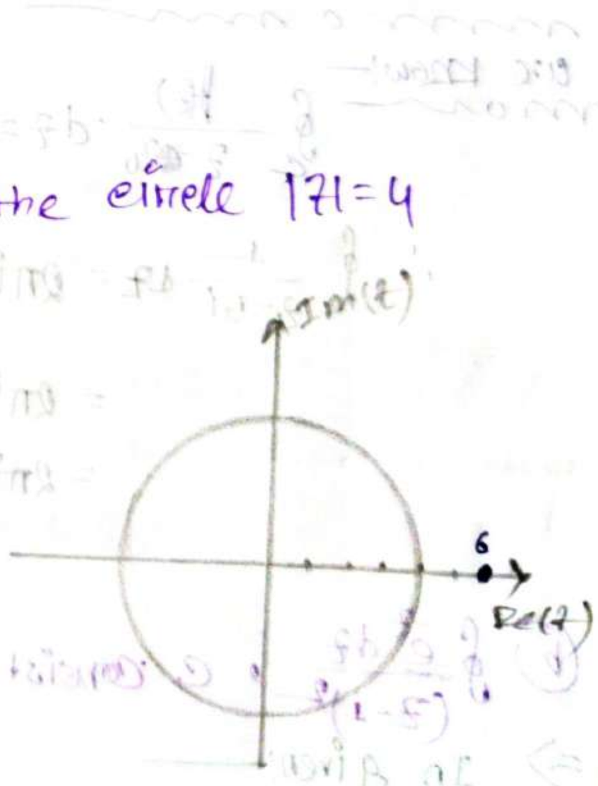
⇒ In given:—

$|z| = 4$

$\therefore x^2+y^2 = 4^2$

So, $(x,y) = (0,0)$

$r = 4$



here,

$z_0 = 6$

$f(z) = 1$

So, the z_0 is not interior.

Then! $\oint_c \frac{dz}{(z-6)^{10}} = 0$

Ans:—

Find the all singular points of the functions, $f(z)$ and show the points in the argand diagram. then find corresponding residues.

(i) (a) $\frac{1}{2z}$, (b) $\frac{1}{z^2-4^2}$, (c) $\frac{\sin z}{z}$, (d) $\frac{1}{z^6+1}$, (e) $\cot z$.

(ii) (a) $\frac{z^2+1}{z^2+z}$, (b) $\frac{1}{z^3+8}$, (c) $\frac{z^2+2}{z-4}$ (d) $\frac{1}{z^6+1}$

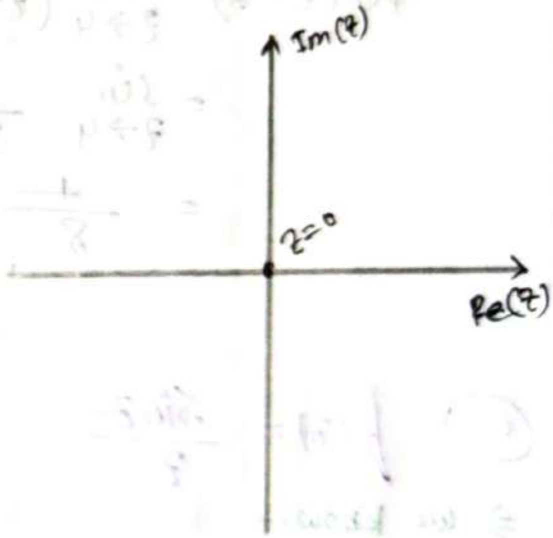
(i)

(a) $f(z) = \frac{1}{2z}$

$\Rightarrow$ we know:—

For singular point:—

$z = 0$



$\therefore$ Residue at $z = 0$:—

$$Res(z_0) = \lim_{z \to z_0} (z - z_0) \times f(z)$$

$$\therefore Res(z=0) = \lim_{z \to 0} (z-0) \times \frac{1}{2z}$$

$$= \frac{1}{2}$$

(b) $f(z) = \frac{1}{z^2-4^2}$

$\Rightarrow \therefore f(z) = \frac{1}{(z+4)(z-4)}$

we know:— For singular point:—

$(z+4)(z-4) = 0$

$\therefore z = 4, -4$

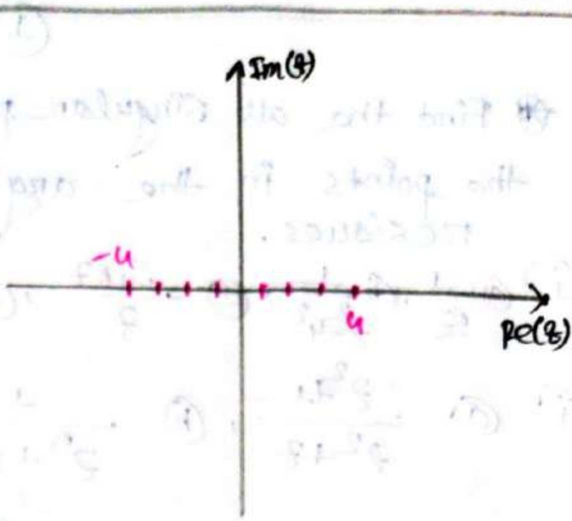
Residue at $z=-4$

$$Res(z=-4) = \lim_{z \to -4} (z+4) \times \frac{1}{(z+4)(z-4)}$$

$$= \lim_{z \to -4} \frac{1}{z-4}$$

$$= \frac{1}{-4-4}$$

$$= \frac{1}{-8}$$



Residue at $z=4$:-

$$Res(z=4) = \lim_{z \to 4} (z-4) \frac{1}{(z+4)(z-4)}$$

$$= \lim_{z \to 4} \frac{1}{z+4}$$

$$= \frac{1}{8}$$

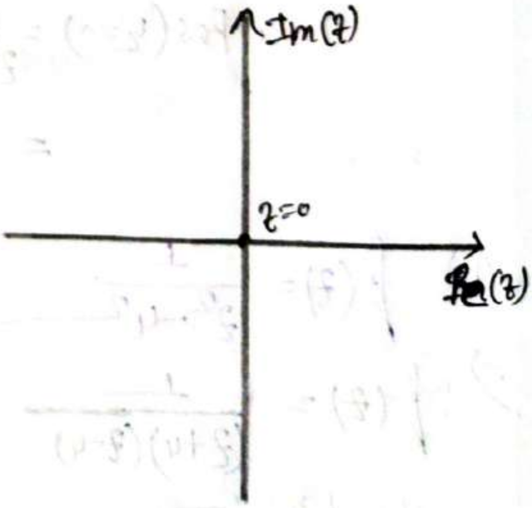
Ans:-

(c) $f(z) = \frac{\sin z}{z}$

⇒ we know:-

For singular point:-

$$z=0$$



∴ Residue at $z=0$:-

$$Res(z=0) = \lim_{z \to 0} (z-0) \times f(z)$$

$$= \lim_{z \to 0} z \times \frac{\sin z}{z}$$

$$= \sin 0$$

$$= 0$$

(d) $f(z) = \frac{1}{z^6+1}$ find the singular point.

$\Rightarrow f(z) = \frac{1}{(z^2+1)(z^4-z^2+1)}$

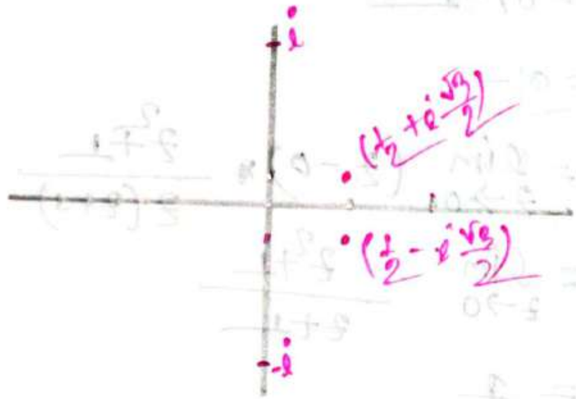
we know:—

For singular point

$z^2+1=0$; $z^4-z^2+1=0$

$z = \pm i$ $\quad (z^2-z+1)(z^2+z+1)=0$

$\therefore z = \frac{1}{2} \pm i\frac{\sqrt{3}}{2}$



(e) $f(z) = \cot z$

$\Rightarrow$ The contangent function has singularities where $\cot z$ is not defined.

thats happen when $\tan z = 0$

$\therefore$ singular point:—

$z = n\pi$ where, n = Integer

(ii) (a) $f(z) = \dfrac{z^2+1}{z^2+z}$

$\Rightarrow f(z) = \dfrac{z^2+1}{z(z+1)}$

We know:-

For singular points:-

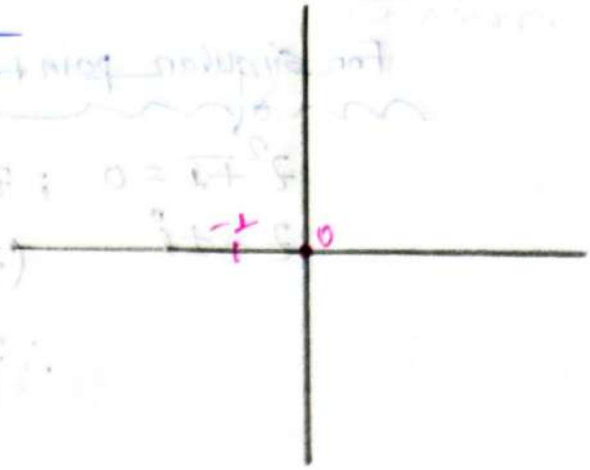
$z(z+1) = 0$

$\therefore z = 0$

$z+1 = 0$

$\therefore z = -1$

So, $z = 0, -1$



residue at $z=0$:-

$$\text{Res}(z=0) = \lim_{z\to 0} (z-0) \cdot \frac{z^2+1}{z(z+1)}$$

$$= \lim_{z\to 0} \frac{z^2+1}{z+1}$$

$$= 1$$

Residue at $z=-1$:-

$$\text{Res}(z=-1) = \lim_{z\to -1} (z+1) \frac{z^2+1}{z(z+1)}$$

$$= \lim_{z\to -1} \frac{z^2+1}{z}$$

$$= -2$$

(b) $f(z) = \dfrac{1}{z^3+i}$

$\Rightarrow$ For singular point:-

$z^3 + i = 0$

$\therefore z^3 = -i$

(c) $f(z) = \dfrac{z^2+2}{z-4}$

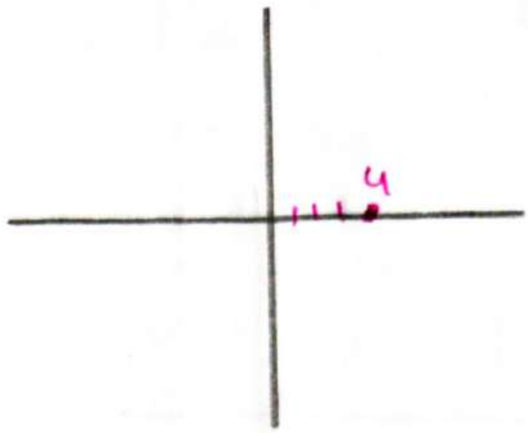
$\Rightarrow f(z) = \dfrac{z^2+2}{z-4}$

We know,—

For singular point:—

$z-4=0$

$\therefore z=4$



$\therefore$ Residue at $z=4$:—

$$\text{Res}(z=4) = \lim_{z\to 4} (z-4)\frac{z^2+2}{(z-4)}$$

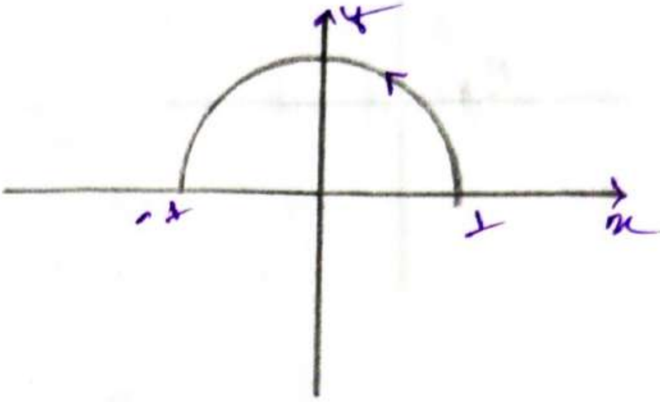
$$= \lim_{z\to 4} z^2+2$$

$$= 18$$

9

⇒ Evaluate the followings applying Cauchy's residue theorem (CRT).

(i) $\oint_C \frac{2z}{(2z-i)^3}\,dz$



$= \oint_C \frac{2z}{8\left(z-\frac{i}{2}\right)^3}\,dz$

we know:—

Singular point:—

$z - \frac{i}{2} = 0$

$\therefore z = \frac{i}{2}$

$\therefore$ For Residue at $z = \frac{i}{2}$:-

$Res\left(z=\frac{i}{2}\right) = \lim_{z\to\frac{i}{2}} \frac{1}{4} \times \frac{1}{(3-1)!} \times \frac{d^2}{dz^2}\left[\left(z-\frac{i}{2}\right)^3 \frac{z}{\left(z-\frac{i}{2}\right)^3}\right]$

$= \lim_{z\to\frac{i}{2}} \frac{1}{8} \times \frac{d}{dz}(1)$

$= \lim_{z\to\frac{i}{2}} \frac{1}{8} \times 0$

$= 0$

$\therefore \oint_C \frac{2z}{(2z-i)^3}\,dz = 2\pi i \times 0$

$= 0$

Ans:-

(11) $\oint_C \frac{dz}{z^2-1} \Rightarrow \oint \frac{1}{(z+1)(z-1)} dz$

we know:-

singular point:-

$z^2 = 1$

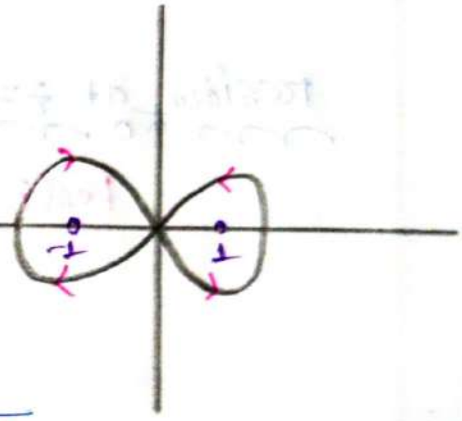
$\therefore z = \pm 1$

residue at $z=1$:-

$Res(z=1) = \lim_{z \to 1} (z-1) \frac{1}{(z+1)(z-1)}$

$= \frac{1}{1+1}$

$= \frac{1}{2}$



residue at $z=-1$:-

$Res(z=-1) = \lim_{z \to -1} (z+1) \frac{1}{(z+1)(z-1)}$

$= \lim_{z \to -1} \frac{1}{z-1}$

$= \frac{1}{-1-1}$

$= \frac{1}{-2}$

Now, $\oint_C \frac{dz}{z^2-1} = 2\pi i \left(Re(z_1) + Re(z_2) \right)$

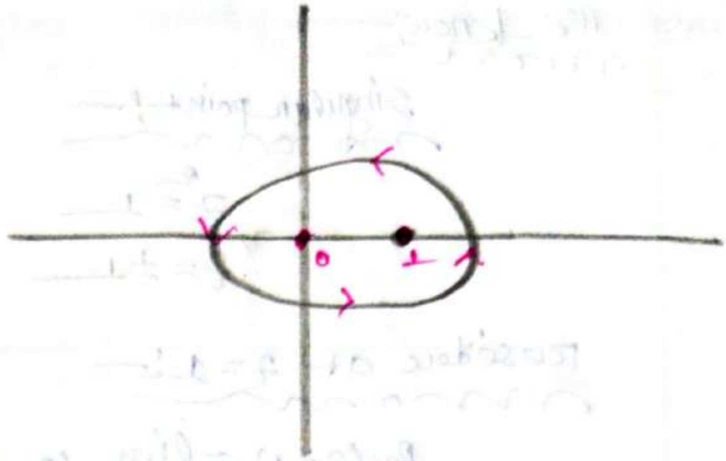
$= 2\pi i \left(\frac{1}{2} - \frac{1}{2} \right)$

$= 2\pi i \times 0$

$= 0$

(iii) $\oint_C \frac{2z-1}{z^2-z}\,dz$

$= \oint_C \frac{2z-1}{z(z-1)}\,dz$



we know—

For Singular point:—

$z(z-1)=0$

$\therefore z=0,1$

residue at $z=0$:—

$$Res(z=0) = \lim_{z\to 0} (z-0)\frac{2z-1}{z(z-1)}$$

$$= \lim_{z\to 0} \frac{2z-1}{z-1}$$

$$= 1$$

Residue at $z=1$:—

$$Res(z=1) = \lim_{z\to 1} (z-1)\frac{2z-1}{z(z-1)}$$

$$= \lim_{z\to 1} \frac{2z-1}{z}$$

$$= 1$$

So, $\oint_C \frac{2z-1}{z^2-z} = 2\pi i\left[R(z=1) + R(z)\right]$

$= 2\pi i\left[1+1\right]$

$= 4\pi i$

Ans.

④

⇒ For the followings sketch the indicated path C and hence evaluate applying Cauchy's residue theorem (CRT)

ⓐ $\oint_C \frac{dz}{z^2+4}$; C in the contour as

ⓘ $|z+2i| = 1$

ⓘⓘ $|z-2i| = 1$

ⓘ

In given:

$|z+2i| = 1$

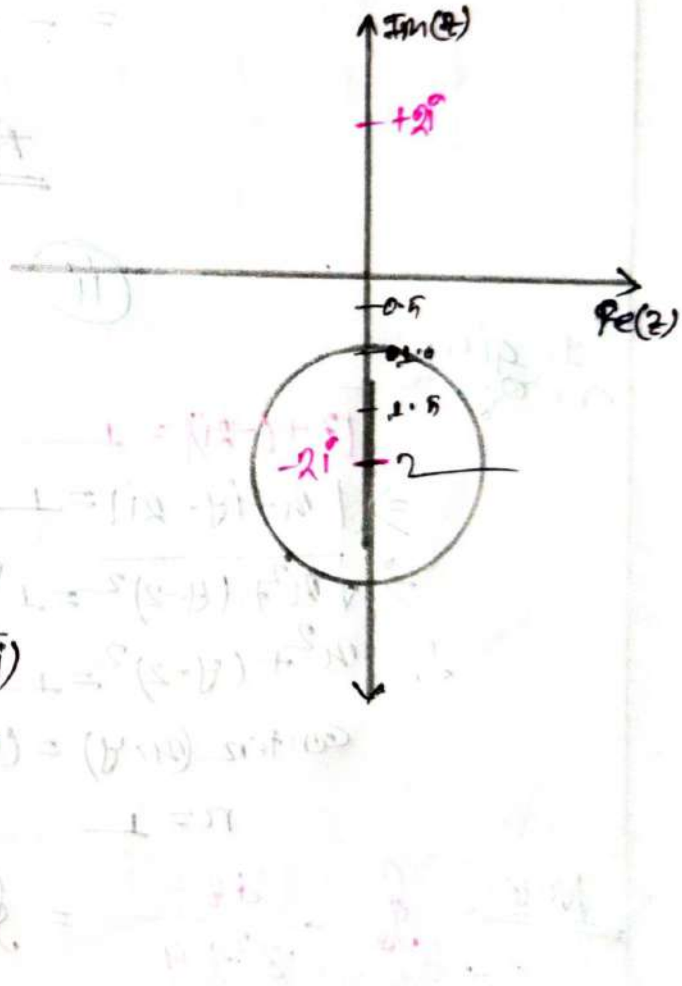
$\therefore |x+iy+2i| = 1$

$\Rightarrow \sqrt{x^2+(y+2)^2} = 1$

$\Rightarrow x^2+(y+2)^2 = 1^2$

Center $(x,y) = (0,-2)$

$r_0 = 1$



Now,
$$\oint_C \frac{dz}{z^2+4} = \oint_C \frac{dz}{(z+2i)(z-2i)}$$

we know

For singular points

$z^2+4=0$

$\Rightarrow z^2 = -4$

$\Rightarrow z^2 = 4i^2$

$\Rightarrow z = \pm 2i$

∴ Residue at $z = -2i$:—

$$Rez(z=-2i) = \lim_{z \to -2i} (z+2i) \frac{1}{(z+i2)(z-i2)}$$

$$= \lim_{z \to -2i} \frac{1}{z-i2}$$

$$= \frac{1}{-4i}$$

Sol $\oint_C \frac{dz}{z^2+4} = 2\pi i\left(Re(z_1)\right)$

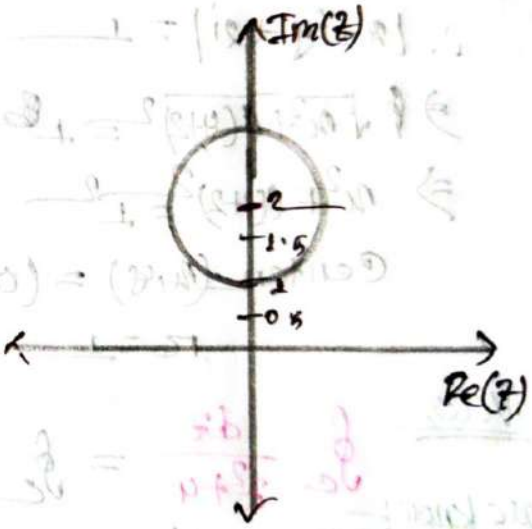
$$= 2\pi i \times -\frac{1}{4i}$$

$$= -\frac{\pi}{2}$$

Ans:—

(ii)

In given—



$$|z+(-2i)| = 1$$

$$\Rightarrow |x+iy-2i| = 1$$

$$\Rightarrow \sqrt{x^2+(y-2)^2} = 1$$

$$\therefore x^2+(y-2)^2 = 1$$

center $(x,y) = (0,2)$

$r = 1$

Now, $\oint_C \frac{dz}{z^2+4} = \oint \frac{dz}{(z+i2)(z-i2)}$

We know— For Singular point:—

$$z^2 = -4$$

$$z = \pm 2i$$

So, Residue at $z=+2i$

$$Re(z) = \lim_{z\to 2i} (z-2i)\frac{1}{(z+2i)(z-2i)}$$

$$= \lim_{z\to 2i} \frac{1}{z+2i}$$

$$= \frac{1}{4i}$$

So, $\oint_C \frac{dz}{z^2+4} = 2\pi i\,[Re(z_1)]$

$$= 2\pi i \times \frac{1}{4i}$$

$$= \frac{\pi}{2}$$

Ans:–

(b) $\oint_C \frac{\cos(\pi z^3)}{(z-1)(z-2)}\cdot dz$; where C is the circle $|z-3| = 4$

⇒ is given:–

$$|z-3| = 4$$

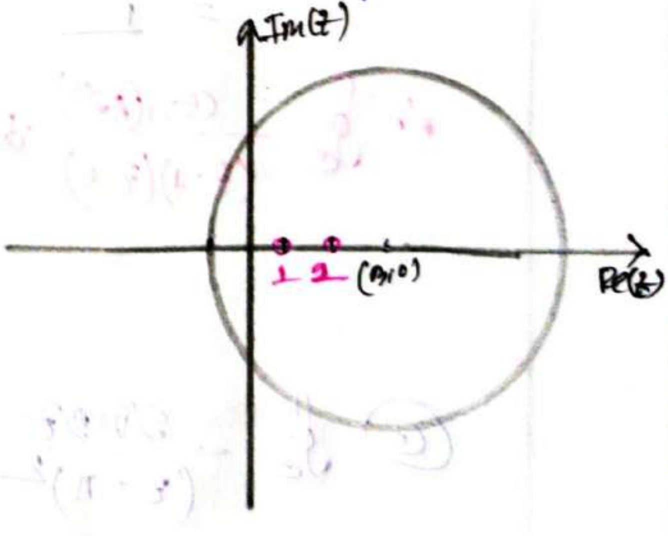
$$\Rightarrow |x-3+iy| = 4$$

$$\Rightarrow \sqrt{(x-3)^2+y^2} = 4$$

$$\therefore (x-3)^2+y^2 = 4^2$$

center $=(3,0)$

$r = 4$



We know:–

For Singular point:–

$$(z-1)(z-2) = 0$$

$$\therefore z = 1, 2$$

Residue at $z=1$:—

$$Res(z_1) = \lim_{z\to 1} (z-1) \cdot \frac{\cos(\pi z^3)}{(z-1)(z-2)}$$

$$= \lim_{z\to 1} \frac{\cos(\pi z^3)}{z-2}$$

$$= \frac{-1}{-1}$$

$$= 1$$

Residue at $z=2$:—

$$Res(z_2) = \lim_{z\to 2} (z-2) \frac{\cos(\pi z^3)}{(z-1)(z-2)}$$

$$= \lim_{z\to 2} \frac{\cos(\pi z^3)}{z-1}$$

$$= 1$$

$$\therefore \oint_C \frac{\cos(\pi z^3)}{(z-1)(z-2)} \cdot dz = 2\pi i \times [Res(z_1) + Res(z_2)]$$

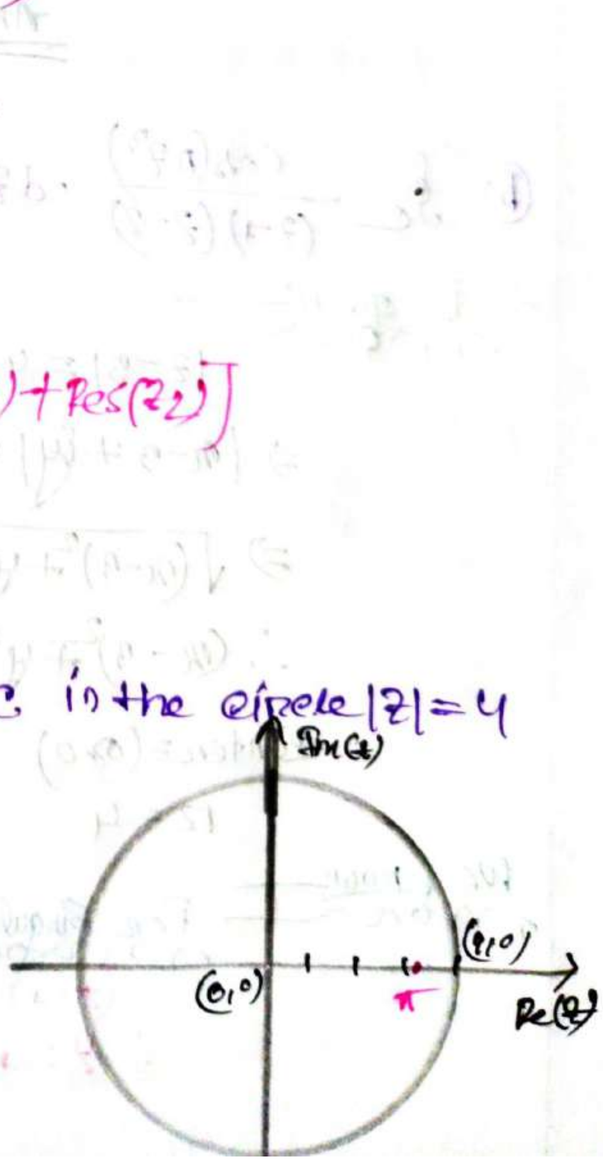
$$= 2\pi i (1+1)$$

$$= 4\pi i$$

(c) $\oint_C \frac{\sin 3z}{(z-\pi)^2} dz$ where C is the circle $|z| = 4$

$\Rightarrow$ In given:—

$$|z| = 4$$
$$\Rightarrow |x+iy| = 4$$
$$x^2+y^2 = 4^2$$
$$(x,y) = (0,0)$$
$$r = 4$$



We know:

For Singular point:

$$z - \pi = 0$$

$$\therefore z = \pi$$

Residue at $z=\pi$:

$$Res(z) = \lim_{z \to \pi} \frac{1}{(2-1)!} \times \frac{d^{2-1}}{dz^{2-1}} \left\{ (z-\pi)^2 \times \frac{\sin 3z}{(z-\pi)^2} \right\}$$

$$= \lim_{z \to \pi} \frac{1}{1} \times \frac{d}{dz}(\sin 3z)$$

$$\Rightarrow \lim_{z \to \pi} 3\cos 3z$$

$$\Rightarrow 3 \times \cos 3 \times \pi$$

$$\Rightarrow -3$$

$$\therefore \oint_c \frac{\sin 3z}{(z-\pi)^2} = 2\pi i \; Res(z)$$

$$= 2\pi i \times -3$$

$$= -6\pi i$$

Ans:-

(d) $\oint \frac{z+2}{z-2} dz$; where c is the circle $|z+1| = 2$

$\Rightarrow$ In given:

$$|z-1| = 2$$

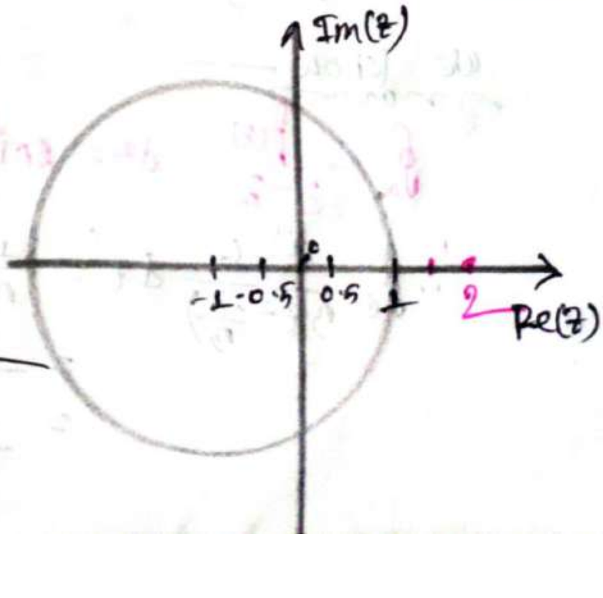
$$\Rightarrow |x - 1 + iy| = 2$$

$$\Rightarrow \sqrt{(x-1)^2 + y^2} = 2$$

$$\therefore (x+1)^2 + y^2 = 2^2$$

$$\therefore (x, y) = (-1, 0)$$

$$r = 2$$



we know:—

For singular point:—

$$(z-2) = 0$$

$$\therefore z = 2$$

So, Since $z=2$ is not an interior point of the given circle: that means

$$\oint_C \frac{z+2}{z-2}\,dz = 0$$

Ans:—

For more practice:—

① $\oint_C \frac{z^3-6}{3z-i}\,dz$; $|z|=1$ using CIF

⇒ is given:—

$$\oint_C \frac{z^3-6}{3z-i}\,dz$$

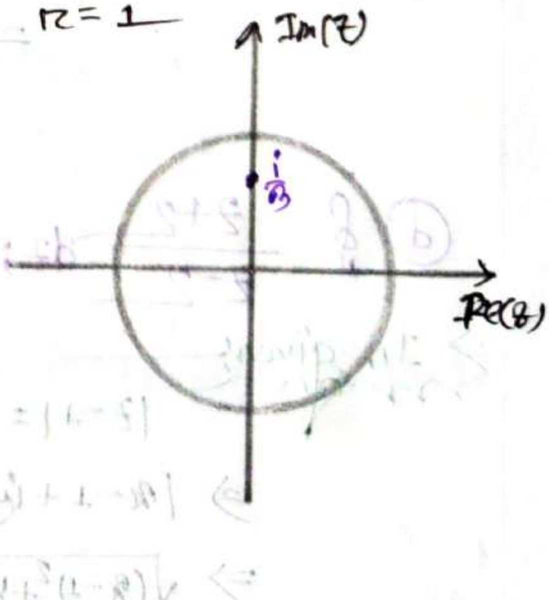
$$= \oint_C \frac{z^3-6}{3\left(z-\frac{i}{3}\right)}\,dz$$

$|z| = 1$

$x^2+y^2 = 1^2$

$(x,y) = (0,0)$

$r = 1$



Here:— $f(z) = z^3-6$

$z_0 = \frac{i}{3}$

we know:—

$$\oint_C \frac{f(z)}{z-z_0}\,dz = 2\pi i f(z_0)$$

$$\therefore \oint_C \frac{z^3-6}{3\left(z-\frac{i}{3}\right)}\,dz = \frac{1}{3} \times 2\pi i \times f(i/3)$$

$$= \frac{2\pi i}{3} \times \left\{\left(\frac{i}{3}\right)^3 - 6\right\}$$

$$= \frac{2\pi i}{3} \times \left(\frac{-i}{27} - 6\right)$$

$$= \frac{2\pi}{81} - 4i$$

$$= \frac{2\pi}{81} - 4i$$

Ans:-

② $\oint_c \frac{\cos 2\pi z}{(2z-1)(z-3)} \cdot dz$ C in the consists of $|z|=1$

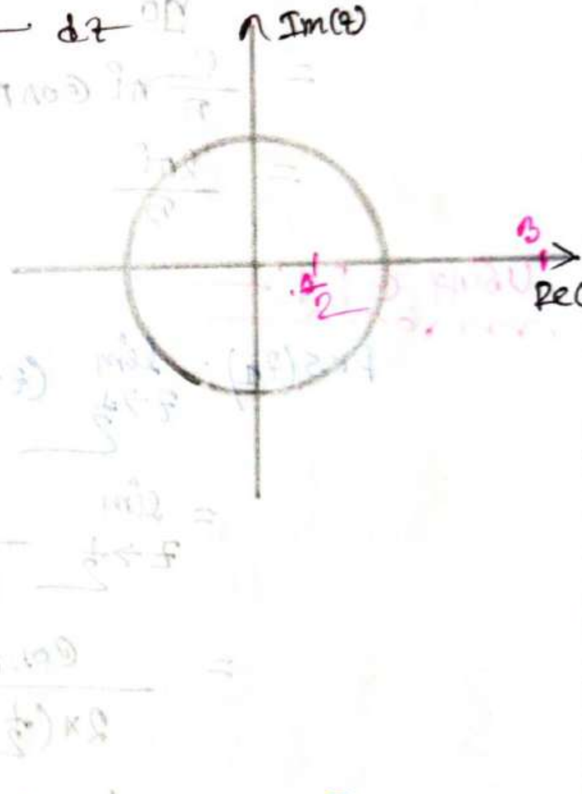
Find singular point. Solve it using CIF and CRT.

⇒ In given:—

$|z| = 1$

$x^2+y^2 = 1^2$

$$\oint_c \frac{\cos 2\pi z}{2(z-\frac{1}{2})(z-3)} \, dz$$



We know:—

For singular point:—

$$\therefore (z-\tfrac{1}{2})(z-3) = 0$$

$$\therefore z = \tfrac{1}{2}, 3$$

Using CIF:—

$$\Rightarrow \oint_c \frac{1}{2} \times \frac{\cos 2\pi z}{(z-\frac{1}{2})(z-3)} \cdot dz$$

here, $f(z) = \cos 2\pi z$

$\therefore$ Now:- $\frac{1}{(z-\frac{1}{2})(z-3)} = \frac{A}{z-\frac{1}{2}} + \frac{B}{z-3}$

using cover up rule:—

$z=\frac{1}{2}$, $A = \frac{1}{z-3} = -\frac{2}{5}$

$z=3$, $B = \frac{1}{z-\frac{1}{2}} = \frac{2}{5}$

We know—

$$= \oint_c \frac{1}{2} \times \left[\frac{-f(z) \times \frac{2}{5}}{(z-\frac{1}{2})} + \frac{\frac{2}{5} f(z)}{(z-3)} \right]$$

$$= \oint_c -\frac{1}{2} \times \frac{2}{5} \frac{\cos \pi z \, dz}{(z-\frac{1}{2})} + \oint_c \frac{1}{2} \times \frac{f(z) \times \frac{2}{5}}{(z-3)} dz$$

$$= -\frac{1}{2} \times \frac{2}{5} \times 2\pi i f(\tfrac{1}{2}) + 0$$

$$= \frac{-2}{10} \times 2\pi i f(\tfrac{1}{2})$$

$$= -\frac{2}{10} \times 2\pi i \times \cos 2 \times \tfrac{1}{2} \times \pi$$

$$= \frac{2}{5} \pi i \cos \pi$$

$$= \frac{2\pi i}{5}$$

Using CRT:—

$$\mathrm{Res}(z_1) = \lim_{z \to \frac{1}{2}} (z-\tfrac{1}{2}) \frac{\cos 2\pi z}{2(z-\frac{1}{2})(z-3)}$$

$$= \lim_{z \to \frac{1}{2}} \frac{\cos 2\pi z}{2 \times (z-3)}$$

$$= \frac{\cos \pi}{2 \times (\frac{1}{2} - 3)}$$

$$= \frac{1}{5}$$

$$\therefore \oint \frac{\cos 2\pi z}{2(z-\frac{1}{2})(z-3)} = 2\pi i \times \mathrm{Res}(z_1)$$

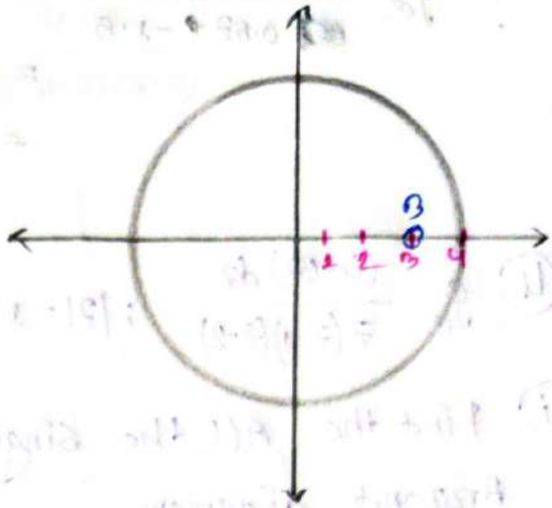
$$= 2\pi i \times \frac{1}{5}$$

$$= \frac{2\pi i}{5}$$

Ans.

(3) $\oint_c \frac{z^2+8}{0.5z-1.5}\,dz$; $|z|=4$ using CRT and CIF

$\Rightarrow$ In given:—



$$|z|=4$$
$$x^2+y^2=4^2$$

$$= \oint_c \frac{z^2+8}{0.5\left(z-\frac{1.5}{0.5}\right)}\cdot dz$$

$$= \oint_c \frac{z^2+8}{0.5(z-3)}\cdot dz$$

CIF:—

we know—

$$\oint_c \frac{f(z)}{z-z_0} = 2\pi i \times f(z_0)$$

$$\oint_c \frac{f(z)}{0.5(z-3)} = 2\pi i\, f(z_0)\times\frac{1}{0.5}$$

here,
$f(x) = z^2+8$
$z_0 = 3$

$$= 4\pi i \times f(3)$$
$$= 4\pi i\,(3^2+8)$$
$$= 68\pi i$$

CRT:—

we know:—

For singular point:—

$$z-3=0$$
$$z=3$$

Residue at $z=3$:—

$$Res(z) = \lim_{z\to 3}\,(z-3)\times\frac{z^2+8}{0.5(z-3)}$$
$$= \frac{3^2+8}{0.5\times 1}$$
$$= 34$$

we know:—

$$\oint_C \frac{z^2+8}{0.5z-1.5} = 2\pi i \times Res(z_0)$$
$$= 2\pi i \times 64$$
$$= 68\pi i$$

(4) $\oint_C \frac{(4-3z)\,dz}{z(z-1)(z-2)}$; $|z| = 1.5$ sketch the circle. using CRT

(i) Find the All the singular points. show the point in Argand diagram.

(ii) Find all corresponding residue.

(iii) Evaluate applying Cauchy's residual theorem also use CIF

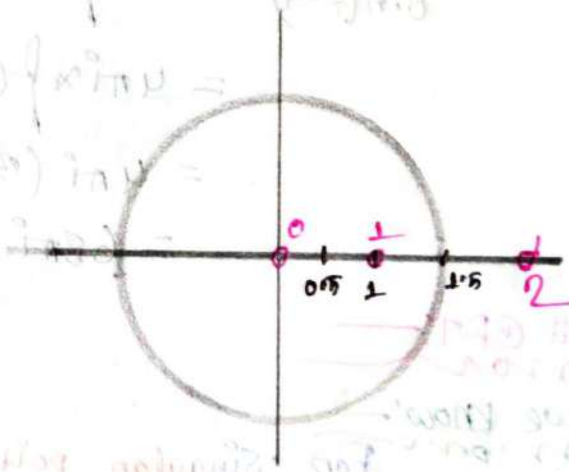
⇒ In given:—

$|z| = 1.5$

$x^2+y^2 = (1.5)^2$ —

$(x,y) = (0,0)$

$r = 1.5$



(1)

$$\oint_C \frac{(4-3z)\,dz}{z(z-1)(z-2)}$$

we know:—

For singular point:—

$z(z-1)(z-2) = 0$

$\therefore z = 0, 1, 2$ —

(ii)

Residue at $z=0$:—

$$Res(z=0) = \lim_{z\to 0} (z-0) \frac{4-3z}{z(z-1)(z-2)}$$

$$= \frac{4-3\times 0}{(0-1)(0-2)}$$

$$= 2$$

Residue at $z=1$:—

$$Res(z=1) = \lim_{z\to 1} (z-1) \frac{4-3z}{z(z-1)(z-2)}$$

$$= \frac{4-3\times 1}{1\times(1-2)}$$

$$= -1$$

Residue at $z=2$:—

$$Res(z=2) = \lim_{z\to 2} (z-2) \frac{4-3z}{z(z-1)(z-2)}$$

$$= \frac{4-3\times 2}{2\times(2-1)}$$

$$= -1$$

(iii)

We know:—

$$\oint_{C_1} \frac{(4z-3z)}{z(z-1)(z-2)} dz = 2\pi i \left[Res(z_1) + Res(z_0)\right]$$

$$= 2\pi i \times (2 - 1)$$

$$= 2\pi i$$

(iii)

$\Rightarrow \oint \frac{4-3z}{z(z-1)(z-2)}$

$\therefore$ in here:—

$4-3z = f(z)$

$\therefore$ Now:—
$$\frac{1}{z(z-1)(z-2)} = \frac{A}{z} + \frac{B}{(z-1)} + \frac{C}{z-2} \quad \text{—(1)}$$

multiply $z(z-1)(z-2)$ on both side:—

$$1 = A(z-1)(z-2) + Bz(z-2) + Cz(z-1)$$

if $z-1=0$

$\therefore z = 1$

$\therefore B = \frac{1}{z^2-2z} = \frac{1}{-1} = -1$

$z-2=0$

$z=2$

$\therefore C = \frac{1}{z^2-z} = \frac{1}{2}$

$\therefore A = \frac{1}{2}$

$\therefore$ put the value in eq (1):—

$$\frac{1}{z(z-1)(z-2)} = \frac{1}{2z} + \frac{-1}{z-1} + \frac{1}{2(z-2)}$$

$$\therefore \oint_C \frac{4-3z}{z(z-1)(z-2)}\,dz = \oint_C \frac{f(z)}{2z}\cdot dz + \oint_C \frac{-f(z)}{z-1}\,dz + \oint_C \frac{f(z)\cdot dz}{2(z-2)}$$

$$= \frac{2\pi i}{2} \times f(0) \quad - 2\pi i\, f(1) + \quad 0$$

$= 4\pi i - 2\pi i$

$= 2\pi i$

Ans.

Application of CRT

Ch: 03

Contour Integrals:—

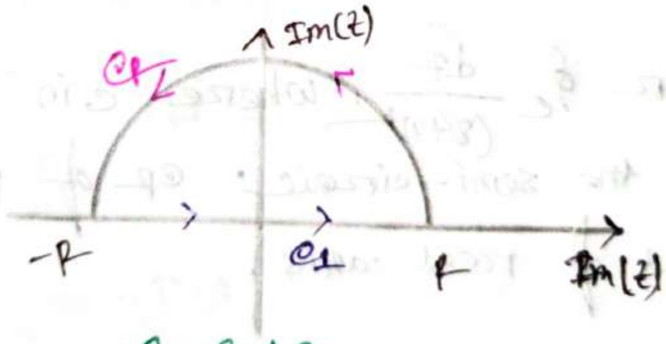
⇒ For finding the integral we take a suitable complex function $f(z)$ and closed curve C.

⇒ Find singular point of the function $f(z)$.

⇒ Calculate residues at the singular point which lie within the curve C.

⇒ Now, evaluate,

$$\oint_C f(z)\cdot dz = 2\pi i \text{ (Sum of residues of } f(z) \text{ at the singular point with in } C)$$



$C = C_R + C_1$

⇒ To evaluate $\int_{-\infty}^{\infty} \frac{f_1(x)}{f_2(x)}\,dx$ consider the contour integrals $\oint \frac{f_1(z)}{f_2(z)}\cdot dz$

where C is the closed contour consisting the real axis from $-R$ to R and the upper half of C_R of the circle $|z|=R$.

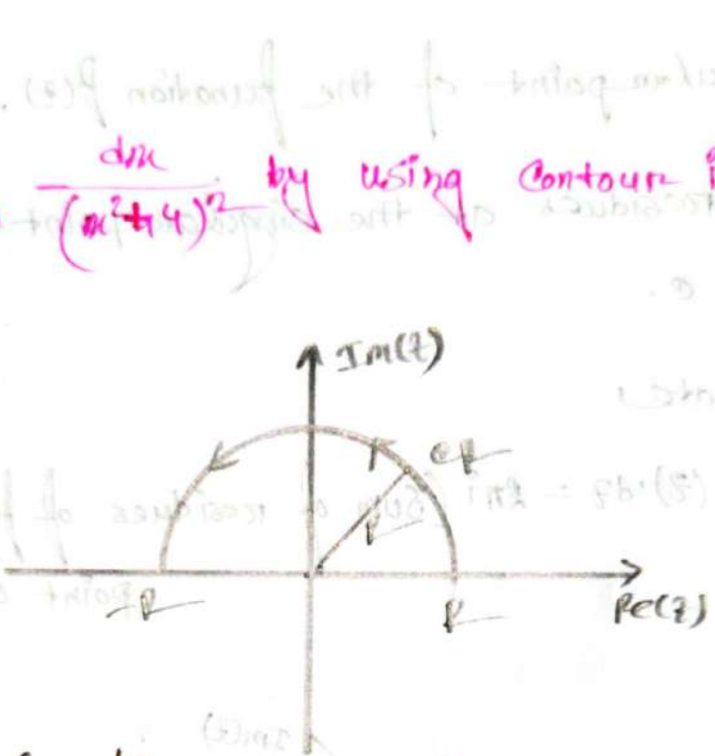
Formula:—

$$\oint_C \frac{f_1(z)}{f_2(z)}\,dz = \int_{-R}^{R} \frac{f_1(x)}{f_2(x)}\,dx + \int_{C_R} \frac{f_1(z)}{f_2(z)}\,dz$$

Example 7.8:—

⇒ Evaluate $\int_{-\infty}^{\infty} \frac{dx}{(x^2+4)^2}$ by using contour integration.

⇒



⇒ we consider $\oint_C \frac{dz}{(z^2+4)^2}$, where C is the closed contour consisting of the semi-circle C_R of Radius R together with the part of real axis.

$$\therefore \oint_C \frac{dz}{(z^2+4)^2} = \int_{-R}^{R} \frac{dx}{(x^2+4)^2} + \int_{C_R} \frac{dz}{(z^2+4)^2} \quad \text{---(1)}$$

Now the 1st integrals has singularities of $(z^2+4)^2=0$, of order 2.

$$\therefore z = \pm 2i$$

The only singular point $z=2i$ is inside the contour C.

$\therefore Res(z=2i) = \lim_{z\to 2i} \frac{1}{(2-1)!} \times \frac{d}{dz}\left[(z-2i)^2 \times \frac{1}{(z+2i)^2(z-2i)^2}\right]$

$= \lim_{z\to 2i} \frac{d}{dz}\left[\frac{1}{(z+2i)^2}\right]$

$= \lim_{z\to 2i} \frac{-2}{(z+2i)^3}$

$= \frac{-2}{(2i+2i)^3}$

$= \frac{-2}{-64i}$

$= \frac{1}{32i}$

$\therefore \oint \frac{dz}{(z^2+4)^2} = 2\pi i \times Res(z_1)$

$= 2\pi i \times \frac{1}{32i}$

$= \frac{\pi}{16}$

put it into the eq (1) $\Rightarrow$

$$\frac{\pi}{16} = \int_{-R}^{R} \frac{dx}{(x^2+4)^2} + \int_{C_R} \frac{dz}{(z^2+4)^2}$$

$\Rightarrow$ By Jordan's Lemma letting $R \to \infty$ and nothing that the 2nd integrals in Right side would become zero.

$$\therefore \frac{\pi}{16} = \lim_{R\to\infty}\left[\int_{-\infty}^{\infty} \frac{dx}{(x^2+4)^2} + \int_{C\infty} \frac{dz}{(z^2+4)^2}\right]$$

$$\therefore \frac{\pi}{16} = \int_{-\infty}^{\infty} \frac{dx}{(x^2+4)^2} + 0$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{dx}{(x^2+4)^2} = \frac{\pi}{16}$$

Ans:-

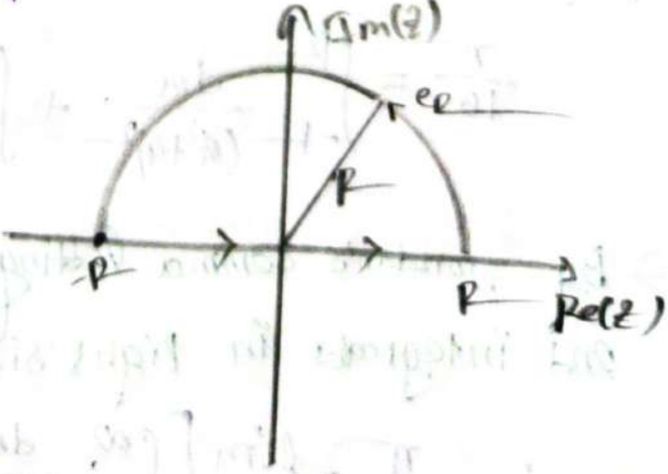
Example: 17.6:-

$\Rightarrow$ Evaluate $\int_0^{\infty} \frac{dx}{(x^4+16)}$ by using contour integration.

$\Rightarrow$ We consider, $\oint_C \frac{dz}{(z^4+16)}$ where C is closed contour consisting the semi-circle C_R, of radius R together with the part of real axis, $-R$ to $+R$.

$$\therefore \oint_C \frac{dz}{(z^4+16)} = \int_{-R}^{R} \frac{dx}{x^4+16} + \int_{C_R} \frac{dz}{(z^4+16)} \quad \text{---} \ (1)$$

Now the 1st integral has singular point at:-



$z^4+16=0$

$\therefore z^4 = -4^4$

$z^4 = -16$

$\therefore z = (-16)^{\frac{1}{4}}$

$\therefore r = \sqrt{(16)^2+0^2} = 16$

$\theta = \pi + \tan^{-1}\left(\frac{0}{16}\right)$

$\therefore \theta = \pi$

$k = 0, 1, 2, 3$

We know:—

$$z_k = r^{\frac{1}{4}} e^{i\left(\frac{\theta+2\pi k}{4}\right)}$$

$$\therefore z_0 = (16)^{\frac{1}{4}} e^{i\left(\frac{\pi}{4}\right)} = 2e^{i\frac{\pi}{4}}$$

$$z_1 = 2e^{i\left(\frac{3\pi}{4}\right)}$$

$$z_2 = 2e^{i\left(\frac{5\pi}{4}\right)}$$

$$z_3 = 2e^{i\left(\frac{7\pi}{4}\right)}$$

So, only z_0 and z_1 lie with in the contour C.

$$\therefore Res(z=z_k) = \lim_{z\to z_k}\left\{(z-z_k)\frac{1}{(z^4+16)}\right\}$$

$$= \lim_{z\to z_k}\left\{\frac{1}{4z_k^3}\right\}$$

$k = 0, 1$

$k=0$,

$$\therefore Res(z_1 = z_0) = \left\{\frac{1}{4(2 \times e^{i\frac{\pi}{4}})^3}\right.$$

$$= \frac{1}{4\times 8} e^{-\frac{3i\pi}{4}}$$

$$= \frac{1}{32} e^{-i\frac{3\pi}{4}}$$

$k=1$

$$Res(z_2 = z_1) = \left\{\frac{1}{32} e^{-i\frac{9\pi}{4}}\right\}$$

$$\therefore \oint_C \frac{dz}{z^4+16} = 2\pi i \times Res(z_1+z_2)$$

$$= 2\pi i \times \left(\frac{1}{32}e^{-i\frac{3\pi}{4}} + \frac{1}{32}e^{-i\frac{9\pi}{4}}\right)$$

$$= \frac{2\pi i}{32}\left(e^{-i3\pi/4} + e^{-i9\pi/4}\right)$$

$$= 2\pi i/32 \times \left[\left(\cos\frac{3\pi}{4} - i\sin\frac{3\pi}{4}\right) + \left(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\right)\right]$$

$$= \frac{\pi i}{16} \times \left[\left(-\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\right)\right]$$

$$= \frac{\pi i}{16} \times \left(-\frac{2i}{\sqrt{2}}\right)$$

$$= \frac{\sqrt{2}\pi}{16}$$

By Jordan Lemma letting $R \to \infty$,

$$\frac{\sqrt{2}\pi}{16} = \lim_{R\to\infty} \int_{-R}^{R} \frac{dx}{x^4+16} + \lim_{R\to\infty} \int_{C_R} \frac{dz}{(z^4+16)}$$

$$\frac{\sqrt{2}\pi}{16} = \int_{-\infty}^{\infty} \frac{dx}{x^4+16} + 0$$

$$\therefore \int_{-\infty}^{\infty} \frac{dx}{x^4+16} = \frac{\sqrt{2}\pi}{16} \Rightarrow 2\int_{0}^{\infty} \frac{dx}{x^4+16} = \frac{\sqrt{2}\pi}{16}$$

$$\therefore \int_{0}^{\infty} \frac{dx}{x^4+16} = \frac{\sqrt{2}\pi}{32}$$ Ans:-

(1.) Evaluate $\int_0^\infty \frac{dx}{x^2+1}$

$\Rightarrow$ Let, consider $\oint_c \frac{dz}{z^2+1}$ where c is the closed path of Contour consisting of the semi circle c_R.

$$\therefore \oint_c \frac{dz}{z^2+1} = \int_{-R}^{R} \frac{dx}{x^2+1} + \int_{c_R} \frac{dz}{z^2+1} \quad \text{——— (1)}$$

Now, Int integral has singularities at

$$z^2+1 = 0$$

$$\therefore z = \pm i$$

only $+i$ lie within the contour c.

$$\therefore Res(z_0 = i) = \lim_{z \to i} (z-i) \times \frac{1}{(z+i)(z-i)}$$

$$= \lim_{z \to i} \frac{1}{z+i}$$

$$= \frac{1}{2i}$$

$$\therefore \oint_c \frac{dz}{z^2+1} = 2\pi i \times Res(z_0)$$

$$= 2\pi i \times \frac{1}{2i}$$

$$= \pi$$

By Jordan Lemma letting $R \to \infty$:

$$\pi = \lim_{R\to\infty} \int_{-R}^{R} \frac{dx}{x^2+1} + \lim_{R\to\infty} \int_{c_R} \frac{dz}{z^2+1}$$

$$\int_{-\infty}^{\infty} \frac{dx}{x^2+1} = \pi$$

P.T.O

$$\Rightarrow 2\int_0^{\infty} \frac{dx}{x^2+1} = \pi$$

$$\Rightarrow \int_0^{\infty} \frac{dx}{x^2+1} = \frac{\pi}{2}$$

Ans:—

(2) Evaluate $\int_{-\infty}^{\infty} \frac{dx}{(x^2-2x+2)^2}$

$\Rightarrow$ Let consider $\oint_C \frac{dz}{(z^2-2z+2)^2}$ where C in the closed path of counter cont consisting of the semi circle C_R

$$\therefore \oint_C \frac{dz}{(z^2-2z+2)^2} = \int_{-R}^{R} \frac{dx}{(x^2-2x+2)^2} + \int_{C_R} \frac{dz}{(z^2-2z+2)^2} \quad \text{—— (1)}$$

1st integral has singularities at:—

$$z^2-2z+2=0$$

$$\therefore z = 1 \pm i$$

Both singularities lie with in the contour C.

$$\text{Res}(z=z_1) = \lim_{z\to 1+i} \left(\frac{1}{(2-1)!}\right) \frac{d}{dz}\left[\{z-(1+i)\}^2 \times \frac{1}{\{z-(1+i)\}^2\{z-(1-i)\}^2}\right]$$

$$= \lim_{z\to 1+i} \frac{d}{dz}\left[\frac{1}{\{z-(1-i)\}^2}\right]$$

$$= \lim_{z\to 1+i} \frac{-2}{(z-1+i)^3}$$

$$= \frac{-2}{(1+i-1+i)^3}$$

$$= \frac{-2}{(2i)^3}$$

$$= \frac{2}{8i}$$

$$\therefore \oint_C \frac{dz}{(z^2-2z+2)^2} = 2\pi i \, Res(z_1)$$

$$= 2\pi i \times \frac{2}{8i}$$

$$= \frac{\pi}{2}$$

By Jordan Lemma letting $R \to \infty$:—

$$\therefore \frac{4\pi}{8} = \lim_{R\to\infty} \int_{-R}^{R} \frac{dx}{(x^2-2x+2)^2} + \int_{C_R} \frac{dz}{(z^2-2z+2)^2}$$

$$\Rightarrow \frac{4\pi}{8} = \int_{-\infty}^{\infty} \frac{dx}{(x^2-2x+2)^2} + 0$$

$$\therefore \int_{-\infty}^{\infty} \frac{dx}{x^2-2x+2} = \frac{4\pi}{8} = \frac{\pi}{2}$$

Ans!

Sample Exercise set on Improper Integrals

① $\int_{-\infty}^{\infty} \frac{dx}{x^2+2x+2}$ Evaluate the using CRT

⇒ let us consider $\oint_C \frac{dz}{z^2+2z+2}$ where C in the closed path of Countor and Consisting the semi-circle of C_R

$$\therefore \oint_C \frac{dz}{z^2+2z+2} = \int_{-r}^{r} \frac{dx}{x^2+2x+2} + \int_{C_R} \frac{dz}{z^2+2z+2} \quad \text{----} (1)$$

So, Int integral has singularities:—

$$z^2+2z+2=0$$

$$\therefore z = -1 \pm i$$

only one point $(-1+i)$ lie with in the Contour C:—

$$\therefore Res(z=-1+i) = \lim_{z\to(-1+i)} (z-(-1+i)) \frac{1}{\{z+(1+i)\}\{z+(1-i)\}}$$

$$= \lim_{z\to(-1+i)} (z+1-i) \frac{1}{(z+1+i)(z+1-i)}$$

$$= \lim_{z\to(-1+i)} \frac{1}{z+1+i}$$

$$= \frac{1}{-1+i+1+i}$$

$$= \frac{1}{2i}$$

We know:—

$$\oint_C \frac{dz}{z^2+2z+2} = 2\pi i \times Res(z)$$

$$= 2\pi i \times \frac{1}{2i} = \pi$$

By Jordan Lemma letting $R \to \infty$:-

$$\pi = \lim_{R\to\infty} \int_{-R}^{R} \frac{dx}{x^2+2x+2} + \lim_{R\to\infty} \int_{C_R} \frac{dz}{z^2+2z+2}$$

$$= \int_{-\infty}^{\infty} \frac{dx}{x^2+2x+2} + 0$$

$$\therefore \int_{-\infty}^{\infty} \frac{dx}{x^2+2x+2} = \pi$$

Ans:-

(2) $\int_0^{\infty} \frac{dx}{x^2+1}$ Evaluate the following integral by using CRT

$\Rightarrow$ We consider $\oint_C \frac{dz}{z^2+1}$ where C in the close path of the contour and consisting the semicircle C_R

$$\therefore \oint_C \frac{dz}{z^2+1} = \int_{-R}^{R} \frac{dx}{x^2+1} + \int_{C_R} \frac{dz}{z^2+1} \quad \text{---} (1)$$

1st integral has singularities at:-

$$z^2+1=0$$

$$\therefore z = \pm i$$

only the $z=+i$ point lie on the countor C:-

$$Res(z_1) = \lim_{z\to i} (z-i) \times \frac{1}{(z+i)(z-i)}$$

$$= \frac{1}{2i}$$

$$\therefore \oint_C \frac{dz}{z^2+1} = 2\pi i \times Res(z_1)$$

$$= 2\pi i \times \frac{1}{2i}$$

$$= \pi$$

By Jordan Lemma letting $R \to \infty$:—

$$\pi = \lim_{R\to\infty} \int_{-R}^{R} \frac{dx}{x^2+1} + \lim_{R\to\infty} \int_{C_R} \frac{dz}{z^2+1}$$

$$\pi = \int_{-\infty}^{\infty} \frac{dx}{x^2+1} + 0$$

$$\therefore 2\int_0^{\infty} \frac{dx}{x^2+1} = \pi$$

$$\therefore \int_0^{\infty} \frac{dx}{x^2+1} = \frac{\pi}{2}$$

Ans.

(5) $\int_0^{\infty} \frac{x^2}{x^6+1}\, dx$ Evaluate using CRT

⇒ We consider $\oint_C \frac{z^2}{z^6+1} \cdot dz$ where C in the closed path of contour and consisting of the semi circle C_R.

$$\oint_C \frac{z^2}{z^6+1} \cdot dz = \int_{-R}^{R} \frac{x^2}{x^6+1} \cdot dx + \int_{C_R} \frac{z^2\, dz}{z^6+1} \quad \text{— (1)}$$

∴ The 1st integral has singularities at

$$z^6+1=0$$

$$z = (-1)^{\frac{1}{6}}$$

$$\therefore r = 1,\ \theta = \pi - \tan^{-1}\left(\frac{0}{-1}\right) = \pi$$

$K = 0, 1, 2, 3, 4, 5$

we know:—

$$z_k = r^{\frac{1}{6}} e^{i\left(\frac{\theta + 2\pi k}{6}\right)}$$

$$\therefore z_0 = (1)^{\frac{1}{6}} e^{i\left(\frac{\theta}{6}\right)} = e^{i\frac{\pi}{6}}$$

$$z_1 = e^{i\frac{3\pi}{6}}$$

$$z_2 = e^{i\frac{5\pi}{6}}$$

$$z_3 = e^{i\frac{7\pi}{6}}$$

$$z_4 = e^{i\frac{9\pi}{6}}$$

$$z_5 = e^{i\frac{11\pi}{6}}$$

here only z_0 and z_1 are lie on the contour:—

$$\therefore \operatorname{Res}(z = z_k) = \lim_{z \to z_k} (z - z_k)\left(\frac{z^2}{z^6 + 1}\right)$$

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$$\frac{\quad}{6} = \frac{1}{6} e^{-\frac{\pi i}{2}}$$

$\therefore \oint_C \frac{z^2}{z^6+1} =$

$\frac{3\pi i}{2})$

$= 2$

$= \frac{2}{6}$ ... $\cos\frac{3\pi}{2} - i\sin\frac{3\pi}{2}]$

$= \frac{\pi}{3}$

$= ($

By Jordan le...

$0 = \lim_{R\to\infty}\int_{-R}^{R}$ ... $\frac{z^2}{z^6+1}$

$= \int_{-\infty}^{\infty} \frac{}{x^6}$

$\therefore 2\int_0^{\infty} \frac{x^2}{x^6+1}$

$\therefore \int_0^{\infty} \frac{x^2}{x^6+1}$

Laurent's Theorem

Laurent's theorem:—

Formulla:—

① $(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots\cdots$ if $|x| < 1$

② $(1-x)^{-1} = 1 + x + x^2 + x^3 + \cdots\cdots$

Example:— 7.7

⇒ obtain Laurent series expansion of $f(z) = \dfrac{1}{(1+z^2)(z+2)}$

when,

(i) $1 < |z| < 2$

(ii) $|z| > 2$

⇒ (i)

given:—

Since $1 < |z| < 2$

$\Rightarrow 1 < |z|$ or $|z| < 2$

$\Rightarrow \dfrac{1}{|z|} < 1$ or $\dfrac{|z|}{2} < 1$

$= \dfrac{1}{|z^2|} < 1$ or $\dfrac{|z|}{2} < 1$

Let $\dfrac{1}{(1+z^2)(z+2)} = \dfrac{A}{z+2} + \dfrac{Bz+C}{1+z^2}$

multiply $(1+z^2)(z+2)$ in both side:

$1 = A(1+z^2) + (Bz+C)(z+2)$ —— ①

$1 = A + Az^2 + Bz^2 + 2Bz + Cz + 2C$

if $z = -2$ put in eq (①):

$\therefore 1 = A((-2)^2+1)$

$A = \dfrac{1}{5}$

$2C = 1 - A$

$C = 2/5$

$B = \dfrac{-C}{2} = -\dfrac{1}{5}$

$$\therefore f(z) = \frac{1}{5(z+2)} + \frac{-z}{5(1+z^2)} + \frac{2}{5(1+z^2)} \quad \text{—— (II)}$$

$$= \frac{1}{5\times 2\left(\frac{z}{2}+1\right)} - \frac{z}{5\times z^2\left(1/z^2+1\right)} + \frac{2}{5\times z^2\left(1/z^2+1\right)}$$

$$= \frac{1}{10}\left(\frac{z}{2}+1\right)^{-1} - \frac{z}{5z^2}\left(\frac{1}{z^2}+1\right)^{-1} + \frac{2}{5z^2}\left(1/z^2+1\right)^{-1}$$

$$= \frac{1}{10}\left(1+\frac{z}{2}\right)^{-1} - \frac{z}{5z^2}\left(1+\frac{1}{z^2}\right)^{-1} + \frac{2}{5z^2}\left(1+\frac{1}{z^2}\right)^{-1}$$

$$= \frac{1}{10}\left(1-\frac{z}{2}+\frac{z^2}{4}-\cdots\right) - \frac{1}{5z}\left(1-\frac{1}{z^2}+\frac{1}{z^4}-\cdots\right)$$

$$+ \frac{2}{5z^2}\left(1-\frac{1}{z^2}+\frac{1}{z^4}-\cdots\right)$$

Ans:—

(II)

In given:—

$$|z| > 2$$

$$\Rightarrow 1 > \frac{2}{|z|}$$

$$\Rightarrow \frac{2}{|z|} < 1, \quad \text{Also } \frac{1}{|z|} < 1$$

Now from the eq (II) =)

$$f(z) = \frac{1}{5\times z\left(1+\frac{2}{z}\right)} - \frac{z}{5\times z^2\left(1+\frac{1}{z^2}\right)} + \frac{2}{5\times z^2\left(1+\frac{1}{z^2}\right)}$$

$$= \frac{1}{5z}\left(1+\frac{2}{z}\right)^{-1} - \frac{z}{5z^2}\left(1+\frac{1}{z^2}\right)^{-1} + \frac{2}{5z^2}\left(1+\frac{1}{z^2}\right)^{-1}$$

$$= \frac{1}{5z}\left(1-\frac{2}{z}+\frac{4}{z^2}-\cdots\right) - \frac{1}{5z}\left(1-\frac{1}{z^2}+\frac{1}{z^4}-\cdots\right) + \frac{2}{5z^2}\left(1-\frac{1}{z^2}+\frac{1}{z^4}-\cdots\right)$$

Sample Exercise on Laurent Series:—

(*) State Laurent Series, Expand $f(z) = \dfrac{3z}{(z-1)\,(2-z)}$ in a Laurent series valid for

(a) $|z|<1$, (b) $1<|z|<2$ (c) $|z|>2$, (d) $|z-1|>2$ (e) $0<|z-1|<1$

⇒ In given:—

$$f(z) = \frac{3z}{(z-1)(2-z)}$$

Let, $f(z) = \dfrac{A}{z-1} + \dfrac{B}{2-z} = \dfrac{3z}{(z-1)(2-z)}$

Using Cover up rule:—

$$z=1 \quad A = \frac{3z}{2-z} = 3$$

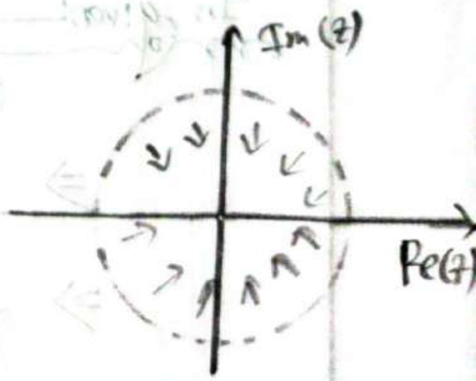
$$z=2 \quad B = \frac{3z}{z-1} = 6$$

$$\therefore f(z) = \frac{3}{z-1} + \frac{6}{2-z} \quad \cdots\cdots \text{(i)}$$

(a)

In given:—

$|z|<1$ $\qquad\qquad \dfrac{|z|}{2} < 1$



Now, from eq (i):—

$$f(z) = \frac{3}{-1(1-z)} + \frac{6}{2\left(1-\frac{z}{2}\right)}$$

$$= -3(1-z)^{-1} + 3\left(1-\frac{z}{2}\right)^{-1}$$

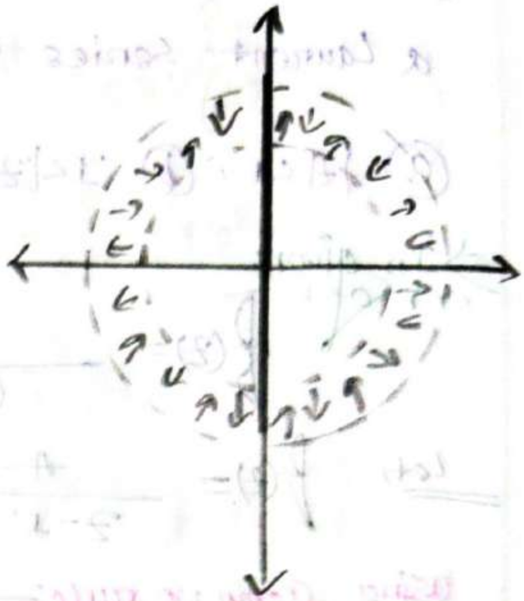
$$= -3\left(1+z+z^2+z^3+\cdots\right) + 3\left(1+z/2+\frac{z^2}{4}+\frac{z^3}{8}+\cdots\right)$$

Ans:—

(b)

In given:-

$$1<|z|<2$$



$$\Rightarrow 1<|z| \text{ or } |z|<2$$

$$\Rightarrow \frac{1}{|z|}<1 \text{ or } \frac{|z|}{2}<\frac{2}{2}$$

$$\Rightarrow \frac{1}{|z|}<1 \text{ or } \frac{|z|}{2}<1$$

$$\therefore f(z) = \frac{3}{z\left(1-\frac{1}{z}\right)} + \frac{6}{2\left(1-\frac{z}{2}\right)}$$

$$= \frac{3}{z}\left(1-\frac{1}{z}\right)^{-1} + 3\left(1-\frac{z}{2}\right)^{-1}$$

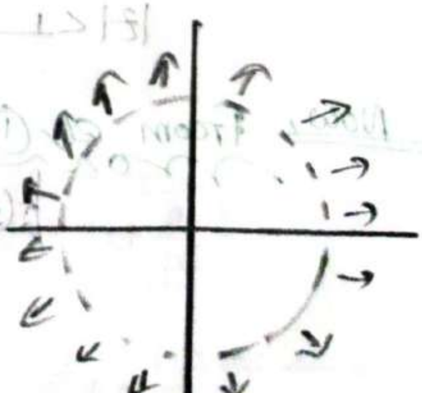
$$= \frac{3}{z}\left(1+\frac{1}{z}+\frac{1}{z^2}+\frac{1}{z^3}+\cdots\right) + 3\left(1+\frac{z}{2}+\frac{z^2}{4}+\cdots\right)$$

Ans:-

(c)

In given:-

$$|z|>2$$



$$\Rightarrow 1>\frac{2}{|z|} \text{ also } \frac{1}{|z|}<1$$

$$\Rightarrow \frac{2}{|z|}<1 \text{ and } \frac{1}{|z|}<1$$

from eq (1) ⇒

$$f(z) = \frac{3}{z\left(1-\frac{1}{z}\right)} + \frac{6}{-z\left(\frac{-2}{z}+1\right)}$$

$$\therefore f(z) = \frac{3}{z}\left(1-\frac{1}{z}\right)^{-1} - \frac{6}{z}\left(1-\frac{2}{z}\right)^{-1}$$

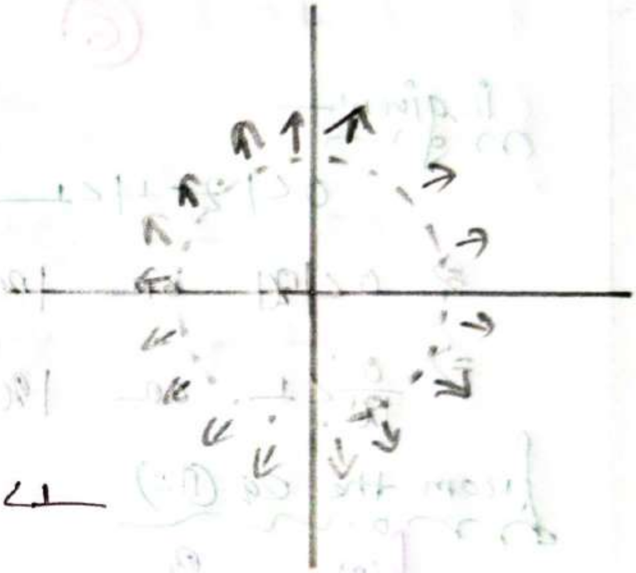
$$= \frac{3}{z}\left(1+\frac{1}{z}+\frac{1}{z^2}+\cdots\right) - \frac{6}{z}\left(1+\frac{2}{z}+\frac{4}{z^2}+\cdots\right)$$

Ans:-

(d)

In given:-

$$|z-1| > 2$$



Let:- $z-1 = x$

$$\therefore z = x+1$$

Now, $|x| > 2$

$$\frac{2}{|x|} < 1 \quad \text{also,} \quad \frac{1}{|x|} < 1$$

From eq (1) =)

$$f(z) = \frac{3}{z-1} + \frac{2+4}{2-z}$$

$$\therefore f(x) = \frac{3}{x+1-1} + \frac{2+4}{2-x-1}$$

$$= \frac{3}{x} + \frac{2+4}{1-x}$$

$$= \frac{3}{x} + \frac{2+4}{-x\left(1-\frac{1}{x}\right)}$$

$$= \frac{3}{x} - \frac{2+4}{x}\left(1-\frac{1}{x}\right)^{-1}$$

$$= \frac{3}{x} - \frac{2+4}{x}\left(1+\frac{1}{x}+\frac{1}{x^2}+\frac{1}{x^3}+\cdots\right)$$

Now subistitute the value of x:—

$$f(z) = \frac{3}{z-1} - \frac{24}{z-1} \times \left(1 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \cdots\cdots\right)$$

$$= \frac{3}{z-1} - \frac{6}{z-1}\left(1 + \frac{1}{z-1} + \frac{1}{(z-1)^2} + \frac{1}{(z-1)^3} + \cdots\cdots\right)$$

Ans:—

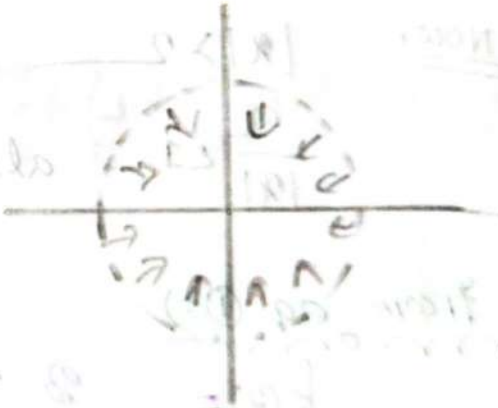
(c)

is given:—

$0 < |z-1| < 1$ ⇒ Let, $z-1 = x$ ⇒ $x+1 = z$

⇒ $0 < |x|$ or $|x| < 1$

⇒ $\frac{0}{|x|} < 1$ or $|x| < 1$



from the eq (1):)

$$f(z) = \frac{3}{z-1} + \frac{6}{2-z}$$

$$\therefore f(x) = \frac{3}{x+1-1} + \frac{6}{2-x-1}$$

$$= \frac{3}{x} + \frac{6}{1-x}$$

$$= \frac{3}{x} + 6(1-x)^{-1}$$

$$= \frac{3}{x} + 6(1+x+x^2+x^3+\cdots)$$

$$= \frac{x}{z-1} + 6\left(1+(z-1)+(z-1)^2+\cdots\cdots\right)$$

Ans:—

② $f(z) = \dfrac{5z}{(z^2+1)(z+2)}$ in a Laurent series valid for

ⓐ $1<|z|<2$

ⓑ $|z|>2$

⇒ In given:—

$$f(z) = \frac{5z}{(z^2+1)(z+2)} = \frac{A}{z+2} + \frac{Bz+C}{z^2+1} \quad \text{—— (I)}$$

Multiply $(z^2+1)(z+2)$ on Both side:—

$$5z = A(z^2+1) + (Bz+C)(z+2) \quad \text{—— (II)}$$

$$5z = Az^2 + A + Bz^2 + 2Bz + Cz + 2C \quad \text{—— (III)}$$

if $(z+2)=0$

$\therefore z=-2$ put in eq (II) ⇒

$$A = \frac{-5\times2}{(-2)^2+1} = -2$$

Equating like term:—

$$Az^2 + Bz^2 = 0$$

$$B = -A = 2$$

$$\therefore A + 2C = 0$$

$$C = 1$$

put A, B, C value in eq (I) ⇒

$$f(z) = \frac{-2}{z+2} + \frac{2z}{z^2+1} + \frac{1}{z^2+1} \quad \text{—— (IV)}$$

(a)

$1<|z|<2$

$\Rightarrow 1<|z|$ or $|z|<2$

$\Rightarrow \frac{1}{|z|}<1$ and, $\frac{|z|}{2}<1$

$\Rightarrow \frac{1}{|z^2|}<1$ and $\frac{|z|}{2}<1$

From the equation (v):—

$$f(z) = \frac{-2}{2\left(1+\frac{z}{2}\right)} + \frac{2z}{z^2\left(1+\frac{1}{z^2}\right)} + \frac{1}{z^2\left(1+\frac{1}{z^2}\right)}$$

$$= -\left(1+\frac{z}{2}\right)^{-1} + \frac{2}{z}\left(1+\frac{1}{z^2}\right) + \frac{1}{z^2}\left(1+\frac{1}{z^2}\right)^{-1}$$

$$= -\left(1-\frac{z}{2}+\frac{z^2}{4}-\cdots\right) + \frac{2}{z}\left(1-\frac{1}{z^2}+\frac{1}{z^4}-\cdots\right) + \frac{1}{z^2}\left(1-\frac{1}{z^2}+\frac{1}{z^4}-\cdots\right)$$

Ans:-

(b)

$|z|>2$

$\Rightarrow 1>\frac{2}{|z|} \Rightarrow \frac{2}{|z|}<1$ also, $\frac{1}{|z|}<1$

From the equation (vi):—

$$f(z) = \frac{-2}{z\left(1+\frac{2}{z}\right)} + \frac{2z}{z^2\left(1+\frac{1}{z^2}\right)} + \frac{1}{z^2\left(1+\frac{1}{z^2}\right)}$$

$$= -\frac{2}{z}\left(1+\frac{2}{z}\right)^{-1} + \frac{2}{z}\left(1+\frac{1}{z^2}\right)^{-1} + \frac{1}{z^2}\left(1+\frac{1}{z^2}\right)^{-1}$$

$$= -\frac{2}{z}\left(1-\frac{2}{z}+\frac{4}{z^2}-\cdots\right) + \frac{2}{z}\left(1-\frac{1}{z^2}+\frac{1}{z^4}-\cdots\right) + \frac{1}{z^2}\left(1-\frac{1}{z^2}+\frac{1}{z^4}-\cdots\right)$$

Ans:-

(3) Find the function $f(z)$

(a) $1+z+z^2+z^3+\cdots$

(b) $1-z+z^2-z^3+\cdots$

(c) $1+2z+3z^2+4z^3+\cdots$

(d) $1-2z+3z^2-4z^3+\cdots$

$\Rightarrow$ we know:—

$$f(z) = (1+z)^{-1} = 1 - z + z^2 - z^3 + \cdots \quad \text{— (I)}$$

$$f'(z) = (1-z)^{-1} = 1 + z + z^2 + z^3 + \cdots \quad \text{— (II)}$$

(a)

In given:—

Compare

$$f(z) = 1 + z + z^2 + z^3 + \cdots$$

Compare it in eq (II) $\Rightarrow$

$$\therefore f(z) = (1-z)^{-1}$$

$$\Rightarrow f(z) = \frac{1}{1-z}$$

Ans:—

(b)

In given:—

$$f(z) = 1 - z + z^2 - z^3 + \cdots$$

Compare it in eq (I) $\Rightarrow$

$$f(z) = (1+z)^{-1}$$

$$f(z) = \frac{1}{1+z}$$

Ans:—

(c)

In given:-

$$f(z) = 1 + 2z + 3z^2 + 4z^3 + \cdots$$

the general term of the series in $a_n = n \cdot z^{(n-1)}$

$$\therefore f(z) = \frac{d}{dz}\left(\sum_{n=0}^{\infty} z^n\right) = \sum_{n=0}^{\infty} n \cdot z^{(n-1)}$$

$$\therefore f(z) = \frac{1}{(1-z)^2}$$

Ans:-

(d)

In given:-

$$f(z) = 1 - 2z + 3z^2 - 4z^3 + \cdots$$

the general term of the series in

$$a_n = (-1)^n \cdot (n+1)$$

$$\therefore f(z) = \sum_{n=0}^{\infty} a_n \cdot z^n$$

$$= \sum_{n=0}^{\infty} (-1)^n \cdot (n+1) \cdot z^n$$

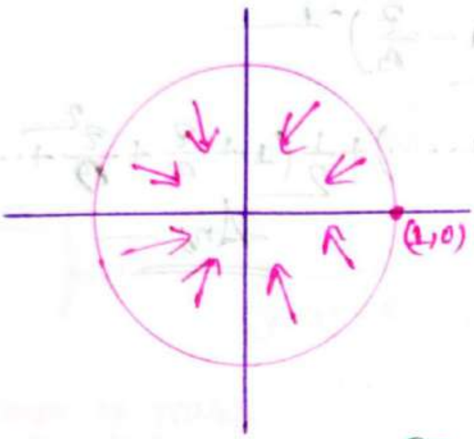
Ans:-

(4) Given function

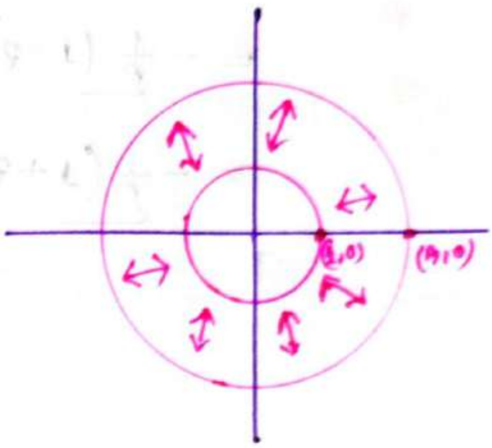
(i) $f(z) = \dfrac{z}{(z-1)(3-z)}$ ...... (I) use for a, b

(ii) $f(z) = \dfrac{z}{(z-1)(2-z)}$ ...... (II) use for c, d

(a)



(b)



$\Rightarrow$ (I)

In given:—

$$f(z) = \frac{z}{(z-1)(3-z)} = \frac{A}{z-1} + \frac{B}{3-z} \quad \text{———— (1)}$$

∴ using cover up rule:—

$$z = 1, \quad A = \frac{z}{3-z} = \frac{1}{3-1} = \frac{1}{2}$$

$$z = 3, \quad B = \frac{z}{z-1} = \frac{3}{3-1} = \frac{3}{2}$$

put it in eq (1):-

$$f(z) = \frac{1}{2(z-1)} + \frac{3}{2(3-z)} \quad \text{———— (II)}$$

(a)

The Condition is:—

$|z| < 1$ also, $\frac{|z|}{3} < 1$

From eq (ii) ⇒

$$f(z) = \frac{1}{2\times -1(1-z)} + \frac{3}{2\times 3\left(1-\frac{z}{3}\right)}$$

$$= -\frac{1}{2}(1-z)^{-1} + \frac{1}{2}\left(1-\frac{z}{3}\right)^{-1}$$

$$= -\frac{1}{2}\left(1+z+z^2+z^3+\cdots\right) + \frac{1}{2}\left(1+\frac{z}{3}+\frac{z^2}{3}+\cdots\right)$$

Ans:—

(b)

The Condition is:—

$1 < |z| < 3$

$\Rightarrow 1 < |z|$ and $|z| < 3$

$\Rightarrow \frac{1}{|z|} < 1$ and $\frac{|z|}{3} < 1$

From eq (ii) ⇒

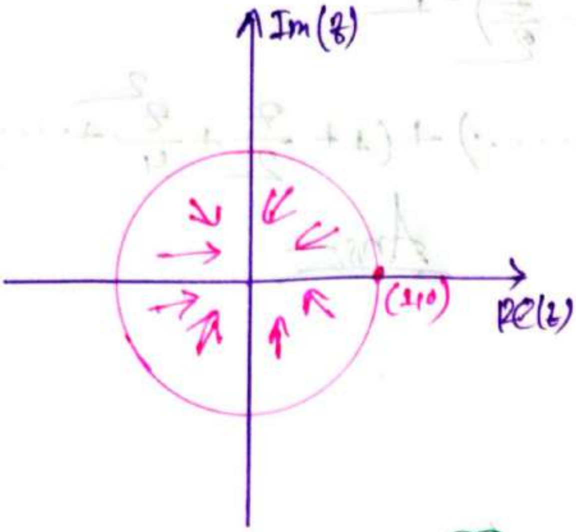
$$f(z) = \frac{1}{2z\left(1-\frac{1}{z}\right)} + \frac{3}{3\times 2\left(1-\frac{z}{3}\right)}$$

$$= \frac{1}{2z}\left(1-\frac{1}{z}\right)^{-1} + \frac{1}{2}\left(1-\frac{z}{3}\right)^{-1}$$

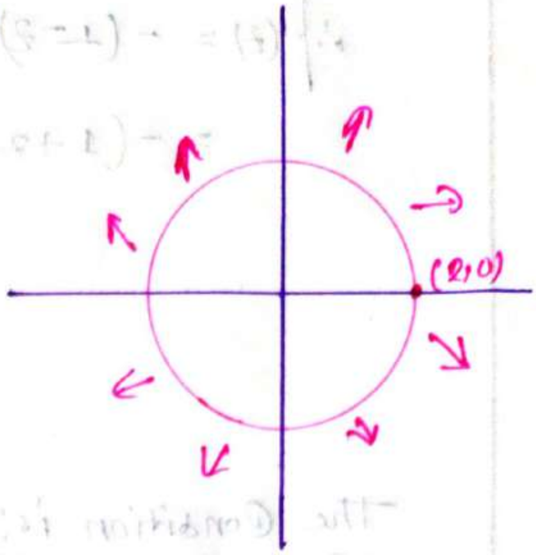
$$= \frac{1}{2z}\left(1+\frac{1}{z}+\frac{1}{z^2}+\cdots\right) + \frac{1}{2}\left(1+\frac{z}{3}+\frac{z^2}{9}+\cdots\right)$$

Ans:—

(c)



(d)



(ii)

In given:—

$$f(z) = \frac{z}{(z-1)(2-z)} = \frac{A}{z-1} + \frac{B}{2-z} \quad \text{—— (I)}$$

using cover up rule:—

$$z=1;\ A = \frac{z}{2-z} = \frac{1}{2-1} = 1$$

$$z=2;\ B = \frac{z}{z-1} = \frac{2}{2-1} = 2$$

put the value of A and B in eq (I) ⇒

$$\therefore f(z) = \frac{1}{z-1} + \frac{2}{2-z} \quad \text{—— (II)}$$

(a)

- the condition is:—

$$|z| < 1 \quad \text{also,} \quad \frac{|z|}{2} < 1$$

From eq (II) ⇒

$$f(z) = \frac{1}{-1(1-z)} + \frac{2}{2\left(1-\frac{z}{2}\right)}$$

$$\therefore f(z) = -(1-z)^{-1} + \left(1-\frac{z}{2}\right)^{-1}$$

$$= -(1+z+z^2+z^3+\cdots\cdots) + \left(1+\frac{z}{2}+\frac{z^2}{4}+\cdots\cdots\right)$$

Ans:—

(d)

The Condition is:—

$$|z| > 2$$

$$\Rightarrow 1 > \frac{2}{|z|}$$

$$\Rightarrow \frac{2}{|z|} < 1 \quad \text{also, } \frac{|z|}{1} < 1$$

From eq (ii) ⇒

$$f(z) = \frac{1}{-1(1-z)} + \frac{2}{-z\left(1-\frac{2}{z}\right)}$$

$$= -(1-z)^{-1} - \frac{2}{z}\left(1-\frac{2}{z}\right)^{-1}$$

$$= -(1+z+z^2+z^3+\cdots) - \frac{2}{z}\left(1+\frac{2}{z}+\frac{4}{z^2}+\cdots\right)$$

Ans:—

#mappings

Choy

Mapping

$$W = z + c$$

Where,

$W = u + iv$

$z = x + iy$

$c = a + ib$

Hence,
$$u + iv = x + iy + a + ib$$
$$= x + a + i(y + b)$$

Translation:

(1) $w = z + c$

Rotation:

(2) $w = iz = ire^{i\theta}$

$w = u + iv$

$z = re^{i\theta}$

Reflectation:

(3) $w = \bar{z} = x - iy$

$w = u + iv, \ \bar{z} = x - iy$

Examples — ~~Translation~~

⇒ The rectangular region R in z-plane which is bounded by the lines.

$$x = 0, \quad x = 2$$
$$y = 0, \quad y = 1$$

Determine the region R' of the w-plane into which R is mapped under the transformation; $w = z + 1$

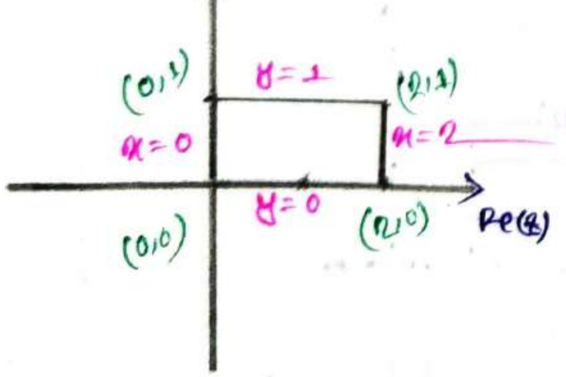
⇒ Is given:–

$$w = z + 1$$
$$= x + iy + 1$$

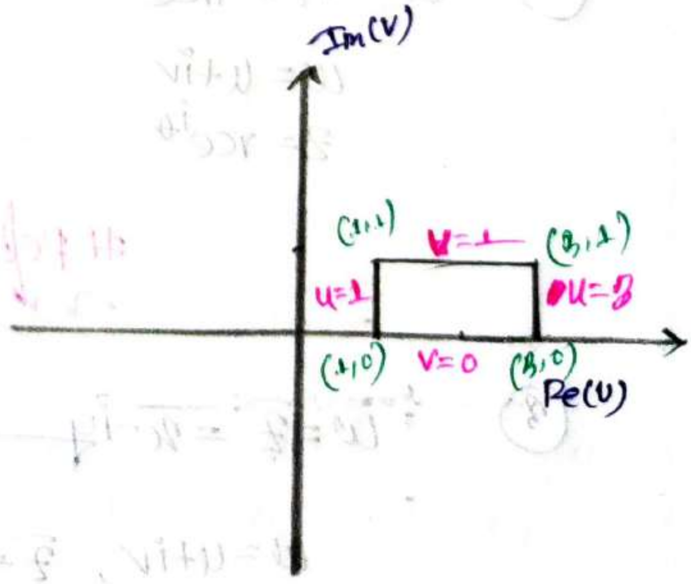
$$\therefore \quad u = x + 1$$
$$v = y$$

NOW,

$$x = 0, \quad u = 1$$
$$y = 0, \quad v = 0$$
$$x = 2, \quad u = 3$$
$$y = 1 \quad v = 1$$



z-plane



w-plane

Example: Rotation:

⇒ Let the rectangular region R in z-plane which is bounded by the lines.

$$x = 0, \quad x = 2$$
$$y = 0, \quad y = 1$$

Determine the region R in z-plane into which is mapped under the transformation $w = iz$

⇒ In given:

$$
\begin{aligned}
w &= iz \\
&= i(x+iy) \\
&= ix + i^2 y \\
&= -y + ix
\end{aligned}
$$

$\therefore \; u = -y$

$v = x$

Now:

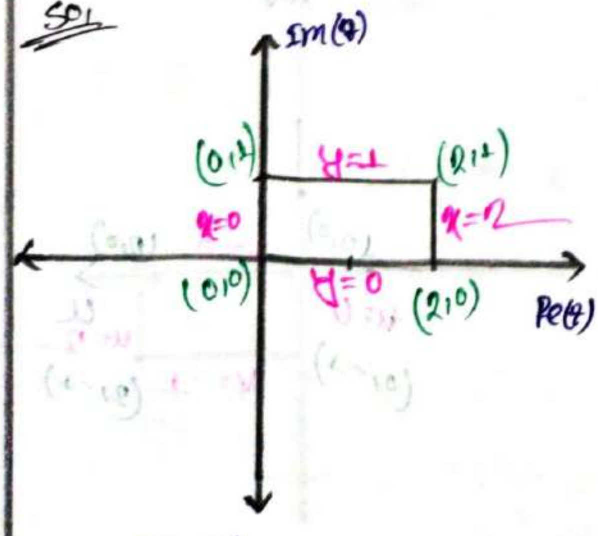
$x = 0, \quad v = 0$

$y = 0 \quad u = 0$

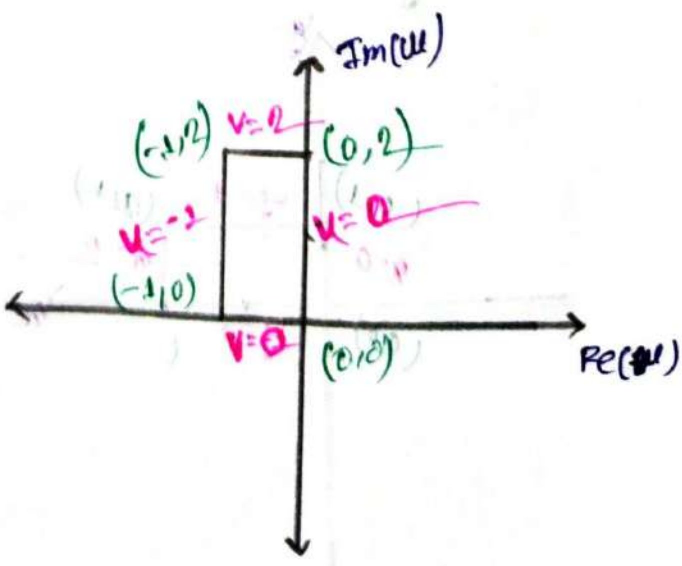
$x = 2, \quad v = 2$

$y = 1 \quad u = -1$

Sol.



Z-Plane



w-plane

Example:- ~~Reflection~~ Reflection:-

⇒ Let the rectangular region R in the z-plane which is bounded by the lines,

$$x=0,\quad y=0$$
$$x=2,\quad y=1$$

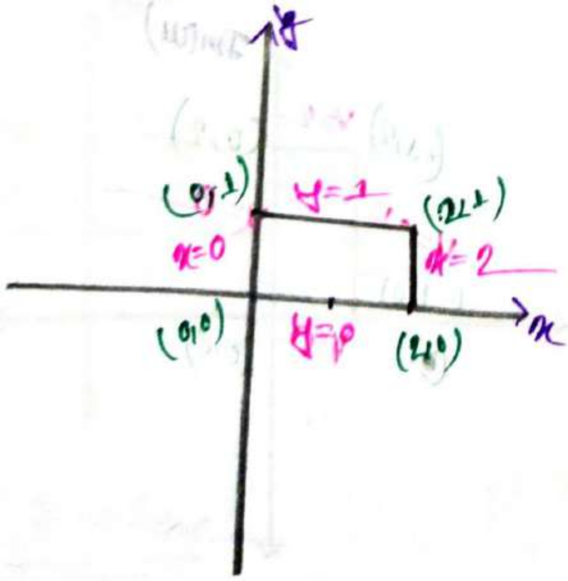
Determine the region R' of the ω-plane into which R is mapped under the transformation:- $\omega = \bar{z}$

⇒ In given:-

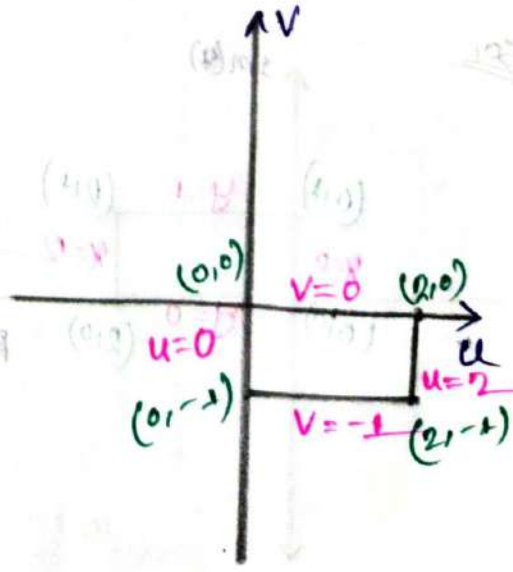
$$\omega = \bar{z}$$
$$\therefore \omega = x - iy$$
$$\therefore u = x$$
$$v = -y$$

Sol

$$x = 0,\quad u = 0$$
$$y = 0,\quad v = 0$$
$$x = 2,\quad u = 2$$
$$y = 1,\quad v = -1$$



z-plane



ω-plane

Example

The given triangle T in the z-plane with vertices at

$-1+2i,\ 1-2i,\ 1+2i$

Determine the triangle T' of the w-plane into which T in mapped under the transformation $w = \sqrt{2}e^{\frac{\pi i}{4}} \times z$

⇒ In given:

$$w = \sqrt{2}e^{\frac{\pi i}{4}} \times z$$

$$= \sqrt{2}\left[\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right] \times (x+iy)$$

$$= \sqrt{2}\left(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\right)(x+iy)$$

$$= (1+i)(x+iy) = (x+iy)+(ix-y)$$

Now,

$$= (x-y)+i(x+y)$$

$$\therefore u = x-y \quad \text{......(I)}$$

$$v = x+y \quad \text{......(II)}$$

(I) + (II) ⇒

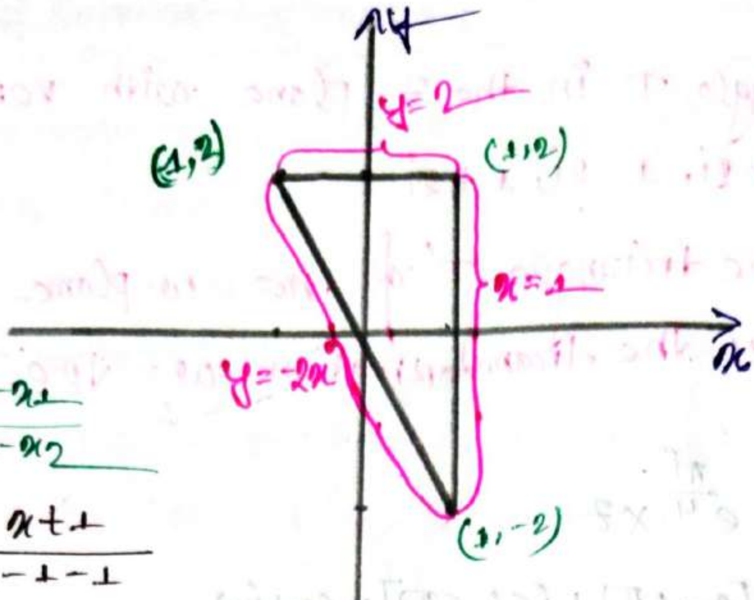
$$u+v = x-y+x+y$$

$$u+v = 2x \quad \text{......(III)}$$

(I) − (II) ⇒

$$u-v = x-y-x-y$$

$$= -2y \quad \text{......(IV)}$$



$$\frac{y-y_1}{y_1-y_2} = \frac{x-x_1}{x_1-x_2}$$

$$\Rightarrow \frac{y-2}{2+2} = \frac{x+1}{-1-1}$$

$$\Rightarrow -2y+4 = 4x+4$$

$$\Rightarrow y = -2x$$

Z - Plane:—

Now:—

put $x=1$ in eq (iii) $\Rightarrow$

$$u+v = 2\times 1$$

$$\boxed{\therefore u+v = 2}$$

put $y=2$ in eq (iv) $\Rightarrow$

$$u-v = -2\times 2$$

$$\boxed{\therefore u-v = -4}$$

From eq (iv) $\Rightarrow$

$$u-v = -2y = -2\times -2x$$

$$= 4x$$

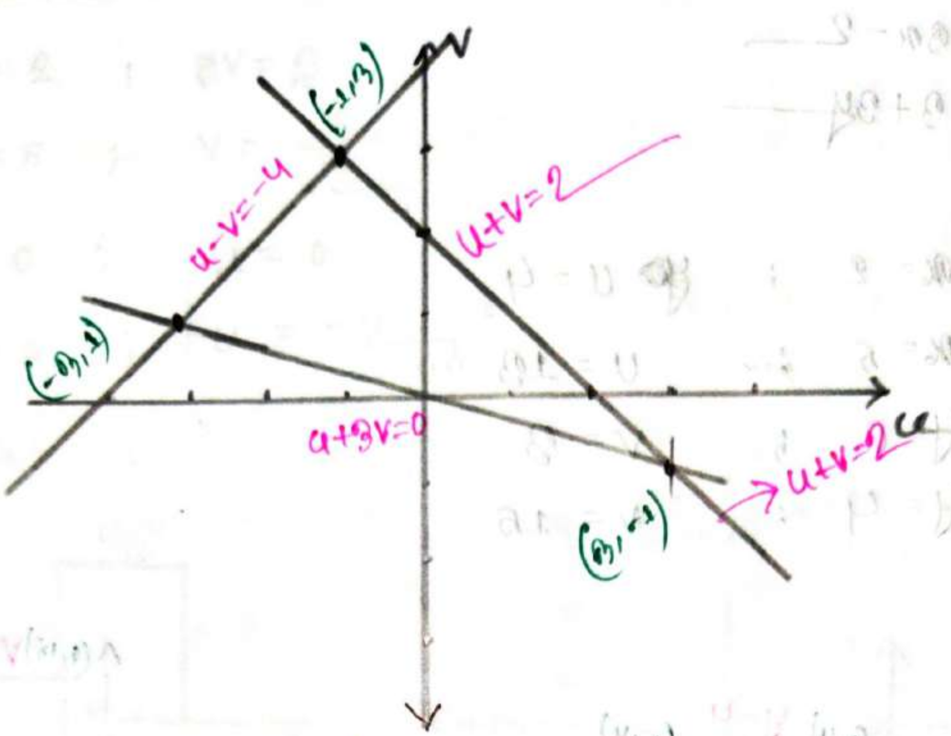
$$\Rightarrow u-v = 2(u+v)$$

$$\Rightarrow u-v = 2u+2v$$

$$\Rightarrow 2u-u+2v+v = 0$$

$$\Rightarrow \boxed{u+3v=0}$$

$\therefore$ $3+3v=0$, $\therefore v=-1$, | $-3+3v=0$, $v=1$ | $0+3v=0$, $v=0$



w - Plane

Exercise set

(1) Let the rectangular region R in z-plane which is bounded by the lines

$x=2, \quad y=0$

$x=5, \quad y=4$

Determine the region R' of the w-plane into which R is mapped under the following transformation.

(i) $w = 3z-(2+3i)$

$\Rightarrow$ In given:

$$w = 3z-(2+3i)$$
$$= 3(x+iy)-2+3i$$
$$= (3x-2)+i(3+3y)$$

$\therefore u = 3x - 2$

$v = 3 + 3y$

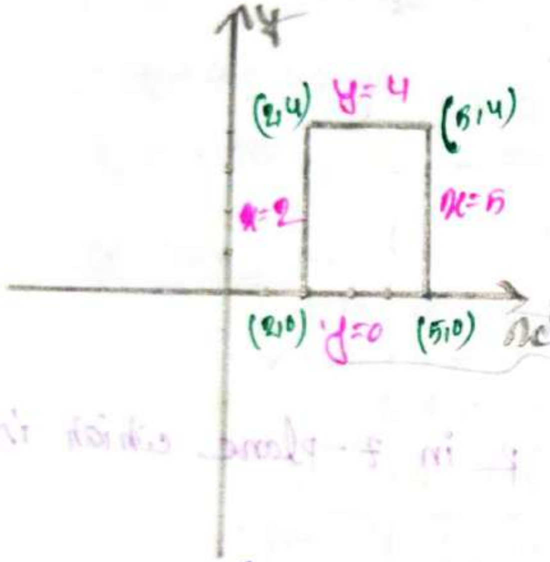
Now:-

$x = 2$; $u = 4$

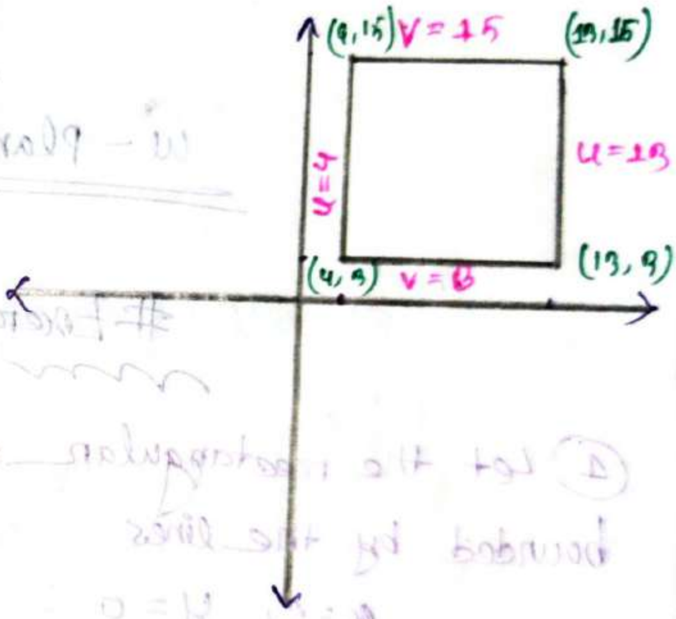
$x = 5$; $u = 13$

$y = 0$; $v = 3$

$y = 4$; $v = 15$



z - Plane



w- Plane:-

(ii) $w = \frac{1}{2} e^{\frac{\pi i}{2}} \times z + 2i$

$\Rightarrow$ In given:-

$$w = \frac{1}{2} e^{\frac{\pi i}{2}} \times z + 2i$$

$$= \frac{1}{2} \times \left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) \times (x + iy) + 2i$$

$$= \frac{1}{2} ix + \frac{1}{2} i^2 y + 2i$$

$$= -\frac{1}{2} y + i\left(\frac{1}{2} x + 2\right)$$

$\therefore u = -\frac{1}{2} y$

$v = \frac{1}{2} x + 2$

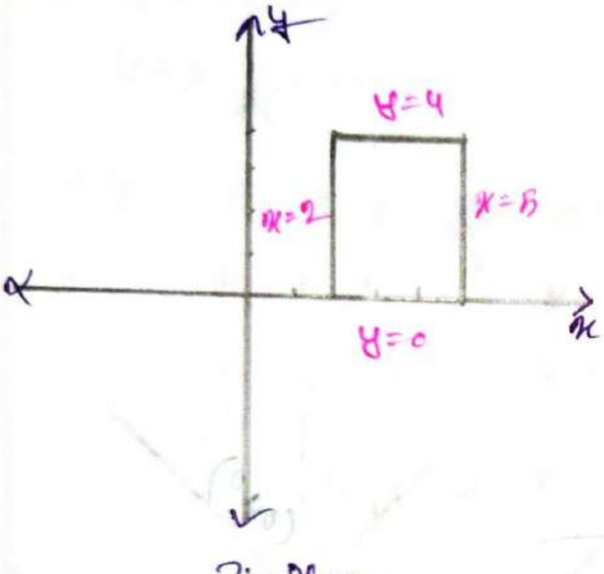
Sol

$x = 2 \quad ; \quad v = 3$

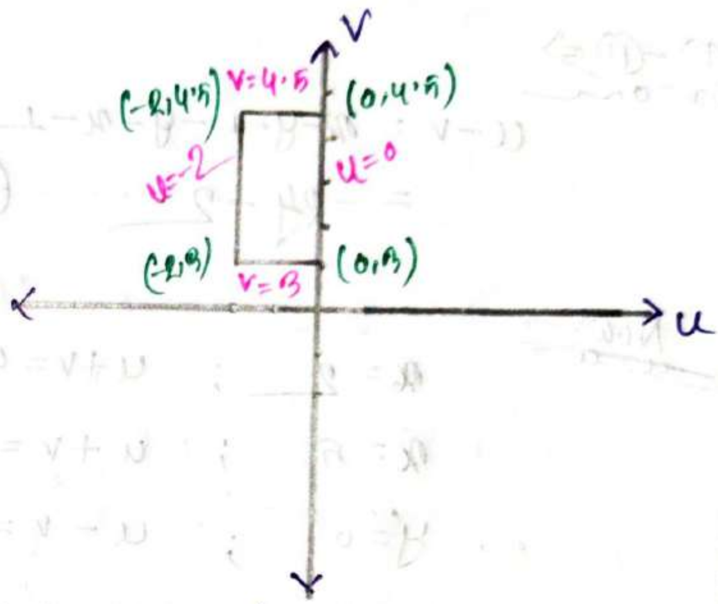
$x = 5 \quad ; \quad v = \frac{9}{2}$

$y = 0 \quad ; \quad u = 0$

$y = 4 \quad ; \quad u = -2$



Z-Plane



w-Plane

(iii) $w = \sqrt{2}e^{\frac{\pi i}{4}} \times z - (1 - i)$

$\Rightarrow w = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) \times (x+iy) - 1 + i$

$\Rightarrow w = (1+i)(x+iy) - 1 + i$

$\Rightarrow x + iy + ix + i^2 y - 1 + i$

$\Rightarrow x + iy + ix - y - 1 + i$

$\Rightarrow (x-y) - 1 + i(y+x+1)$

$\Rightarrow (x-y-1) + i(y+x+1)$

Now:

$u = x - y - 1$ ...... (i)

$v = y + x + 1$ ...... (ii)

(i) + (ii) ⇒

$u + v = x - y - 1 + y + x + 1$

$= 2x$ ...... (iii)

(i) − (ii) ⇒

$u - v = x - y - 1 - y - x - 1$

$= -2y - 2$ ...... (iv)

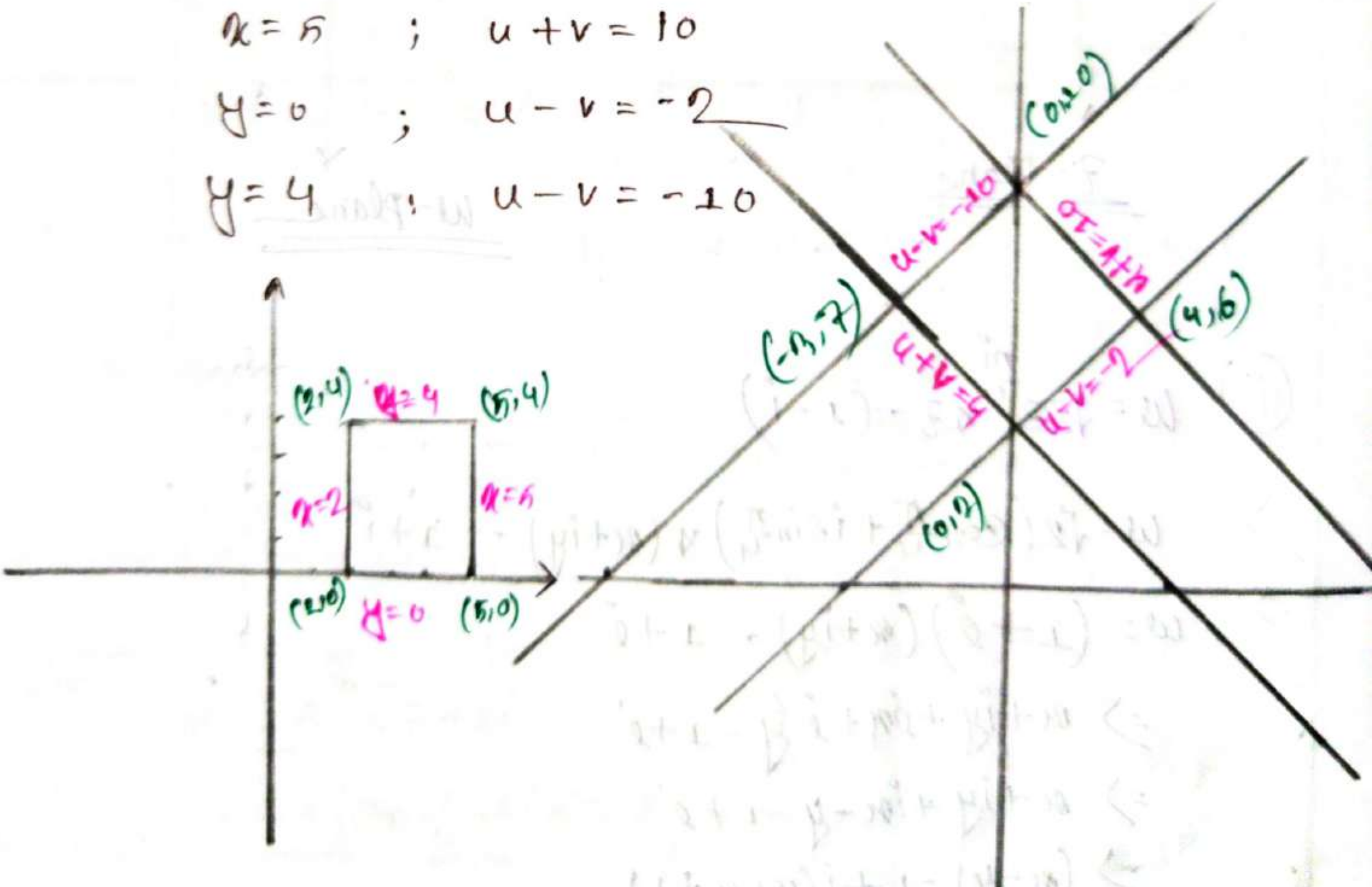
Now:

$x = 2$; $u + v = 4$

$x = 5$; $u + v = 10$

$y = 0$; $u - v = -2$

$y = 4$; $u - v = -10$



z - Plane

w - Plane

(iv) $w = e^{i\pi} \times z + 3 + i$

$\Rightarrow w = (\cos\pi + i\sin\pi)(x+iy) + 3 + i$

$\Rightarrow \quad = -1(x+iy) + 3 + i$

$= -x + 3 + i(1-y)$

$\therefore u = -x+3$

$v = 1-y$

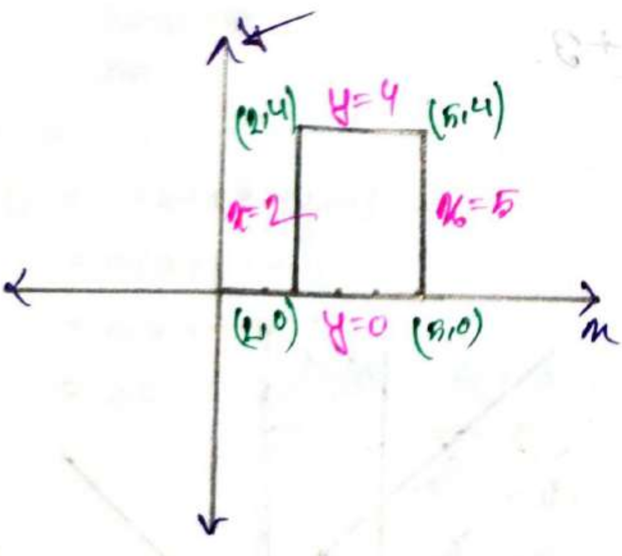
Now:–

$x = 2$; $\quad u = 1$

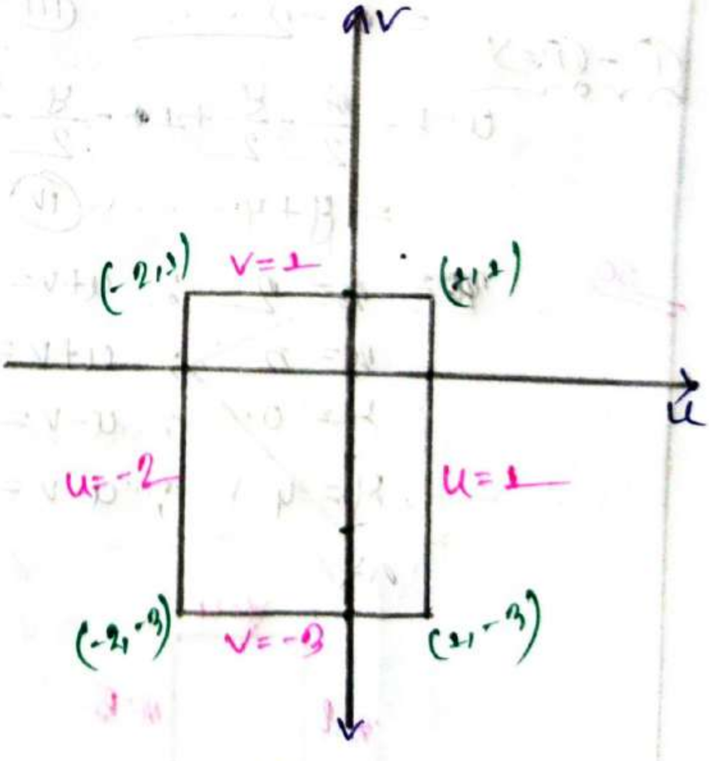
$x = 5$; $\quad u = -2$

$y = 0$; $\quad v = 1$

$y = 4$; $\quad v = -3$



z - Plane



w- plane

(v) $w = \frac{1}{\sqrt{2}} e^{\pi/4\, i} \times z + 1 - 3i$

$\Rightarrow w = \frac{1}{\sqrt{2}} \left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4} \right) \times (x+iy) + 1 - 3i$

$= \frac{x}{2} + \frac{iy}{2} + \frac{xi}{2} + \frac{i^2 y}{2} + 1 - 3i$

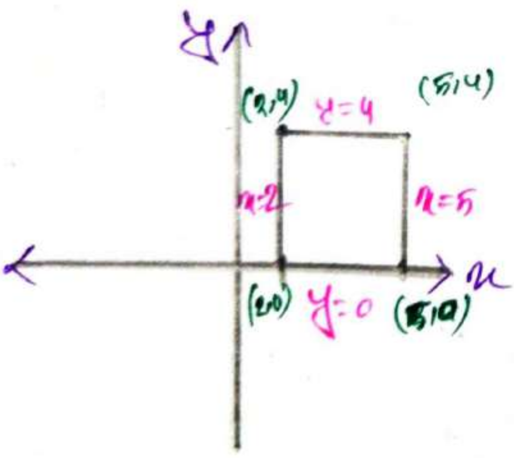
$= \frac{x}{2} - \frac{y}{2} + 1 + \frac{xi}{2} + \frac{yi}{2} - 3i$

$\therefore u = \frac{x}{2} - \frac{y}{2} + 1$

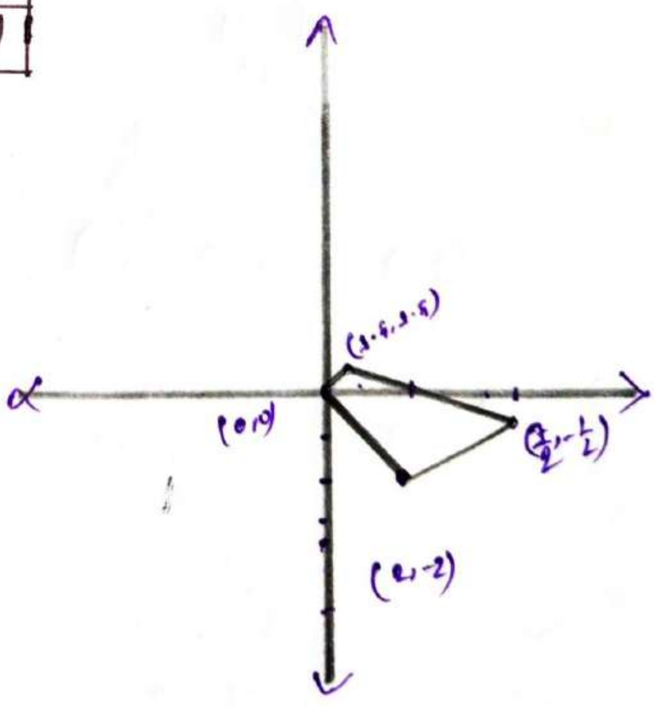
$v = \frac{x}{2} + \frac{y}{2} - 3$

Now:-

(x,y)	u	v	(u,v)
$(2,0)$	2	-2	$(2,-2)$
$(2,4)$	0	0	$(0,0)$
$(5,0)$	$\frac{7}{2}$	$-\frac{1}{2}$	$(\frac{7}{2}, -\frac{1}{2})$
$(5,4)$	$\frac{3}{2}$	$\frac{3}{2}$	$(\frac{3}{2}, \frac{3}{2})$



z - plane



(2) Given triangle T in the z-plane with vertices at 1, 1−3i and 3−i. Determine the triangle T′ of the w-plane into which mapped under the following

(i) $w = 3z+1-3i$

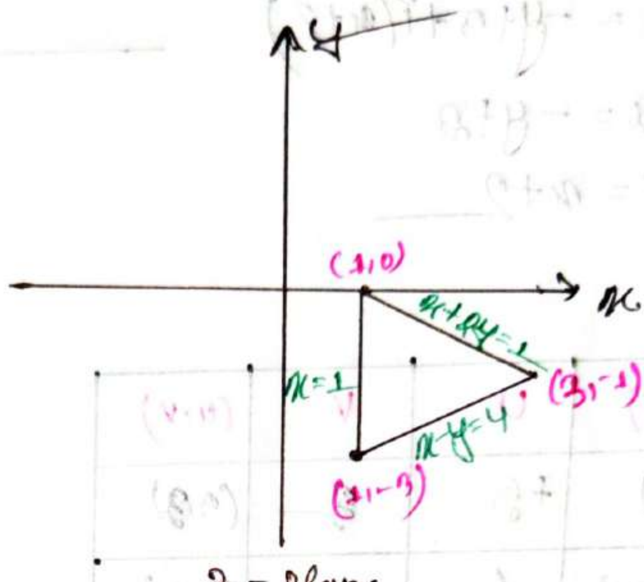
$\Rightarrow w = 3(x+iy)+1-3i$

$= 3x+3iy+1-3i$

$= (3x+1)+i(3y-3)$

$\therefore u = 3x+1$

$v = 3y-3$



z - plane

Now:

(x,y)	u=3x+1	v=3y−3	(u,v)
(1,0)	4	−3	(4,−3)
(1,−3)	4	−12	(4,−12)
(3,−1)	10	−6	(10,−6)

For (4,−3) to (10,−6)

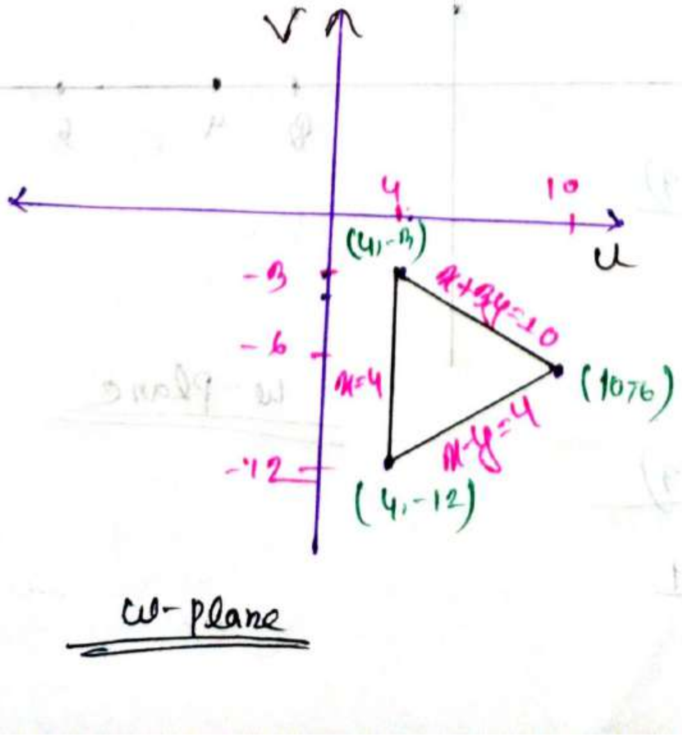
$$\frac{y+3}{-3+6} = \frac{x-4}{4-10}$$

$$x+3y = 10$$

For (10,−6) to (4,−12)

$$\frac{y-6}{-6+12} = \frac{x-10}{10-4}$$

$$x-y = 4$$



w- plane

(ii) $w = iz + 3 + 2i$

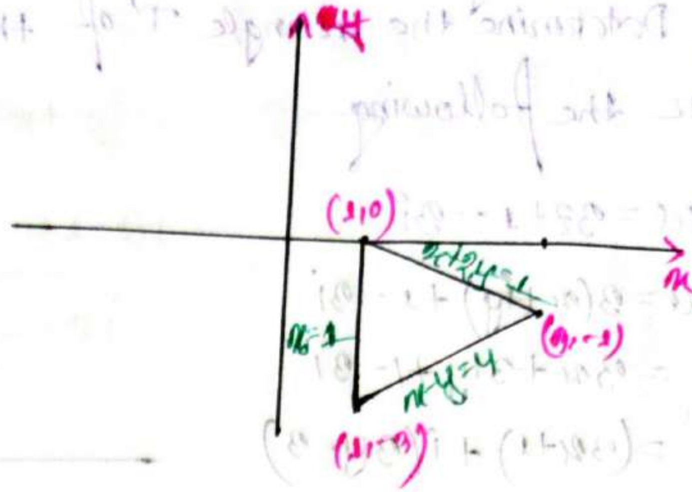
$\Rightarrow w = i(x+iy) + 3 + 2i$

$= ix + i^2 y + 3 + 2i$

$= -y + 3 + i(x+2)$

$\therefore \; u = -y + 3$

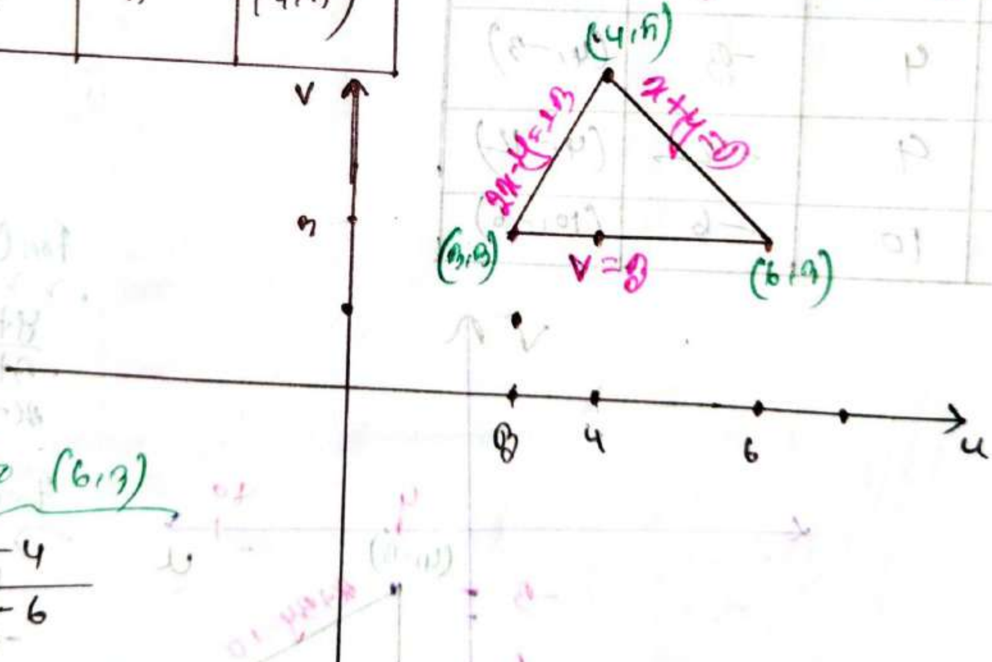
$v = x + 2$



Now:-

(x,y)	u	v	(u,v)
$(1,0)$	$+3$	3	$(3,3)$
$(1,-3)$	6	3	$(6,3)$
$(3,-1)$	4	5	$(4,5)$

So



w - Plane

For $(4,5)$ to $(6,3)$

$$\frac{y-5}{5-3} = \frac{x-4}{4-6}$$

$$x + y = 9$$

For $(4,5)$ to $(3,3)$

$$\frac{y-5}{5-3} = \frac{x-4}{4-3}$$

$$2x - y = 3$$

(iii) $w=(1+2i)\times z - i$

$\Rightarrow w=(1+2i)(x+iy)-i$

$= x+iy+i2x-2y-i$

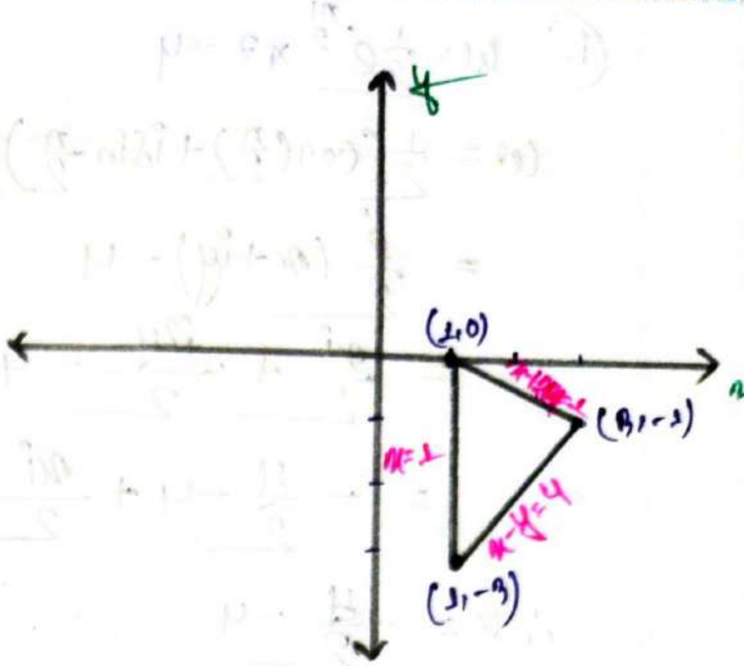
$=(x-2y)+i(2x+y-1)$

$\therefore u = x-2y$

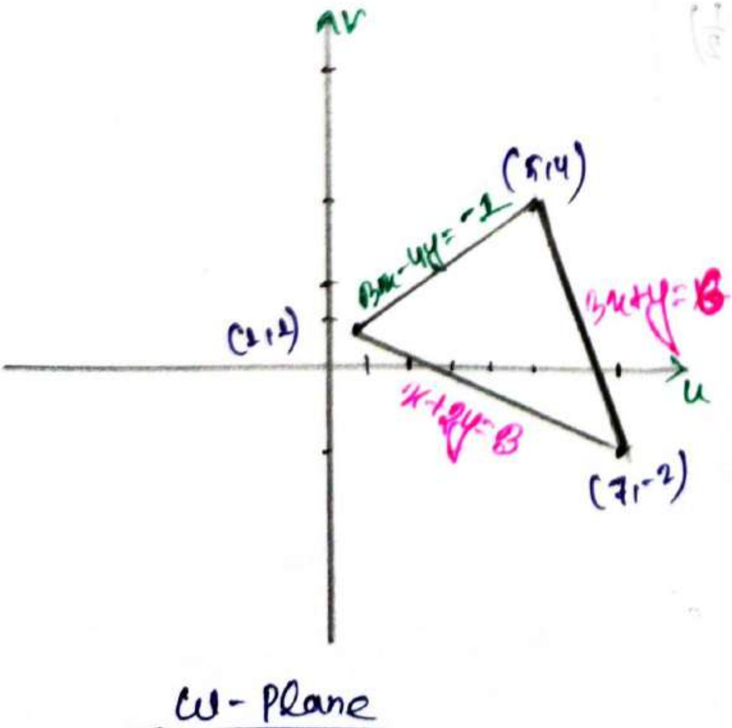
$v = 2x+y-1$

Now:-

(x,y)	$u=x-2y$	$v=2x+y-1$	(u,v)
$(1,0)$	1	1	$(1,1)$
$(1,-3)$	7	-2	$(7,-2)$
$(3,-1)$	5	4	$(5,4)$



Z-Plane:-



W-Plane

For (1,1) to (5,4):-

$\frac{y-1}{1-5}=\frac{x-1}{1-5}$

$\therefore 3x-4y=-1$

For (1,1) to (7,-2)

$\frac{y-1}{1+2}=\frac{x-1}{1-7}$

$x+2y=3$

For (5,4) to (7,-2)

$\frac{y-4}{4+2}=\frac{x-5}{5-7}$

(iv) $w = \frac{1}{2} e^{\frac{\pi i}{2}} z - 4$

$w = \frac{1}{2}\left(\cos\left(\frac{\pi}{2}\right) + i\sin\frac{\pi}{2}\right)(x+iy) - 4$

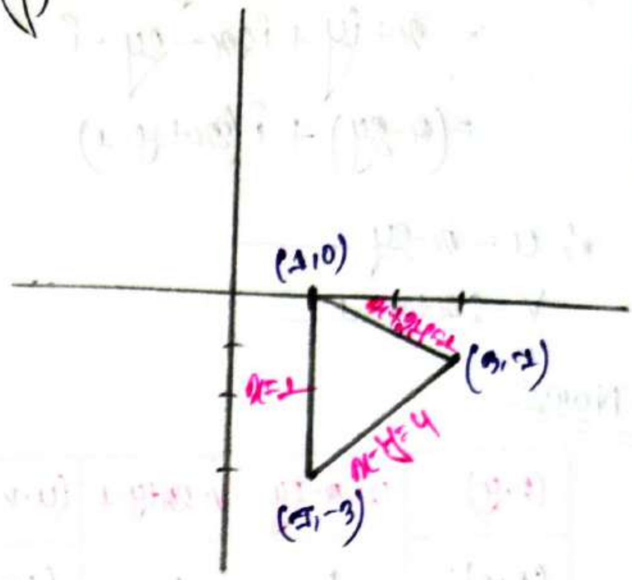
$= \frac{i}{2}(x+iy) - 4$

$= \frac{xi}{2} + \frac{i^2 y}{2} - 4$

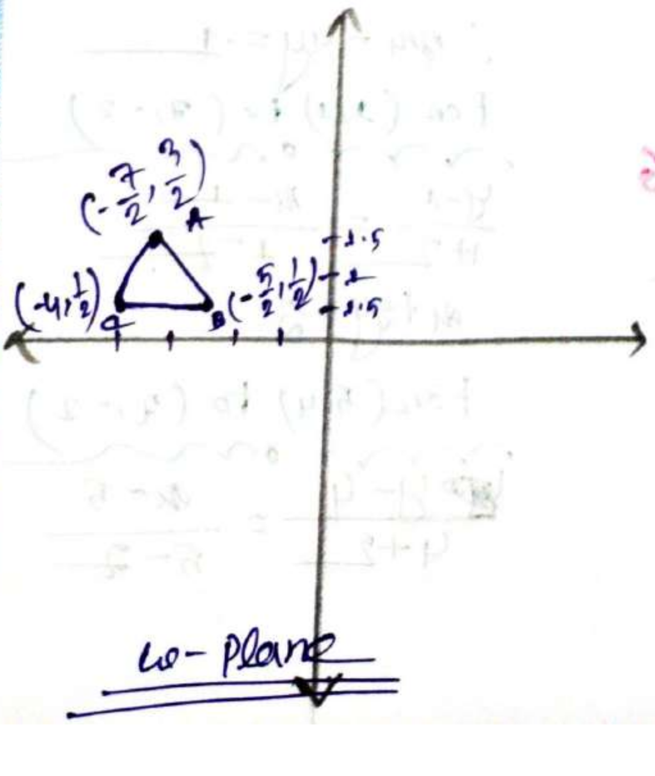
$= -\frac{y}{2} - 4 + \frac{xi}{2}$

$\therefore u = -\frac{y}{2} - 4$

$v = \frac{x}{2}$



(x,y)	u	v	(u,v)
(1,0)	-4	$\frac{1}{2}$	$(-4, \frac{1}{2})$
(3,-1)	$-\frac{7}{2}$	$\frac{3}{2}$	$(-\frac{7}{2}, \frac{3}{2})$
(1,-3)	$-\frac{5}{2}$	$\frac{1}{2}$	$(-\frac{5}{2}, \frac{1}{2})$



w- plane